

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)

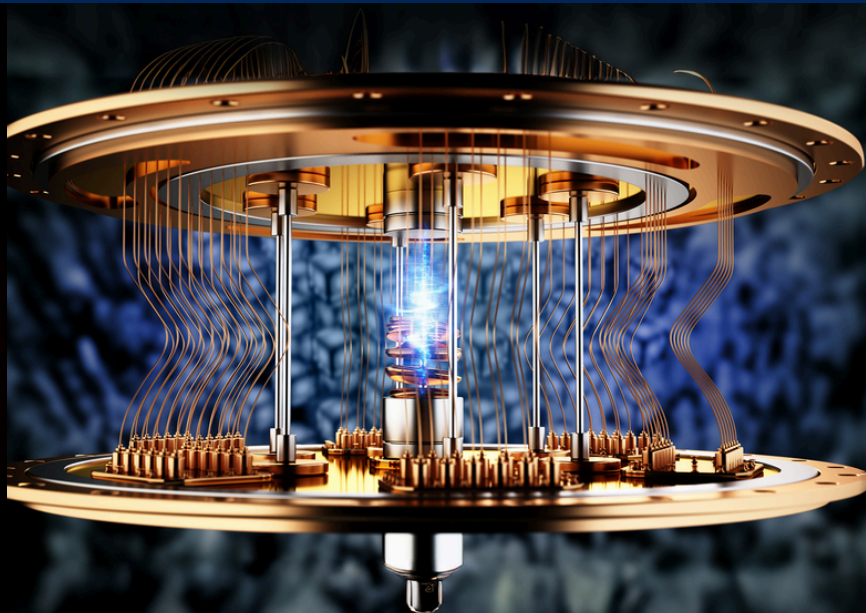
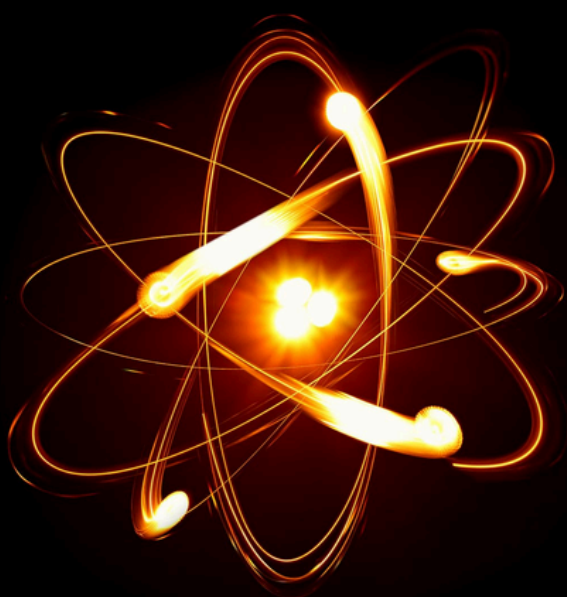
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LEARNING MATERIAL OF ADVANCED ENGINEERING PHYSICS (2600BS05)

B.TECH. I YEAR (I&II SEM COMMON TO ALL)
[BATCHES ADMITTED FROM THE ACADEMIC YEAR 2026 – 2027]

First Edition



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DR. UPENDER CH
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DEPARTMENT OF PHYSICS, H&S

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(2600BS05) ADVANCED ENGINEERING PHYSICS**I B.Tech. I SEMESTER ECE, CSE & IT****L T P C****II SEMESTER EEE, CSE(CS), CSE(AIML) & CSE(DS)****3 0 0 3****Course Objectives**

1. To study crystal structures, defects, and material characterization techniques like XRD.
2. To understand fundamental concepts of quantum mechanics and their applications in solids and nanomaterials.
3. To introduce quantum computing principles, quantum gates, and basic quantum algorithms.
4. To learn the properties and applications of magnetic and dielectric materials.
5. To explore the working and applications of lasers and fibre optics in modern technology.

Course Outcomes: Upon graduation:

CO1: To understand the crystal structures, identify defects, and apply XRD techniques for material characterization.

CO2: To apply quantum mechanical principles to explain particle behaviour and energy band formation in solids.

CO3: To analyze the advance concepts linear algebra for quantum computation along operators', Dirac's Bra and Ket notations and their properties.

CO4: To understand quantum computing concepts, use quantum gates and explain basic quantum algorithms.

CO5: To classify magnetic and dielectric materials and explain their properties, synthesis, and applications.

CO6: To explain the principles of lasers and fibre optics and their applications in communication and sensing.

PROGRAM OUTCOMES-POs

PO1	Engineering knowledge	An ability to apply knowledge of mathematics (including probability, statistics and discrete mathematics), science, and engineering for solving Engineering problems and modeling
PO2	Problem analysis	An ability to design, simulate and conduct experiments, as well as to analyze and interpret data including hardware and software components
PO3	Design / development of solutions	An ability to design a complex electronic system or process to meet desired specifications and needs
PO4	Conduct investigations of complex problems	An ability to identify, formulate, comprehend, analyze, design synthesis of the information to solve complex engineering problems and provide valid conclusions.
PO5	Modern tool usage	An ability to use the techniques, skills and modern engineering tools necessary for engineering practice
PO6	The engineer and society	An understanding of professional, health, safety, legal, cultural and social responsibilities
PO7	Environment and sustainability	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental and demonstrate the knowledge need for sustainable development.
PO8	Ethics	Apply ethical principles, responsibility and norms of the engineering practice
PO9	Individual and team work	An ability to function on multi-disciplinary teams.
PO10	Communication	An ability to communicate and present effectively
PO11	Project management and finance	An ability to use the modern engineering tools, techniques, skills and management principles to do work as a member and leader in a team, to manage projects in multi-disciplinary environments
PO12	Life-long learning	A recognition of the need for, and an ability to engage in, to resolve contemporary issues and acquire lifelong learning

COURSE OUTCOMES MAPPING WITH PROGRAMME OUTCOMES

Course out comes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	H	M	-	M	M	-	-	-	-	-	-	L
CO 2	H	M	-	M	M	-	-	-	-	-	-	L
CO 3	H	M	-	M	M	-	-	-	-	-	-	L
CO 4	H	M	-	M	M	-	-	-	-	-	-	L
CO 5	H	M	-	M	M	-	-	-	-	-	-	L
CO 6	H	M	-	M	M	-	-	-	-	-	-	L

H - High

M - Medium

L - Low

SYLLABUS

UNIT – I: CRYSTALLOGRAPHY & MATERIALS CHARACTERIZATION

Introduction: Unit cell, space lattice, basis, lattice parameters; crystal structures, Bravais lattices, packing factor: SC, BCC, FCC; Miller indices, inter-planar distance. Defects in crystals (Qualitative): point defects, line defects, surface defects and volume defects. Concept of nanomaterials: surface to volume ratio, X-ray diffraction: Bragg's law, powder method, calculation of average crystallite size using Debye Scherrer's formula.

UNIT – II: QUANTUM MECHANICS

Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box, Bloch's theorem (qualitative), Kronig-Penney model (Qualitative): E-k diagram, formation of energy bands, origin of bandgap, classification of solids, concept of discrete energy levels and quantum confinement in nanomaterials.

UNIT – III: QUANTUM COMPUTING

Introduction, linear algebra for quantum computation, Dirac's Bra and Ket notation and their properties, Hilbert space, Bloch's sphere, concept of quantum computer, classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, evolution of quantum systems, quantum measurements, entanglement, quantum gates, challenges and advantages of quantum computing over classical computation, quantum algorithms: Deutsch-Jozsa, Shor, Grover (Qualitative).

UNIT – IV: MAGNETIC AND DIELECTRIC MATERIALS

Magnetic Materials: Introduction to magnetic materials, origin of magnetic moment-classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, synthesis of ferrimagnetic materials using sol-gel method, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

Dielectric Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

UNIT – V: LASERS AND FIBER OPTICS

Lasers: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, CO₂ laser, semiconductor diode laser, applications: Bar code scanner, LIDAR for autonomous vehicle.

Fiber optics: Introduction to fiber optics, total internal reflection, construction of optical fiber, acceptance angle, numerical aperture, classification of optical fibers, losses in optical fiber, applications: optical fiber for communication system, sensor for structural health monitoring.

Textbooks :

1. Walter Borchardt-Ott, Crystallography: An Introduction, Springer.
2. Charles Kittel, Introduction to Solid State Physics, John Wiley & Sons, Inc.

3. Thomas G. Wong, Introduction to Classical and Quantum Computing, Rooted Grove.
4. A Textbook of Advanced Engineering Physics, Dr. T.V.S Arun Murthy, Dr. M. N. Avadhanulu, S. Chand, First edition, 2026.
5. Engineering Physics by P.K. Palanisamy, SciTech Publications.

Reference Books :

1. Jozef Gruska, Quantum Computing, McGraw Hill
2. Michael A. Nielsen & Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.
3. John M. Senior, Optical Fiber Communications Principles and Practice, Pearson Education Limited.

Useful Links :

- ♠ <https://shijuinpallotti.wordpress.com/wp-content/uploads/2019/07/optical-fiber-communications-principles-and-pr.pdf>
- ♠ https://www.geokniga.org/bookfiles/geokniga-crystallography_0.pdf
- ♠ <https://dpbck.ac.in/wp-content/uploads/2022/10/Introduction-to-Solid-State-PhysicsCharles-Kittel.pdf>
- ♠ <https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf>
- ♠ <https://www.fi.muni.cz/usr/gruska/qbook1.pdf>
- ♠ <https://profmcruz.wordpress.com/wp-content/uploads/2017/08/quantum-computation-and-quantum-information-nielsen-chuang.pdf>

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UNIT - I: CRYSTALLOGRAPHY & MATERIALS CHARACTERIZATION

Contents: - Introduction: Unit cell, space lattice, basis, lattice parameters; crystal structures, Bravais lattices, packing factor: SC, BCC, FCC; Miller indices, inter-planar distance. Defects in crystals (Qualitative), Concept of nanomaterials: surface to volume ratio, X -ray diffraction: Bragg's law, powder method, calculation of average crystallite size using Debye Scherrer's formula.

1.1. INTRODUCTION

Crystallography is the branch of science, which deals with the study of geometrical form and physical properties of crystalline solids, is called crystallography. Crystallography is the experimental science of determining the arrangement of atoms in crystalline solids. The word "crystallography" is derived from the Greek words crystallon "cold drop, frozen drop", with its meaning extending to all solids with some degree of transparency.

A solid composed of a closure collection of atoms. In solids, atoms are very strongly held together. On the basis of arrangement of constituent particles solids are divided into, they are

- 1) Crystalline solids
- 2) Amorphous (non-crystalline solids)

1. Crystalline Solids:

The crystalline state of solid is characterized by **regular and periodic arrangement of atoms or molecules**. Most of the solids are crystalline in nature. The crystalline solid may be further divided into,

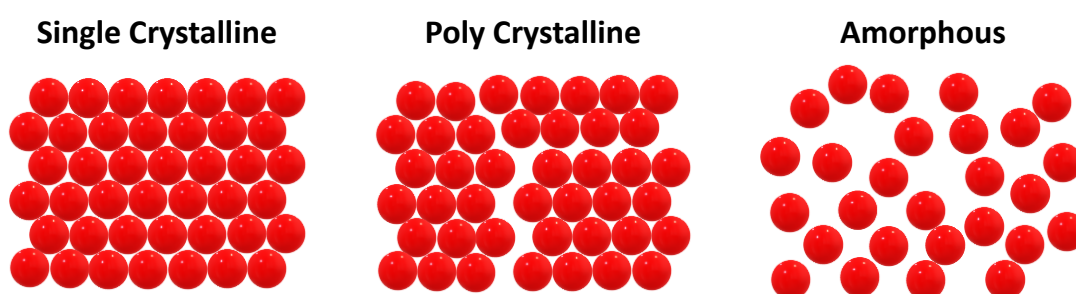


Figure 1.1: The atomic structures of the single crystals, polycrystals and amorphous solids in a solid material.

- a) **Single crystalline:** Atoms are arranged in a continuous, ordered 3D lattice (Eg: Diamond, quartz, mica etc.)
- b) **Polycrystalline:** Composed of many small crystals (grains) in random orientations, separated by grain boundaries (Metals, ceramics etc.).

2. Amorphous Solids

Amorphous solids are characterized by the complete **random arrangement of atoms or molecules**. Amorphous solids are formed when the atoms do not get sufficient time to undergo a periodic arrangement (Glass, plastics, rubber, wood etc.).

❖ Difference between crystalline and amorphous solids

S. No	Crystalline	Amorphous
1	The fundamental atoms (molecules) are arranged in regular fashion.	The fundamental atoms (molecules) are not arranged in any regular fashion
2	There is one repeating unit	There no repeating unit
3	Have a long-range order	Have a short-range order
4	They are anisotropic	They are isotropic
5	They have sharp melting point i.e. all the bonds break at particular temperature as they are of equal strength	They melt over a range of temperature as all bonds are of not equal strength.

1.2. LATTICE POINTS AND SPACE LATTICE

An imaginary point in the array representing a particle position is called “**lattice point**”. The 3D arrangement of lattice points in space is called as “**space lattice**”.

Assume a crystal is a three-dimensional body, which is made up of regular and periodic arrangement of atoms or molecules in space. The periodicity in the arrangement generally varies in different directions. It is very convenient to imagine points in space about which these atoms are located. Such points in space are called **lattice points** and totality of such points form a **crystal lattice** or **space lattice**.

- If all the atoms at the lattice points are identical, the lattice is called **Bravias lattice**.

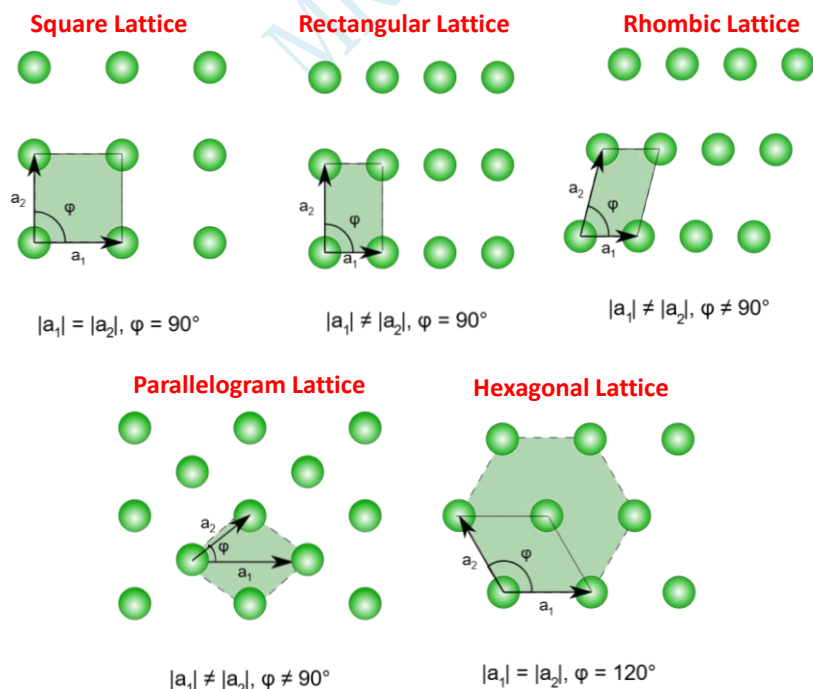


Figure 1.2: Five types of two-dimensional lattices.

Let us consider the case of a two-dimensional square array of points as shown in figure 1.2. By repeated translation of the two vectors **a** and **b** on the plane of the paper, we can generate

the square array. The magnitudes of $\mathbf{a}$ and $\mathbf{b}$ are equal and can be taken to be unity. The angle between them is 90° . $\mathbf{a}$ and $\mathbf{b}$ are called the fundamental translation vectors that generate the square array. Let us choose any arbitrary point O as origin. If we choose a lattice point P at position $\mathbf{r}$ it can be represented by translation vectors as,

$$\bar{\mathbf{r}} = \ell \bar{\mathbf{a}} + m \bar{\mathbf{b}}, \text{ where } \ell, m \text{ are integers.}$$

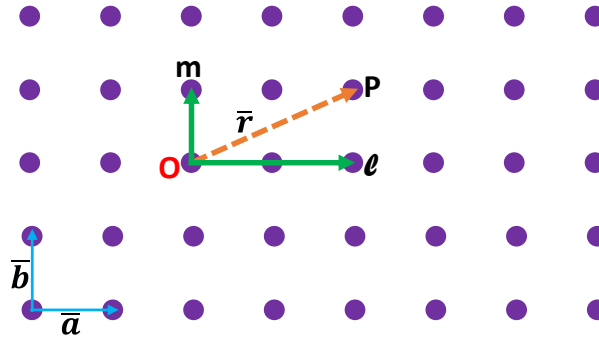


Figure 1.3: Illustration of space lattice.

Similarly for three-dimensional lattice, $\bar{\mathbf{r}} = \ell \bar{\mathbf{a}} + m \bar{\mathbf{b}} + n \bar{\mathbf{c}}$. Thus, a 3D space lattice is generated by repeated translation of three noncoplanar vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

1.3. THE BASIS

An atom or group of atoms must replace with the lattice points by more complex objects, called **the basis (motif)**. When the basis is repeated with correct periodicity in all directions it gives the actual **crystal structure**. The crystal structure is real, while the lattice is imaginary.

$$\text{Space lattice} + \text{basis} = \text{Crystal Structure}$$

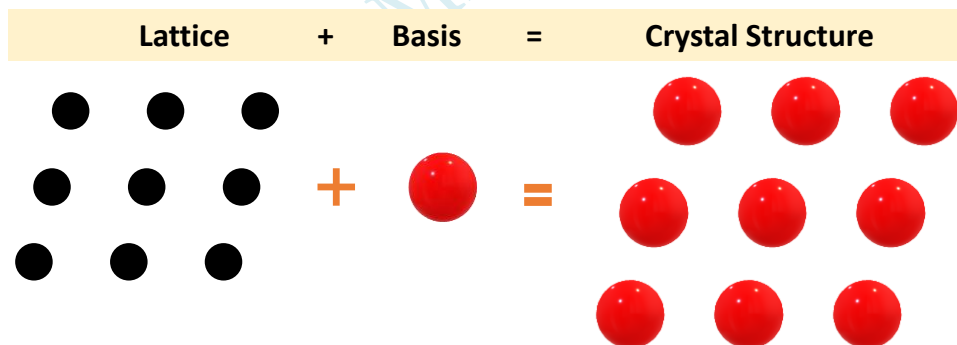


Figure 1.4: A crystal structure is defined as a lattice with a basis added to each lattice site. Usually, the basis consists of an atom, a group of atoms or a molecule.

The number of atoms in a basis varies from one to several thousands. In crystalline solids like Cu and Na the basis is Single atom, In NaCl & CsCl the basis is diatomic, where as in crystals, like CaF_2 the basis is tri-atomic.

1.4. UNIT CELL

The unit cell is the smallest block (geometrical figure) from the entire crystal, built up by repetition in three dimensions. The unit cell consists of lattice points that represent the locations of atoms or ions. The entire structure then consists of this unit cell repeating in three dimensions, as illustrated in Figure 1.5 (a).

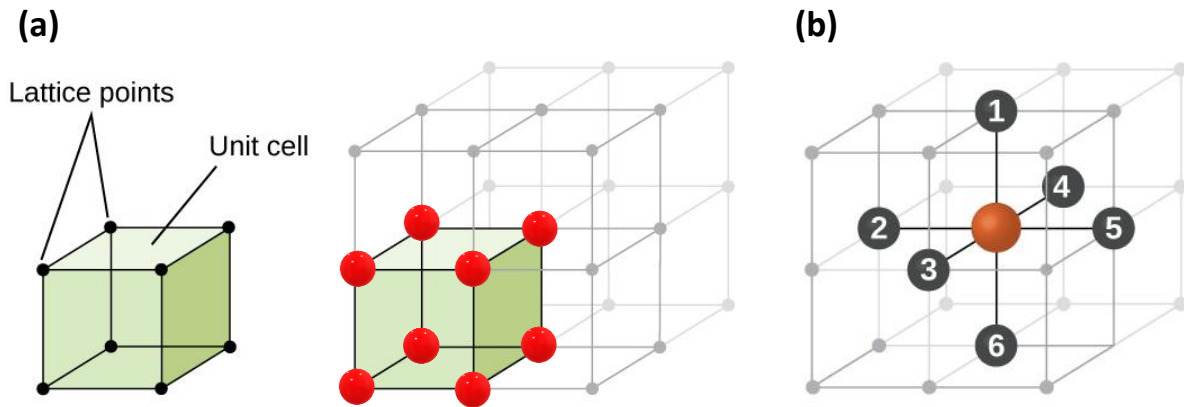


Figure 1.5: (a) A unit cell shows the locations of lattice points repeating in all directions. (b) Simple cubic lattice structure coordinates with six other atoms.

There are two kinds of unit cells. They are,

- i. **Primitive cell:** It is the smallest unit cell in the crystal, usually there are four points at the corners in (2-D) (possess lattice points at its corners only), it is the simplest type of unit cell which contain one lattice point per unit cell. There are 8 points at the corners in (3-D) also one point per unit cell (each corner point is shared among the eight cell).
- ii. **Non-Primitive cell:** It is the larger unit cell than the primitive cell; there are additional points to that at the corners. The area of the non-primitive cell must be an integral multiple of primitive cell.

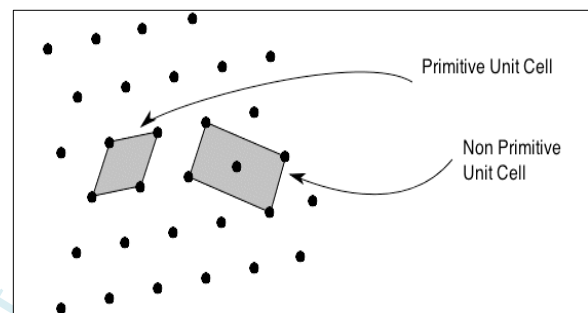


Figure 1.6: Primitive and non-primitive unit cells.

1.5. LATTICE PARAMETERS

In general, a unit cell is defined by the lengths are known as primitives which are represented by a , b , c where the interfacial angles are represented by α , β , γ , as shown in figure 1.7. These axes drawn along with the primitives are called crystallographic axes which do not lie the same plane. They are represented by X , Y and Z . These measuring lengths and axes are called Lattice parameters.

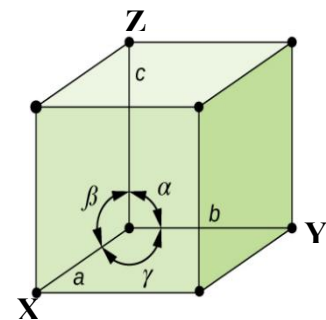


Figure 1.7: A unit cell length (a , b , and c) and interfacial angles (α , β , and γ).

1) Parameters determine the crystal structure of materials:

Let us discuss some of the important parameters which are used to describe the crystal structure.

1. **Number of atoms per unit cell:** The total number of atoms present in (or) shared by a unit cell is known as number of atoms per unit cell.

2. **Co-ordination number:** It is the number of nearest neighboring atoms to a particular atom.
3. **Atomic radius:** Atomic radius is defined as half of the distance between any two nearest neighbor which have direct contact with each other, in a crystal of pure element. It is usually expressed in terms of cube edge 'a'.
4. **Atomic Packing Factor (APF):** The fraction of volume occupied by atoms in a unit cell is known as atomic packing factor or packing factor or density of packing. It is the ratio of volume occupied by the atoms in unit cell (v) to the volume of the unit cell relating to that structure.

$$\text{Atomic Packing Factor [APF]} = \frac{\left(\frac{\text{Number of atoms}}{\text{unit cell}} \right) \times \text{Volume of each atom}}{\text{Volume of the unit cell}}$$

1.6. CRYSTAL STRUCTURES

A crystal structure is the ordered, repeating three-dimensional arrangement of atoms, ions, or molecules in a solid material. It is defined by a unit cell (the smallest repeating building block), a lattice (the spatial point pattern), and a basis (the specific group of atoms attached to each point).

Most of the metal crystals have different types of unit cells, they are,

1.6.1. Simple Cubic (SC):

- The primitive unit cell of the simple cubic (SC) crystal structure contains eight atoms, one on each corner of the cube.
- SC contains one lattice point (atom) per unit cell ($8 \times \frac{1}{8} = 1$).
- Each corner atom touches the six neighboring corner atoms; hence co-ordination is six ($N = 6$).
- The distance between the centers of two neighboring atoms 'a = 2r'.
- The atomic radius of an atom is (r) $\frac{a}{2}$.
- Atomic Packing Fraction (APF):

Calculation: Knowing this and the formula of volume of sphere ($\frac{4}{3}\pi r^3$), it becomes possible to calculate the APF as follows:

$$\text{APF} = \frac{N_{\text{atom}} \times V_{\text{atom}}}{V_{\text{unit cell}}} = \frac{1 \times \frac{4}{3}\pi r^3}{a^3} = \frac{4\pi r^3}{3(2r)^3} = \frac{\pi}{6} = 0.52$$

- Examples: Polonium (Po).

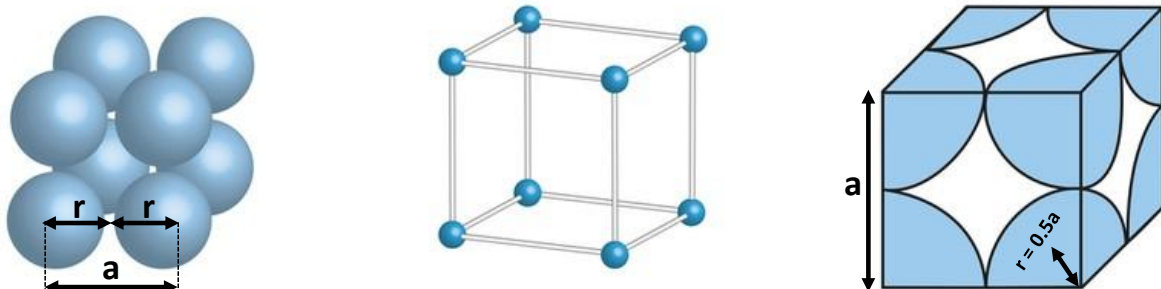


Figure 1.8: SC crystal structure.

Body-Centered Cubic (BCC):

- The primitive unit cell of body-centered cubic (BCC) crystal structure contains nine atoms. Eight atoms at each corner and one atom is at the centre of the cube.
- The BCC unit cell contains of net total of two atoms. Therefore, number of atoms per unit cell is two (one at each corner and one at center- $8 \times \frac{1}{8} + 1 = 2$).
- The co-ordination number of BCC unit cell is 8, that means each atom touches **eight** nearest neighbour atoms ($N = 8$).
- The distance between the centers of two neighbouring atoms is ' $2r = \frac{\sqrt{3}a}{2}$ '.
- Atomic radius $r = \frac{\sqrt{3}a}{4}$.

Calculation: A line is drawn from one corner of the cube through the center and to the other corner passes through $4r$, where r is radius of an atom, as shown in Fig. 1.9 (c).

From $\triangle ABC$,

$$\begin{aligned} AC^2 &= AB^2 + BC^2 = a^2 + a^2 \\ AC^2 &= 2a^2 \quad \text{----- (1)} \end{aligned}$$

From $\triangle ACD$,

$$AD^2 = AC^2 + CD^2 \quad \text{----- (2)}$$

Substituting equation (1) in (2), we get,

$$\begin{aligned} AD^2 &= 2a^2 + a^2 = 3a^2 \\ AD^2 &= \sqrt{3}a \quad \text{----- (3)} \end{aligned}$$

$$\text{If, } r \text{ is radius of an atom, } AD = 4r \quad \text{----- (4)}$$

Hence, from equations (3) and (4), we get,

$$4r = \sqrt{3}a$$

$$r = \frac{\sqrt{3}a}{4} \quad (\text{or}) \quad a = \frac{4r}{\sqrt{3}}$$

- Atomic Packing Fraction (APF):

Calculation: Knowing this and the formula of volume of sphere ($\frac{4}{3}\pi r^3$), it becomes possible to calculate the APF of BCC using above equation as follows:

$$\begin{aligned} \text{APF} &= \frac{N_{\text{atom}} \times V_{\text{atom}}}{V_{\text{unit cell}}} = \frac{2 \times \frac{4}{3} \pi r^3}{a^3} \\ &= \frac{2 \times \frac{4}{3} \pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3} \\ &= 2 \times \frac{4}{3} \pi r^3 \times \frac{3\sqrt{3}}{64r^3} \\ \text{APF} &= \frac{\pi\sqrt{3}}{8} = 0.68 \end{aligned}$$

- Examples: Na, K, Cr, Fe, Li, etc.

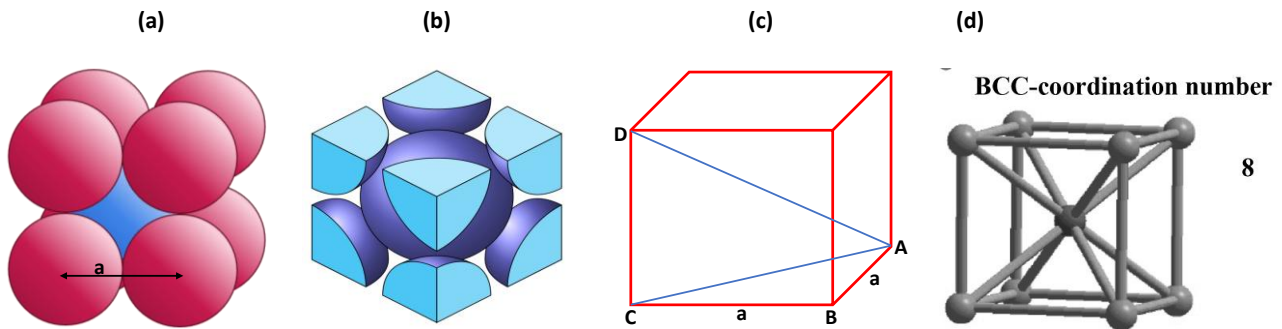


Figure 1.9: (a, b) BCC crystal structure and (c) face body diagonals of BCC system and (d) co-ordination number of BCC.

Face-Centered Cubic (FCC):

- One of the close packed structures is face centered cubic structure. The face centered cubic system has atoms located at the eight corners and the centers of all cubic faces, as shown in figure 1.10 (a).
- Therefore, face centered cubic structure contains four atoms (one at each corner and one in the center of each face - $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$).
- Each atom touches **twelve** nearest neighbour atoms, that means the co-ordination number of FCC unit cell 12 ($N = 12$).
- The distance between the centers of two neighbouring atoms is ' $2r = a/\sqrt{2}$ '.
- Atomic radius $r = \frac{a}{2\sqrt{2}}$.

Calculation: If two diagonal atoms on the bottom faces are joined as shown in Fig. 1.10 (c).

$$AC = 4r$$

$$AC^2 = 16r^2 \quad \text{----- (1)}$$

$$\text{From } \triangle ABC, AC^2 = AB^2 + BC^2 = a^2 + a^2 = 2a^2 \quad \text{----- (2)}$$

Hence, from the equations (1) and (2), we get,

$$16r^2 = 2a^2$$

$$r^2 = a^2/8$$

$$r = \frac{a}{\sqrt{8}} \quad (\text{or}) \quad a = 2\sqrt{2} r$$

- Atomic Packing Fraction (APF):

Calculation: Knowing this and the formula of volume of sphere ($\frac{4}{3}\pi r^3$), it becomes possible to calculate the APF of FCC using above equation as follows:

$$APF = \frac{N_{atom} \times V_{atom}}{V_{unit\ cell}} = \frac{4 \times \frac{4}{3}\pi r^3}{a^3} = \frac{4 \times \frac{4}{3}\pi r^3}{8 \times 2\sqrt{2} r^3}$$

$$APF = \frac{\pi}{3\sqrt{2}} = 0.74$$

- Examples: Al, Ag, Au, Pb, Cu, etc.

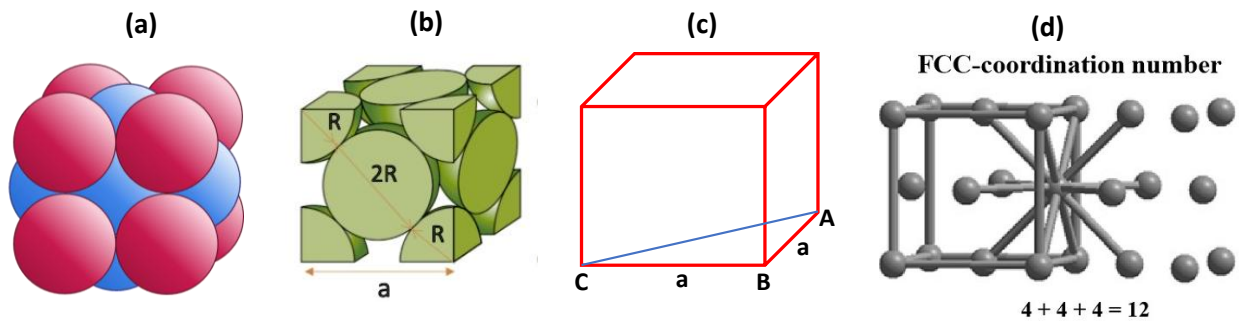


Figure 1.10: (a, b) FCC crystal structure and (c) face diagonal of FCC system and (d) co-ordination number of FCC.

❖ **Summary:** Comparison of unit cell properties of some crystal structures

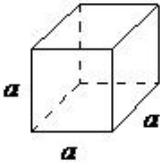
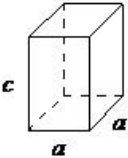
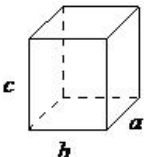
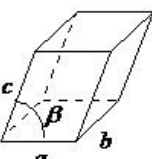
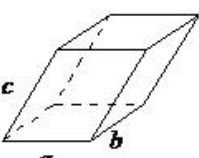
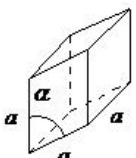
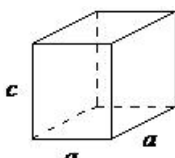
Cubic Cell Name	Structure	Atoms per Unit Cell	Nearest neighbor distance ($2r$)	Atomic radius (r)	Atomic packing factor	Coordination Number (N)
Simple Cubic		1	a	$\frac{a}{2}$	0.52	6
Body-Centered Cubic		2	$\frac{\sqrt{3}a}{2}$	$\frac{\sqrt{3}a}{4}$	0.68	8
Face-Centered Cubic		4	$\frac{a}{\sqrt{2}}$	$\frac{a}{2\sqrt{2}}$	0.74	12

Table 1.1: A summary of the characteristics of three main types of unit cells.

1.7 BRAVAIS LATTICES

The Bravais lattice is the basic pattern used to build every crystal structure. The concept originated as a topological problem of finding the number of different ways to arrange points in space where each point would have identical “surroundings”. Mathematician **Auguste Bravais** discovered in 1846 that there were 14 different collections of the groups of points, which are known as **Bravais lattices**.

These lattices fall into seven different "crystal systems" 14 “Bravais lattice” types, as differentiated by the relationship between the interfacial angles and the distance between points in the unit cell.

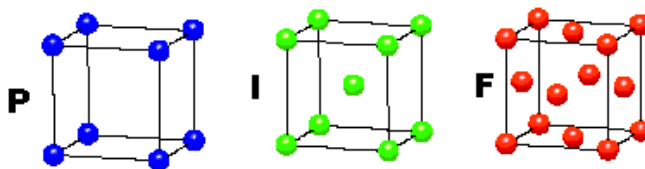
Crystal system	Bravais Lattice Type	Axes	Angles	Unit Cell Geometry	Examples
C ubic	SC, BCC, FCC	$a = b = c$	$\alpha = \beta = \gamma = 90$		Po, Cu, Ag, NaCl
T etragonal	SC, BCC	$a = b \neq c$	$\alpha = \beta = \gamma = 90$		SnO ₂ , NiSO ₄ , TiO ₂
O rthorhombic	SC, BCC, FCC, ECC	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90$		KNO ₃ , BaSO ₄
M onoclinic	SC, ECC	$a \neq b \neq c$	$\alpha = \beta = 90 \neq \gamma$		CaSO ₄ .2H ₂ O, FeSO ₄
T riclinic	SC	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90$		K ₂ Cr ₂ O ₇ , CuSO ₄ .5H ₂ O
R hombohedral (or Trigonal)	SC	$a = b = c$	$\alpha = \beta = \gamma \neq 90$		As, Sb, Bi
H exagonal	SC	$a = b \neq c$	$\alpha = \beta = 90, \gamma = 120$		SiO ₂ , Zn, Mg, Cd

SC: Simple Cubic, BCC: Body-Centered Cubic, FCC: Face-Centered Cubic, ECC: Edge centered Cubic

CUBIC

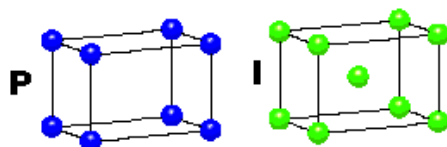
$$a = b = c$$

$$\alpha = \beta = \gamma = 90^\circ$$

**TETRAGONAL**

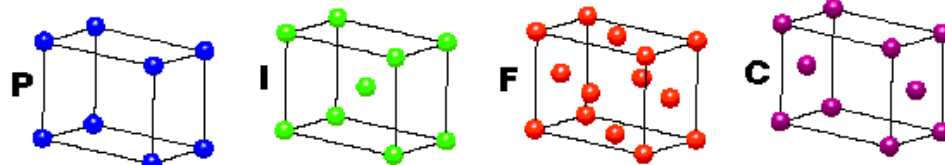
$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

**ORTHORHOMBIC**

$$a \neq b \neq c$$

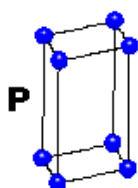
$$\alpha = \beta = \gamma = 90^\circ$$

**HEXAGONAL**

$$a = b \neq c$$

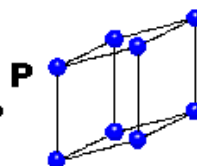
$$\alpha = \beta = 90^\circ$$

$$\gamma = 120^\circ$$

**TRIGONAL**

$$a = b = c$$

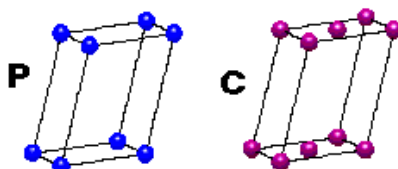
$$\alpha = \beta = \gamma \neq 90^\circ$$

**MONOCLINIC**

$$a \neq b \neq c$$

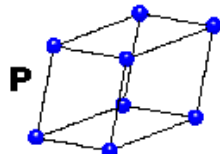
$$\alpha = \gamma = 90^\circ$$

$$\beta \neq 120^\circ$$

**TRICLINIC**

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$

**4 Types of Unit Cell**

P = Primitive

I = Body-Centred

F = Face-Centred

C = Side-Centred

+

7 Crystal Classes**→ 14 Bravais Lattices**

Figure 1.11: Schematic representation of the 14 different Bravais lattices possible found in nature.

1.8. MILLER INDICES

In a crystal lattice, there are sets of planes which have the layer concentration of atoms (or group of atoms) at the lattice points. Miller developed a method to represent a plane by three number (hkl) which are known as miller indices.

Miller-indices - A shorthand notation to describe certain crystallographic directions and planes in a material.

“Miller indices are the three smallest possible integers h, k and l, which have the same ratios as the reciprocals of the intercepts of the concerned plane on the three axes”

Procedure to find the miller indices:

- Find the intercepts of the plane on the three crystallographic axes in terms of lattice parameters a, b, c.
Eg: a, 2b, c/2
- Express the intercepts as multiples of unit cell direction or lattice parameters.

$$\begin{matrix} 1 & 2 & 1/2 \\ p & q & r \end{matrix}$$
- Find the ratio reciprocals of these three numbers.

$$1/1 : 1/2 : 2/1$$

$$1/p : 1/q : 1/r$$

4. Convert these reciprocals into whole numbers by multiplying with their LCM to get smallest whole number.

$$2 \times 1/1 : 2 \times 1/2 : 2 \times 2/1 = 214$$

$$1/p : 1/q : 1/r = hkl$$

5. Write the integers with in the parentheses which is the miller indices of the plane (214) in general (hkl).

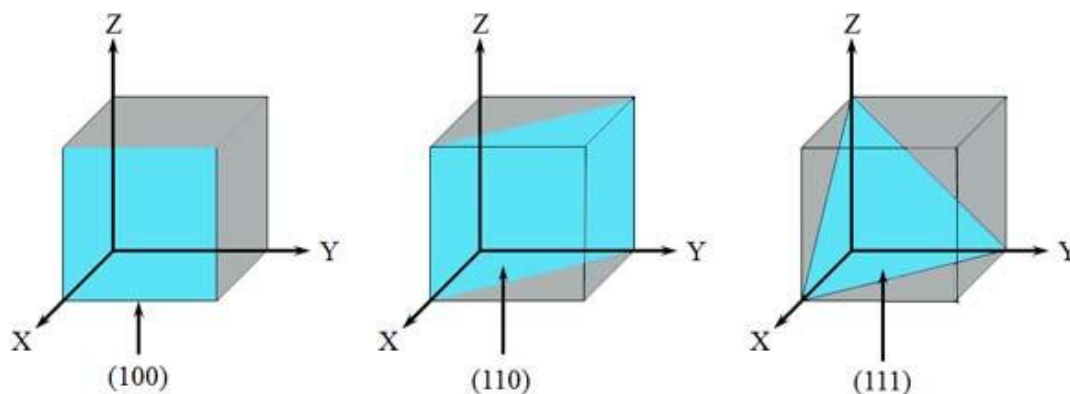


Figure 1.12: Crystal planes in cubic cell.

Importance of Miller indices:

1. When planes are parallel to one of the coordinate axes, it is said to meet that axis at infinity i.e. $1/\infty = 0$, the miller index for that axis is zero.
2. When the intercept of a plane is in the negative part of any axis the miller index is distinguished by a minus sign directly over it.
3. All equally spaced parallel planes have the same index number (hkl)
4. If a plane passes through origin, it is defined in terms of || planes having non zero intercept
5. If (hkl) is the miller indices of a crystal plane then the intercepts made by the plane with crystallographic axis are given by a/h , b/k , c/l . where a, b and c are primitives.
6. Inter planer distance is given by

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \text{ (for cubic system)}$$

7. The direction [hkl] is perpendicular to plane (hkl).
8. The angle θ between any two crystallographic direction $[u_1 v_1 w_1]$ and $[u_2 v_2 w_2]$ can be calculated from the relations

$$\cos \theta = \frac{u_1 u_2 + v_1 v_2 + w_1 w_2}{\sqrt{u_1^2 + v_1^2 + w_1^2} \sqrt{u_2^2 + v_2^2 + w_2^2}}$$

Example: - Draw the (112) & (120) planes.

1. (112) plane:

- 1) Given miller indices (h k l) = (112).
- 2) Reciprocal of Miller indices = (1/1, 1/1, 1/2)
- 3) Take reciprocals as intercepts i.e., 1, 1, 1/2
- 4) (1, 1, 1/2) are marked on the x, y, z axes and joining them, then (112) plane is formed.

2. (120) plane:

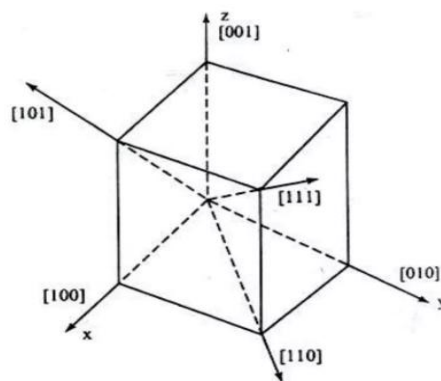
- 1) Given millers indices $(h\ k\ l) = (120)$
- 2) Reciprocals of miller's indices $= 1/1, 1/2, 0/1 = (1, 1/2, \infty)$.
- 3) Now intercepts are $(1, 1/2, \infty)$.
- 4) $(1, 1/2, \infty)$ are marked on the X, Y, Z axes & joining them, then (120) plane is formed.

Miller Directions:

- A crystallographic direction or Miller direction is a line or vector connecting two lattice points in a crystal. A crystallographic direction is represented by three integers enclosed in square brackets: $[u, v, w]$.
- Consider a crystal lattice with three mutually independent crystallographic axes:
- x, y, z. The direction of any line in the crystal can be described in terms of its components along these three axes.
- If a vector has components u,v,w along the x, y, and z axes, respectively, the direction is represented as $[u, v, w]$. The indices u,v,w are always reduced to the smallest possible integers.

General rules for defining a crystallographic direction:

- They pass through the origin of a coordinate system.
- Determine length of the vector projection in the unit cell dimensions a, b, and c.
- Remove the units $[Ua, Vb, Wc] \Rightarrow [U, V, W]$
- Example $[2a, 3b, 5c] \Rightarrow [2\ 3\ 5]$
- UVW are multiplied and divided by a common factor to reduce them to smallest integer value.
- Denote the direction by $[u, v, w]$
- $\langle u, v, w \rangle$ represents the family of equivalent directions.
- Material properties along any direction in a family are the same. Example: $[100]$, $[010]$, $[001]$ in a simple cubic are same.
- Negative index is represented by a bar over the index.

**Figure 1.13:** Miller directions**Important directions in a crystal****Example:**

Determination of Direction along the X-axis

Consider a direction from $P = (0, 0, 0)$ to $Q = (1, 0, 0)$

The direction components are

$$\Delta x = 1 - 0 = 1$$

$$\Delta y = 0 - 0 = 0$$

$$\Delta z = 0 - 0 = 0$$

Therefore, $\Delta x: \Delta y: \Delta z = 1:0:0$

Hence, $[100]$ is the direction along the X-axis.

Similarly, $[010]$ is the direction along the Y-axis and $[001]$ along the Z-axis.

Direction and vector representation:

A crystallographic direction can also be considered as a vector. For a cubic crystal, the direction $[u \ v \ w]$ corresponds to the vector $\mathbf{r} = u\hat{i} + v\hat{j} + w\hat{k}$, where: a, b, c are lattice parameters, $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the crystallographic axes.

For a cubic crystal: $a = b = c$,

Therefore: $\mathbf{r} = a(u\hat{i} + v\hat{j} + w\hat{k})$.

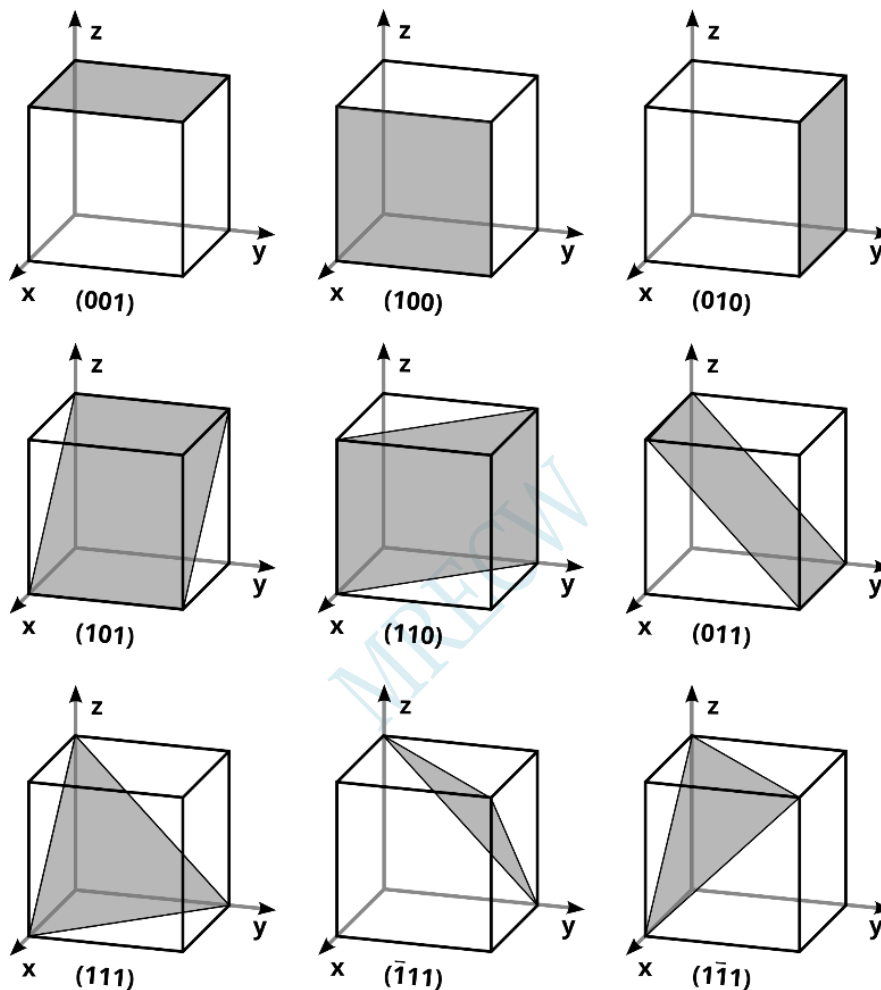
Some Examples of Miller Planes and Directions:

Figure 1.14: Planes with different Miller indices in cubic crystals.

1.9. INTER PLANAR DISTANCE (d_{hkl})

Interplanar distance (d) is the perpendicular distance between two adjacent, parallel planes of atoms in a crystal lattice. It depends on the size of the unit cell and the orientation of the planes, which are defined by Miller indices (h, k, l).

Consider a plane ABC of an orthogonal crystal lattice as shown in Fig 1.15. This plane belongs to a family of planes whose miller indices are $(h \ k \ l)$ because the miller indices represent a set of parallel planes. Let $ON = d$ be the perpendicular distance of the plane ABC from the origin.

Let, α, β, γ be the angles between the coordinate axes x, y and z respectively.

The intercepts of the plane on the three axes are given by, $OA = a/h$; $OB = b/k$ and $OC = c/l$, where a, b , and c are the unit cell edge lengths along the X, Y and Z axes, respectively.

Then the direction cosines of ON are,

$$\begin{aligned}\cos \alpha' &= \frac{ON}{OA} = \frac{d}{OA} = \frac{dh}{a} \\ \cos \beta' &= \frac{ON}{OB} = \frac{d}{OB} = \frac{dk}{b} \\ \cos \gamma' &= \frac{ON}{OC} = \frac{d}{OC} = \frac{dl}{c}\end{aligned}$$

From the figure 1.15,

$$ON^2 = (x^2 + y^2 + z^2)$$

$$d^2 = d^2 \cos^2 \alpha + d^2 \cos^2 \beta + d^2 \cos^2 \gamma$$

$$d^2 = d^2 [\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma]$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Substituting the values of $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ in the above equation, we get

$$\Rightarrow \left(\frac{dh}{a}\right)^2 + \left(\frac{dk}{b}\right)^2 + \left(\frac{dl}{c}\right)^2 = 1$$

$$\Rightarrow d^2 \left[\left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 + \left(\frac{l}{c}\right)^2 \right] = 1$$

$$d = ON = \frac{1}{\sqrt{\left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 + \left(\frac{l}{c}\right)^2}}$$

The above expression represents the distance between two successive planes of a cubic system.

Interplanar distance between two adjacent parallel planes belongs to orthorhombic unit cell with miller indices (h k l) is given by

$$d_{hkl} = \frac{1}{\sqrt{\left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 + \left(\frac{l}{c}\right)^2}}$$

Special Cases:

1. Cubic System: In Cubic system, we know that $a = b = c$, then

$$\begin{aligned}d_{hkl} &= \frac{1}{\sqrt{\left(\frac{h}{a}\right)^2 + \left(\frac{k}{a}\right)^2 + \left(\frac{l}{a}\right)^2}} = \frac{1}{\sqrt{\frac{h^2 + k^2 + l^2}{a^2}}} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \\ \therefore d_{hkl} &= \frac{a}{\sqrt{h^2 + k^2 + l^2}}\end{aligned}$$

It is the interplanar distance of Cubic crystal system.

i. In Simple Cubic system: $d_1 : d_1 : d_1 = \frac{a}{1} : \frac{a}{\sqrt{2}} : \frac{a}{\sqrt{3}}$

$$= \sqrt{6} : \sqrt{3} : \sqrt{2}$$

$$\frac{1}{d_1} : \frac{1}{d_1} : \frac{1}{d_1} = 1 : \sqrt{2} : \sqrt{3}$$

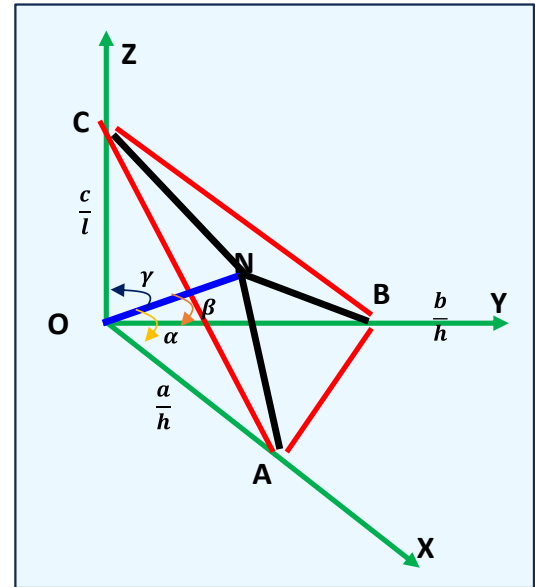


Figure 1.15: Separation between successive planes in orthogonal unit cell.

ii. In BCC structure: $d_1 : d_1 : d_1 = \frac{a}{2} : \frac{a}{\sqrt{2}} : \frac{a}{2\sqrt{3}}$ (or) $1 : \frac{2}{\sqrt{2}} : \frac{2}{\sqrt{3}}$

$$\frac{1}{d_1} : \frac{1}{d_1} : \frac{1}{d_1} = 1 : \frac{\sqrt{2}}{2} : \sqrt{3}$$

iii. In FCC structure: $d_1 : d_1 : d_1 = \frac{a}{2} : \frac{a}{2\sqrt{2}} : \frac{a}{\sqrt{3}}$ (or) $1 : \frac{1}{\sqrt{2}} : \frac{2}{\sqrt{3}}$

$$\frac{1}{d_1} : \frac{1}{d_1} : \frac{1}{d_1} = 1 : \sqrt{2} : \frac{\sqrt{3}}{2}$$

2. Tetragonal System: $a = b \neq c \rightarrow d_{hkl} = \frac{1}{\sqrt{\left(\frac{h}{a}\right)^2 + \left(\frac{k}{a}\right)^2 + \left(\frac{l}{c}\right)^2}} = \frac{1}{\sqrt{\left(\frac{h+k}{a}\right)^2 + \left(\frac{l}{c}\right)^2}}$

Expression for Lattice Constant of a Cubic Crystal (a):

Let us consider, 'a' is the edge length or primitive of a cubic unit cell and 'ρ' is the density of the crystal.

Density $\rho = (\text{total mass of molecules belonging to unit cell}) / (\text{volume of unit cell})$ ----- (i)

Total mass of molecule belonging to unit cell = $n \cdot M / N_A$ ----- (ii)

Where, n = number of molecules belonging to unit cell, M = Molecular weight, N_A = Avagadro Number.

The volume of cube = a^3 ----- (iii)

$\Rightarrow$ Substitute (ii) and (iii) in (i)

Therefore, $\rho = n M / a^3 \cdot N_A$

$$a^3 = nM / \rho N_A$$

Lattice Constant for Cubic Lattice, $a = (n M / \rho N_A)^{1/3}$

1.10. DEFECTS IN CRYSTALS

Crystal is regular and periodic arrangement of atoms in three dimensions. But no crystal is perfectly regular which means crystal possesses some defects.

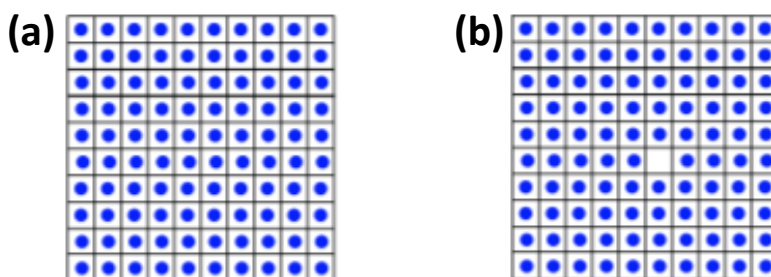


Figure 1.16: Illustration of (a) a perfect crystal and (b) crystal with the defect.

Classification of Crystal Defects:

Crystal Defects (or) Imperfections can be classified mainly into 4 types on the basis of their geometry. They are i) Point defects, ii) Line defects, iii) Surface defects and iv) Volume defects.

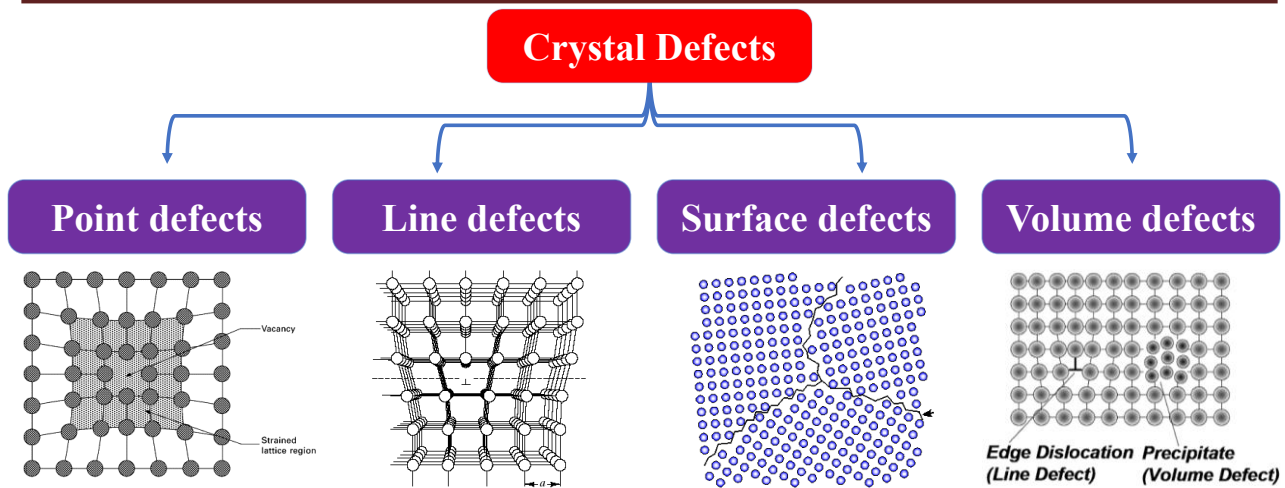


Figure 1.17: Types of crystal defects.

1. Point Defects:

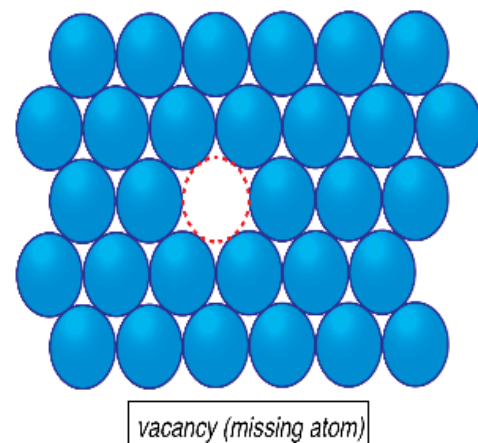
- Point imperfections are also called zero-dimensional imperfections. One or two atomic diameters is the typical size of a point imperfection.
- In a crystal lattice, point defect is one which is completely local in its effect, e.g., a vacant lattice site.
- The introduction of point defect into a crystal increases its internal energy as compared to that of the perfect crystal. Further they produce distortions inside the crystal structure.
- They change the electrical resistance of a crystal.
- Point defects are created during crystal growth and application of thermal energy, mechanical stress or electric field. Further they are created by irradiating the crystal by x-rays, microwaves and light.

Point defects in crystalline solids are generally sub-divided into different categories. These categories describe different ways atoms or ions can go missing, move around, or add extra elements inside a crystal lattice.

a. Vacancy Defect:

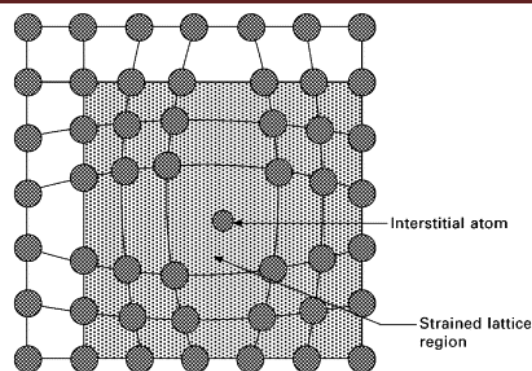
A vacancy or vacant site implies an unoccupied atom position within the crystal lattice. In other words, vacancies are simply empty atom sites. Vacancies may be single (as shown in figure) or two or more of them may condense into a divacancy or tri-vacancy.

Vacancies may occur as a result of imperfect packing during the original crystallization or they may arise from thermal vibrations of atoms at elevated temperatures, because as thermal energy is increased there is a higher probability that individual atoms will jump out of their position of lowest energy.

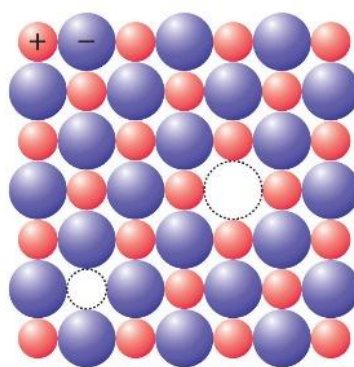
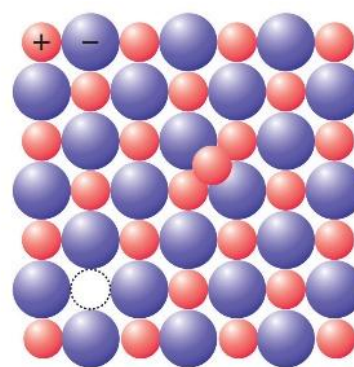


b. Interstitial Defect:

An interstitial defect is a type of point defect in a crystal structure where an extra atom or ion squeezes into the open space, or interstitial site, between the regular atoms. This type of defect can produce atomic distortion or strain because the interstitial atom tends to push the surrounding atoms further apart, unless the interstitial atom is smaller than the rest of the atoms in the crystal. Small interstitial atoms (like carbon in iron) block movement inside the structure, making the metal harder and stronger.

**c. Stoichiometric Defects (Ionic Crystals)****c1. Schottky Defect:**

An equal number of cations (positive ions) and anions (negative ions) completely leave their regular places in the crystal lattice, creating empty holes called vacancies (see Fig. a). Schottky defect arises due to ions leaving the crystal completely, the overall

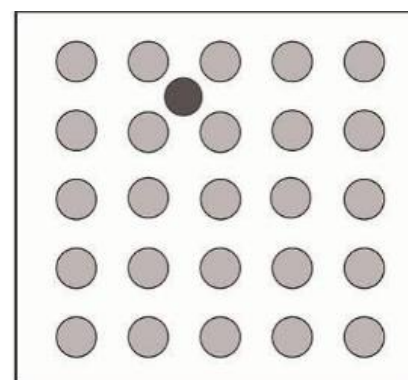
**(a) Schottky defect****(b) Frenkel defect**

density decreases. It happens in ionic solids where cations and anions are similar in size and the coordination number is high (e.g., Sodium Chloride [NaCl], Potassium Chloride [KCl], and Cesium Chloride [CsCl]).

c2. Frenkel Defect: A smaller ion (cation) leaves its normal spot and moves into an interstitial space (the empty gap between regular atoms), creating a vacancy at its old spot and an interstitial defect at the new one (see Fig. b). Frenkel defect arises due to the ions staying inside the solid structure; the overall density stays the same. It happens when there is a large-size difference between the small cation and the large anion, and the coordination number is low (e.g., Silver Chloride [AgCl], Zinc Sulfide [ZnS], and Silver Bromide [AgBr]).

d. Impurity Defects:

The spaces between the parent atoms in a crystal are known as interstitial spaces. Small sized (lower atomic number) atoms such as hydrogen etc., may fit into these interstitial spaces. These atoms are known as interstitial atoms and the defect formed due to the presence of interstitial atoms is known as interstitial defect. This is shown in the fig. If 'r' is the radius of a parent atom, then an octahedral and a tetrahedral space can accommodate an interstitial atom of radius $0.414r$ and $0.225r$, respectively.

**Interstitial Defect**

2. Line Defects:

- Line defects are called dislocations. These are one-dimensional imperfections in the geometrical sense.
- A dislocation may be defined as a disturbed region between two substantially perfect parts of a crystal. It arises due to a part— plane of atoms is displaced from its symmetrically stable positions in the array.
- Dislocations arise in crystals as a result of
 1. Growth accidents (during crystallization)
 2. Thermal stresses
 3. Phase transformations
 4. External stresses causing plastic flow (during plastic deformation)
 5. Segregation of solute atoms causing mismatches, etc.
- The two basics types of dislocations are:
 - i. Edge dislocation
 - ii. Screw dislocation

i. Edge dislocation:

- An edge dislocation is a linear crystal defect caused by the insertion of an extra half-plane of atoms into an otherwise perfect lattice.
- Figure 1.18(a) shows a perfect crystal, the top-view sketch showing the atoms on the front face.
- From the sketch of Fig. 1.18(b), it can be clearly seen that the atoms above the edge of the extra plane are squeezed together and are in a state of compression.
- The bond lengths have been compressed to smaller than the equilibrium value just below the edge of the extra plane the atoms are pulled apart and are in a state of tension. Here the bond lengths have been stretched to normal values.
- This distorted configuration extends all along the edge into the crystal. There is an extra energy due to the distortion in the region immediately surrounding the edge of the incomplete plane.
- If one of the planes of atoms glides over another separated by an integral multiple of interatomic distance is called slip, and slip plane is the plane in which slip has taken place. In Fig. 1.18(c), XY- is the slip plane.
- If the incomplete extra plane is above the slip plane, then the edge dislocation is positive and is represented by symbol ' $\perp$ '.
- If the incomplete extra plane is below the slip plane, then the edge dislocation is negative and is represented by symbol ' T '. As shown in Fig. 1.13(c).

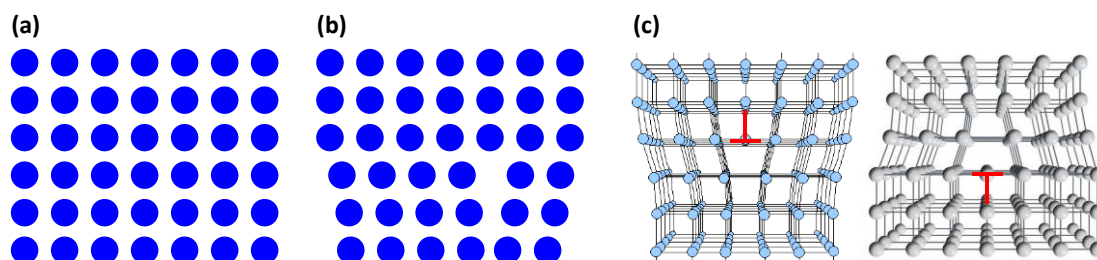
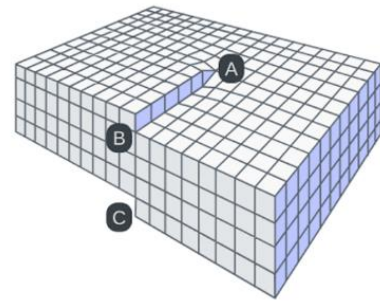


Figure 1.18. Edge dislocation propagation inside a film crystal. (a) a perfect crystal, (b) edge dislocated crystal. (c) Slip plane in ' $\perp$ ' and ' T ' representation.

ii. Screw Dislocation:

Screw dislocation results from a displacement of the atoms in one part of a crystal relative to the rest of the crystal, forming a spiral ramp around the dislocation line as shown in fig. The plane ABCD is the slipped area. AD dislocation line. The name screw follows the new character of atomic planes which are transferred. Such that as one moves around the dislocation line in circuit. Speed of movement of a screw dislocation is less than the edge dislocation.

**3. Surface Defects:**

- The defects, which takes place on the surface of a material are known as surface defects or plane defects. They are also called as two-dimensional surface imperfections.
- They are caused due to the change in the stacking of atomic planes across boundary.
- They occur inside the crystal during the recrystallization process. In recrystallization, the grains of crystal grow in different orientations.
- Key types of surface defects are:
 - Grain boundary: Grain boundaries are interfacial/surface defects separating individual crystal grains with different crystallographic orientations in a polycrystalline material.
 - Tilt boundary: A Tilt Boundary, between two slightly mis-aligned grains appears as an array of edge dislocations. Rotation axis is parallel to the boundary plane.
 - Twist Boundary: Rotation axis is perpendicular to the boundary plane Low angle grain boundaries, that appear as an array of Screw dislocations.
 - Twin boundary: These are the boundaries in the grains at which the atomic arrangement on one side of the boundary is the mirror image of the atoms on the other side. These are produced by shear deformation. The region between the pair of boundaries is called the twinned region
 - Stacking faults: It is a planar crystallographic defect where the normal, repeating sequence of atomic layers in a crystal is disrupted.

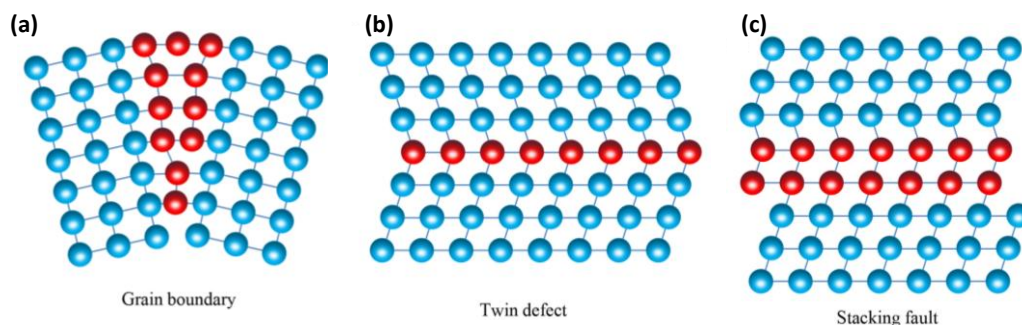
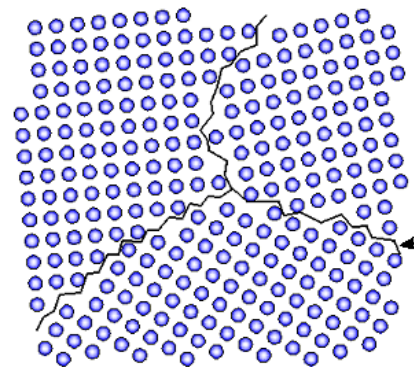


Figure 1.19. Schematic illustrations of the different types of surface defects, (a) grain

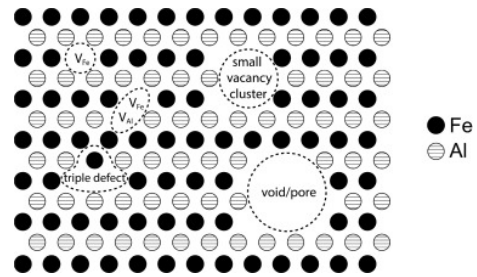
boundaries, (b) twin and (c) stacking fault.

4. Volume defects:

- Volume defects are large, three-dimensional (3D) imperfections in a crystal structure that span a significant macroscopic space, including voids, pores, and inclusions.
- It is common to divide them into four classes in an imprecise classification that is based on a combination of the size and effect of the particle.

The four categories are:

- i. *Precipitates*, which are a fraction of a micron in size and decorate the crystal.
- ii. *Dispersants*, which vary in size from a fraction of a micron to the normal grain size (10-100 μm), but are intentionally introduced into the micro structure.
- iii. *Inclusions*, which vary in size from a few microns to macroscopic dimensions, and are relatively large, undesirable particles that entered the system as dirt or formed by precipitation;
- iv. *Voids*, which are holes in the solid formed by trapped gases or by the accumulation of vacancies.



1.11. CONCEPT OF NANOMATERIALS

Introduction: Nanomaterials are materials that have at least one dimension in the nanometer range, typically between 1 and 100 nanometers (1 nm = 10^{-9} m). This size range is crucial because it imparts unique physical, chemical, and biological properties to the materials that are not observed in their bulk counterparts. At the nanoscale, materials exhibit increased surface area to volume ratios, quantum effects, and other phenomena that can significantly alter their behavior and interactions.

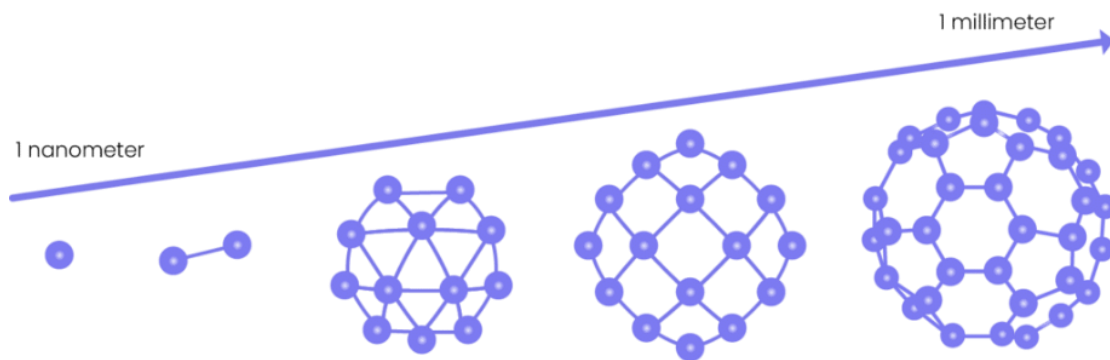


Figure 1.20: A general description of micro to nano-sized particles.

In general, the nanomaterials are classified into various types based on their dimensions: zero-dimensional (0D) nanoparticles, one-dimensional (1D) nanowires and nanotubes, two-dimensional (2D) nanofilms and nanocoatings, and three-dimensional (3D) nanocomposites. Each type exhibits distinct characteristics and offers specific advantages for different technological applications.

- i. **Zero-Dimensional (0D) Nanomaterials:** These are materials where all three dimensions

are in the nanoscale. Examples include nanoparticles, nanospheres, and nanoclusters. They are often used in biomedical applications, including drug delivery and imaging, due to their small size and high surface area.

- ii. **One-Dimensional (1D) Nanomaterials:** These materials have one dimension outside the nanoscale. Examples include nanowires, nanotubes, and nanorods. One-dimensional nanomaterials are used in electronics and photonics for their ability to conduct electricity and light efficiently along their length.
- iii. **Two-Dimensional (2D) Nanomaterials:** These materials have two dimensions outside the nanoscale. Examples include graphene, nanofilms, and nanocoatings. Two-dimensional nanomaterials are known for their excellent mechanical, electrical, and thermal properties and are used in applications ranging from flexible electronics to barrier coatings.
- iv. **Three-Dimensional (3D) Nanomaterials:** These materials have three dimensions outside the nanoscale but contain nanoscale features, such as nanocomposites and nanostructured bulk materials. Three-dimensional nanomaterials combine the properties of their nanoscale components with those of bulk materials, making them suitable for structural applications and advanced material design.

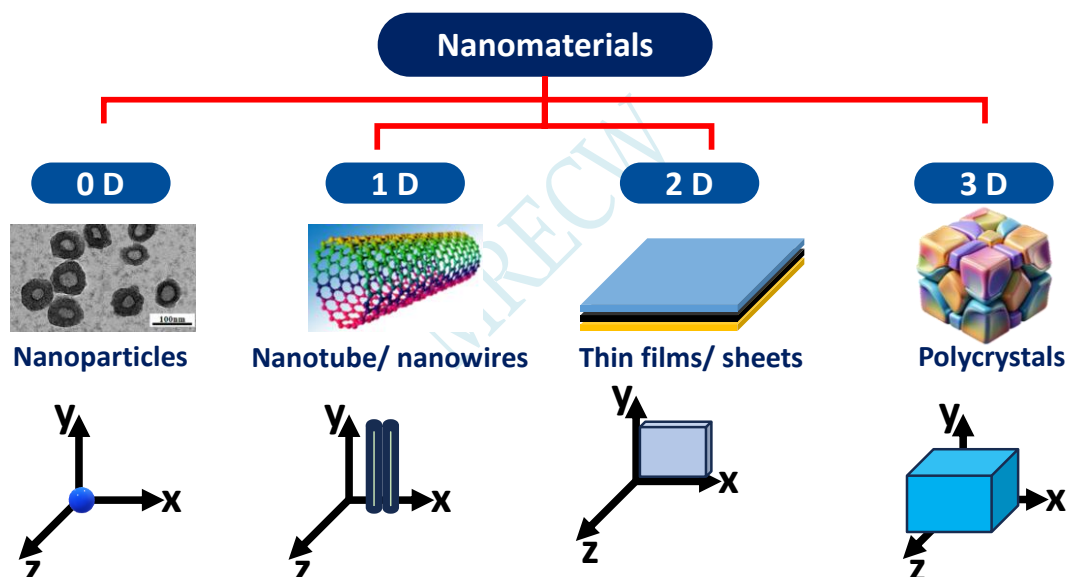


Figure 1.21: Schematic representation of 0D, 1D, 2D and 3D nanomaterials.

Surface Area to Volume Ratio:

When ordinary materials are broken down to this ultra-fine (nano) scale, they stop behaving like bulk materials and begin displaying drastically different physical, chemical, and biological properties. In bulk materials (like a bar of iron or a block of wood), almost all the atoms sit inside the core, safe and inactive. Only a tiny fraction of atoms sits on the outer surface where they can interact with the surrounding environment.

As you divide a material into smaller pieces: The total volume stays constant, but, unexposed or hidden internal atoms, massively increasing the total surface area. Because surface atoms have fewer neighboring atoms to bond with than internal atoms, they possess higher surface energy and unbound chemical bonds. This makes surface atoms far more reactive. In nanomaterials, a dominant percentage of all atoms lie on the surface—making the

Surface-to-Volume (S/V) ratio the single most decisive factor driving nanoscale properties.

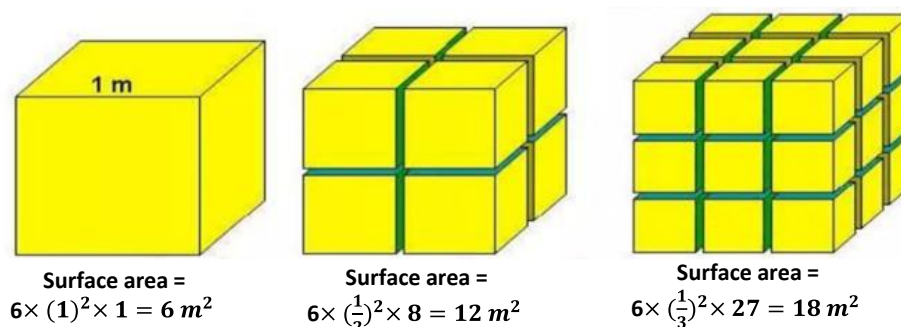


Figure 1.22: Surface Area Growth through Particle Subdivision.

To understand the mathematics, let's compare a single large cube against smaller subdivisions of the same total material volume.

For a cube with edge length a ,

- Total Surface Area (SA) = $6a^2$
- Total Volume (V) = a^3
- Surface-to-Volume Ratio (S/V) = $\frac{6a^2}{a^3} = \frac{6}{a}$

The surface-to-volume ratio is inversely proportional to particle size ($S/V \propto \frac{1}{a}$). As the side length a decrease, S/V increase dramatically.

Comparative Calculation:

Scale	Edge Length	Surface Area	Volume	S/V Ratio
Macro	1 cm = 10^{-2} m	6 cm ²	1 cm ³	6 cm ⁻¹
Micro	1 μ m = 10^{-6} m	6×10^{-8} m ²	10^{-18} m ³	6×10^6 m ⁻¹
Nano	1 nm = 10^{-9} m	6×10^{-18} m ²	10^{-27} m ³	6×10^9 m ⁻¹

When going from a 1-cm bulk cube to 1-nm cubes, the surface area increases. Essentially, this S/V ratio will show the effect on i) chemical reactivity, ii) Lower the melting point of the material, iii) increase tensile strength, enhance the electrical & optical properties.

❖ Real-World Applications:

- **Catalytic Converters:** Exhaust systems use platinum-palladium nanoparticles. A tiny mass of precious metal provides thousands of square meters of active surface area to neutralize harmful vehicle emissions into CO₂ and N₂.
- **Energy Storage (Batteries & Supercapacitors):** Lithium-ion battery electrodes built with nanostructured materials allow rapid ion diffusion over vast surface interfaces, drastically shortening recharge times.
- **Targeted Drug Delivery:** Nanoparticles carrying cancer therapeutics offer huge surface areas for functionalizing targeting molecules (ligands). This allows them to bind selectively to tumor cells and release medication precisely where needed.
- **Sunscreens:** Titanium dioxide (TiO₂) and Zinc oxide (ZnO) at the nanoscale absorb UV light effectively due to increased surface interaction, while remaining completely transparent on human skin.

1.12. X-RAY DIFFRACTION

Structures of solids can be determined by various methods. These are listed below

1. X-ray diffraction technique
2. Microscopic examination
3. Electron diffraction patterns
4. Spectroscopy
5. Nuclear magnetic resonance
6. Mossbauer studies
7. Neutron diffraction
8. Holography



Out of these techniques X rays and microscopic studies have helped immensely in studying crystallography of metals. Therefore, these two techniques are very popular. However, only x rays will be discussed in accordance with the requirement of syllabus.

X ray diffraction:

Diffraction effects can be observed only if the spacing between the lines ruled on the grating is of the order of magnitude of the wavelength of light used. In the diffraction of light, the grating element is in the order of 10^{-6} m. X-rays are electromagnetic waves with wavelength in the order of an angstrom ($1/1000^{\text{th}}$ of wavelength of visible light). Therefore, to observe diffraction pattern of X rays the grating should have a spacing $1/1000^{\text{th}}$ of the grating used for the diffraction of light. Von laue suggested that the crystal can be used for the diffraction of X rays because the spacing between the atoms of a crystal is of the order of $1 \text{ \AA}$. The diffracting centers in the crystal are not in one plane thus the crystal acts as a three-dimensional grating.

Production of X- rays

Roentgen discovered the x rays. X rays are produced with a beam of fast-moving electrons strike a solid target material (tungsten or Molybdenum). The X rays are electromagnetic waves having wavelengths from $0.5 \text{ \AA}$ to $15 \text{ \AA}$ and are not deflected by electric and magnetic fields. The X rays produced may be either (1) Hard X-rays (short-wavelength) and 2) Soft X-rays (long-wavelength).

Hard X rays require voltages above 50KV for their productions Hard X rays are mainly employed in the study of solid structures and in industrial applications. Soft X rays are common in medical applications. W.H Bragg and his son W.L. Bragg while studying the properties of X rays studied the x ray diffraction patterns of few crystals. According to them the diffraction is due to the reflection of some incident X rays from the various set of parallel crystal planes (called Bragg planes) which contain a large number of atoms.

1.13. BRAGG'S LAW

W.H. Bragg and W.L Bragg derived a simple equation relating the wavelength of X rays to the angular positions of the scattered beams and separation of atomic plane in the crystal. *Bragg's law states that the path difference between two reflected X rays by the crystal planes should be an integral multiple of wavelength of incident X rays for producing maximum or*

constructive interference.

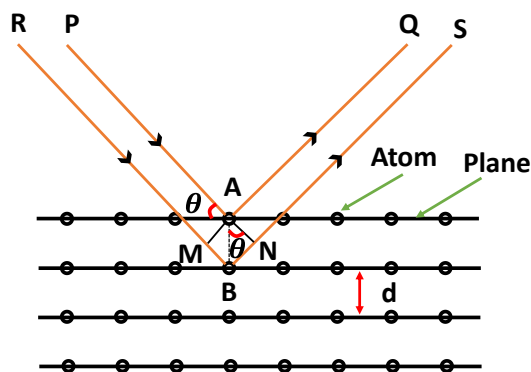


Figure 1.23: Schematic illustration of Bragg condition and Bragg's law. The path difference between reflected waves is equal to $MB + BN$.

When the two consecutive planes scatter waves in phase with each other, then the path difference must be equal to integral multiple of the incident wavelength.

Consider two parallel rays PA and RB of the incident beam which are scattered by two atoms A and B in adjacent layers as shown in Fig. Constructive interference takes place only between those scattered rays that are parallel and whose path differ exactly $\lambda, 2\lambda, 3\lambda$ and so on. i.e. the path difference Δ must be $n\lambda$, where 'n' is an integer. Draw AM and AN perpendicular to the rays RB and BS. Path difference between the X rays reflected along AQ and BS are given by

$$\Delta = MB + BN \Rightarrow AB \sin\theta + AB \sin\theta$$

$$\Delta = 2d \sin\theta$$

Where $AB=d$ is the interplanar distance. i.e. spacing between the atomic planes.

For constructive interference to take place, the path difference, $\Delta = n\lambda$. Where $n=1,2,3$ etc. for first order, second order, third order..... etc., maxima respectively. This relation is known as Bragg's law for X ray diffraction.

Different directions in which intense reflections will be produced can be obtained by giving different values to θ i.e.

$$\text{For 1}^{\text{st}} \text{ maxima } \sin\theta_1 = \lambda/2d$$

$$2^{\text{nd}} \text{ maxima } \sin\theta_2 = 2\lambda/2d$$

$$3^{\text{rd}} \text{ maxima } \sin\theta_3 = 3\lambda/2d$$

The intensities of maxima's go on decreases with increase in order of the spectrum.

When a beam of monochromatic X rays falls on a crystal, each atom becomes a source of scattering radiation. In the crystal there are certain planes which are particularly rich in atoms. The combined scattering of X rays from these planes can be looked up on as reflections from these planes. Generally, the Bragg scattering is regarded as bragg reflections and hence these set of parallel planes are known as Bragg planes. At certain glancing angles, reflections from these planes are in phase with each other and hence they reinforce each other to produce maximum intensity. For other angles, the reflections from different planes are out of phase and hence reinforce to produce either Zero or extremely feeble intensity. Since the maximum

possible value for θ is 1.

$$n\lambda/2d \leq 1$$

This sets limitation on wavelength. The wavelength λ should not exceed twice the interplanar distance for diffraction to occur.

Applications of X-Ray diffraction:

The method of examinations by X rays is called radiography. X ray diffraction procedures will reveal details about

- 1) Grain size or crystal size
- 2) Disorders in the unit cell
- 3) Orientations of poly crystalline crystals
- 4) Arrangement of atoms
- 5) Revealing of imperfections
- 6) Type of unit cell

X-ray diffraction methods

Conventionally three standard methods are used in the studies of crystal structure using X rays, they are

- a) The Laue method (applicable to single crystals)
- b) The powder method (applicable to finely divided crystals or polycrystalline specimen powders)
- c) The rotating crystal method (applicable to single crystals).

1.14. POWDER DIFFRACTION METHOD

Powder crystal method is a standard technique to study the structure of micro crystals (substances which are available in the form of very small crystals or in powder form). This method gives the information regarding the size of the crystal, presence of imperfections, distortion. Preferred orientations of crystals.

This method is due to Debye scherrer and hull. It consists of a cylindrical camera called debye scherrer camera whose length is small compared to diameter. The finely powdered sample is filled in a thin capillary tube and mounted at the center of camera. The capillary tube should be a non-diffracting material. A collimated beam of monochromatic X rays is produced by passing the X rays through a filter and collimator. Then the monochromatic X rays enter the camera through a small hole and strike the specimen. Since the specimen contains large number of small crystallites with random orientations. The diffraction takes place for those values of d and θ .

Which satisfy the Bragg's condition i.e. $2d \sin\theta = n\lambda$

If θ is the glancing angle corresponding to a value of 'd' for one set of planes the reflected ray will be inclined at an angle ' 2θ '. The open of the cone is the point of contact of X ray with the specimen.

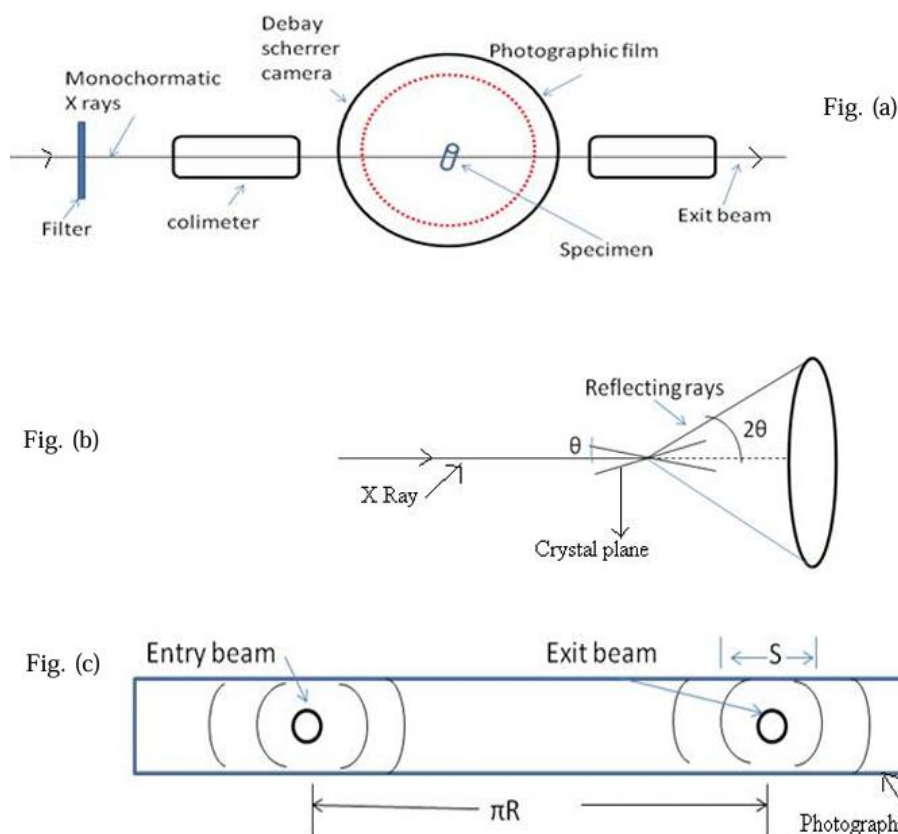


Figure.1.24. Powder method (a) Debye Scherrer Camera (b and c) Powder diffraction pattern.

Different cones are observed for different sets of d and θ for a particular value of 'n' and also for different combinations of θ and n for a particular value of 'd'. The transmitted ray moves out of the camera through an exit hole located diametrically opposite to the entrance hole. A part of each reflected cone is recorded on the film which is positioned around the camera. The recorded lines are in the form of arc's as shown in figure 1.24.

The angle between the surface of cone i.e. reflected ray and exit beam is 2θ or we can say that open angle is 4θ . In case Bragg angle is 45° the cone will take the form of a circle. This circle will intersect the film strip at midpoints between entrance and exit hole. When the Bragg angle is greater than 45° , the back reflection takes place, the reflection cones are formed towards incidence hole.

The first pair of arcs on each side of exit hole is due to smaller value of θ . The other pairs of arcs beyond this pair are due to longer value of θ . The distance S on the film between two diffraction lines corresponding to a particular plane is related to the Bragg angle by the equation.

$$4\theta = \frac{S}{R} \text{ radians}$$

$$4\theta = \frac{1}{R} S \text{ radians}$$

where R is the specimen to film distance usually the radius of the camera housing the film.

A list of θ values can thus be obtained from the measured values of S . Since the wavelength is known, substitution of θ and λ gives a list of spacing d . Each spacing is the

distance between neighboring planes (h k l). So, the reflections can be indexed and the unit cell parameters are evaluated.

Determination of crystal structure:

Let us consider the 1st order spectrum from three planes one after the other reflecting a given monochromatic X-ray beam of wavelength λ at glancing angles $\theta_1, \theta_2, \theta_3$.

According to Bragg law,

$$2d \sin\theta = n\lambda$$

$$2d \sin\theta = \lambda \quad (\text{let } n=1)$$

$$\frac{1}{d} = \frac{2\sin\theta}{\lambda}$$

Now calculate d values for the given planes (h k l) as shown below.

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = \sin\theta_1 : \sin\theta_2 : \sin\theta_3$$

The ratio of $\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}}$ for different cubic lattices are given below.

Simple cubic: $\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}}$ is $1 : \sqrt{2} : \sqrt{3}$

Body centered cubic: $\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}}$ is $1 : \frac{1}{\sqrt{2}} : \sqrt{3}$

Face centered cubic: $\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}}$ is $1 : \sqrt{2} : \frac{\sqrt{3}}{2}$

- For example, for NaCl crystal find order reflection maxima occurs at $5.9^\circ, 8.4^\circ, 5.2^\circ$ for (100), (110) and (111) planes respectively.

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = \sin 5.9 : \sin 8.4 : \sin 5.2$$

$$d_{100} : d_{110} : d_{111} = 9.3 : 6.84 : 11.04$$

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = 1 : \sqrt{2} : \frac{\sqrt{3}}{2}$$

Hence NaCl has a face centered cubic structure.

By knowing the values of glancing angles ratio, the ratio of Inter planer spacing and hence the type of the lattice can identify.

Advantages of Powder diffraction method:

- ❖ The powder technique has the advantage that it does not require large single crystals and almost any substance can be ground into powder.
- ❖ The experiment is easy to perform.
- ❖ Small crystals of any substance can be easily obtained.

Problems

Problem1: A plane cuts the axes at $2a, 3b$ and ∞c . Find its Miller indices.

Solution: Reciprocals of intercepts: $1/2, 1/3, 0$. Multiply by 6: 3, 2, 0.

Answer: (320).

Problem2: For a cubic crystal with lattice constant (a) $4.20 \text{ \AA}$, calculate Miller indices of a plane $d110$.

Solution: $d(hkl) = a/\sqrt{(h^2+k^2+l^2)}$. Therefore $d110 = 4.20/\sqrt{2} = 2.97 \text{ \AA}$.

Answer: $2.97 \text{ \AA}$.

Problem3: The wavelength of radiation (λ) $1.54 \text{ \AA}$ gives first-order reflection at an glancing angle (θ) 30° . Find inter planar spacing (d).

Solution: $n\lambda = 2d \sin\theta$. $d = 1.54/(2 \sin 30^\circ) = 1.54 \text{ \AA}$.

Answer: $1.54 \text{ \AA}$.

Problem4: An X-Ray diffraction peak occurs at an angle (2θ) is 44.7° using wavelength (λ) of $1.5406 \text{ \AA}$. Find inter planar distance (d) for first order.

Solution: $\theta = 22.35^\circ$. $d = 1.5406/[2\sin 22.35^\circ] \approx 2.03 \text{ \AA}$.

Answer: $\approx 2.03 \text{ \AA}$.

Problem5: A cubic crystal gives inter planar spacing for the plane $d111 = 2.10 \text{ \AA}$. Find the lattice constant.

Solution: $d111 = a/\sqrt{3}$, so $a = d\sqrt{3} = 2.10 \times 1.732 = 3.64 \text{ \AA}$.

Answer: $a \approx 3.64 \text{ \AA}$.

Problem6: Molybdenum has a BCC structure. Its density is $10.2 \times 10^3 \text{ kg/m}^3$ and its atomic weight is 95.94. determine the radius of molybdenum atom?

Solution: Density of a cubic unit cell is given by $\rho = \frac{ZM}{N_A a^3}$

$$a^3 = \frac{ZM}{\rho N_A} = \frac{2 \times 95.94 \text{ kg/k.mol}}{10.2 \times 10^3 \text{ kg/m}^3 \times 6.02 \times 10^{26} \text{ /k.mol}} = 31.25 \times 10^{-30} \text{ m}^3$$

Answer: $a = 3.15 \text{ \AA}$

The atomic radius in BCC structure is given by $R = \frac{a\sqrt{3}}{4} = \frac{3.15 \text{ \AA} \times 1.732}{4} = 1.364 \text{ \AA}$

Problem7: At a diffraction angle 2θ is 60° with full width half maximum (FWHM) $\beta = 0.30^\circ$, $\lambda = 1.5406 \text{ \AA}$, $K = 0.9$. Find crystallite size D .

Solution: $\theta = 30^\circ$, $\beta = 0.005236 \text{ rad}$. $D = 0.9(1.5406 \times 10^{-10})/(0.005236 \times 0.8660) = 3.06 \times 10^{-8} \text{ m}$.

Answer: $D \approx 30.6 \text{ nm}$.

Problem8: A cubic material shows a first-order peak at $2\theta = 36^\circ$ using $\lambda = 1.54 \text{ \AA}$. If it is the (111) reflection, find inter planar spacing $d111$ and a .

Solution: $\theta = 18^\circ$. $d = 1.54/[2\sin 18^\circ] = 2.49 \text{ \AA}$. Since $d111 = a/\sqrt{3}$, $a = 2.49\sqrt{3} = 4.31 \text{ \AA}$.

Answer: $d \approx 2.49 \text{ \AA}$, $a \approx 4.31 \text{ \AA}$.

Problem9: Calculate inter planar spacing for the planes $d100$, $d110$ and $d111$ with a lattice constant $5 \text{ \AA}$. Which has the smallest spacing?

Solution: $d100 = 5.00 \text{ \AA}$; $d110 = 5/\sqrt{2} = 3.54 \text{ \AA}$; $d111 = 5/\sqrt{3} = 2.89 \text{ \AA}$.

Answer: $d111$ is smallest.

Problem10: An X-Ray diffraction peak occurs at an angle (2θ) is 50° with full width half maximum (FWHM, β) 0.25° of wavelength (λ) $1.5406 \text{ \AA}$ (constant $K = 0.94$ given). Find crystallite size.

Solution: $\theta = 25^\circ$, $\beta = 0.004363 \text{ rad}$. $D = 0.94(1.5406 \times 10^{-10}) / (0.004363 \times \cos 25^\circ) = 3.80 \times 10^{-8} \text{ m}$.
Answer: $\approx 38.0 \text{ nm}$.

Problem11: A peak occurs at an angle (2θ) 50° with full width half maximum (β) 0.25° , of wavelength (λ) $1.5406 \text{ \AA}$, $K = 0.94$. Assume it is (200) of a cubic crystal. Find inter planar spacing (d), lattice constant (a) and crystallite size (D).

Solution: $\theta = 25^\circ$. $d = 1.5406 / [2 \sin 25^\circ] = 1.82 \text{ \AA}$. For (200), $d = a/2$, so $a = 3.63 \text{ \AA}$.
 $\beta = 0.25\pi/180 = 0.004363 \text{ rad}$. $D = 0.94(1.5406 \times 10^{-10}) / (0.004363 \cos 25^\circ) = 3.80 \times 10^{-8} \text{ m}$.
Answer: $d \approx 1.82 \text{ \AA}$, $a \approx 3.63 \text{ \AA}$, $D \approx 38.0 \text{ nm}$.

Problem12: The density of copper is 8980 kg/m^3 and unit cell dimension is $3.61 \text{ \AA}$. Atomic weight of copper is 63.54 . Determine crystal structure? Calculate atomic radius and interplanar spacing of (110) plane?

Solution: i) The effective number of atoms per unit cell is given by $Z = \frac{\rho N_A a^3}{M}$

$$Z = \frac{8980 \text{ kg/m}^3 \times 6.02 \times 10^{26} / \text{k.mol} \times (3.61 \times 10^{-10} \text{ m})^3}{63.54 \text{ kg/k.mol}} = \frac{254.32}{36.54} = 4 \text{ atoms /unit cell}$$

Hence Copper exhibits FCC structure

ii) The lattice constant is related to the atomic radius in FCC structure through $a = 2\sqrt{2}r$

$$r = \frac{a}{2\sqrt{2}} = \frac{3.61 \text{ \AA}}{2 \times 1.4142} = 1.276 \text{ \AA}$$

iii) The inter planar spacing is given by $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{3.61 \text{ \AA}}{\sqrt{1^2 + 1^2 + 0}} = 2.553 \text{ \AA}$

Review Questions

Short Answer Questions

1. Define a unit cell and state its significance in crystallography.
2. What are Bravais lattices?
3. Define the Atomic Packing Factor (APF).
4. What are Miller indices?
5. Differentiate between point defects and line defects in a crystal structure.
6. Why does the surface-to-volume ratio increase significantly at the nanoscale?
7. State Bragg's law of X-ray diffraction.
8. What is the primary purpose of using the Debye-Scherrer formula in XRD analysis?
9. Write the relationship between inter-planar spacing (d) and lattice parameter (a) for a cubic system.
10. Give an example of a surface defect in crystalline solids.

Long Answer Questions

1. Describe the crystal parameters, unit cell characteristics, and calculate the Atomic Packing Factor (APF) for Simple Cubic (SC), Body-Centered Cubic (BCC), and Face-Centered Cubic (FCC) crystal structures.
2. Explain Miller indices with suitable examples. Derive the expression for inter-planar

distance (d) between parallel (hkl) planes in a cubic crystal lattice.

3. Provide a detailed classification of crystal defects (point, line, surface, and volume defects) with appropriate diagrams and descriptions.
4. Discuss the concept of nanomaterials, focusing on how the surface-to-volume ratio affects physical and chemical properties at the nanoscale.
5. Describe the principle, experimental setup, and working of the Powder XRD method for crystal structure determination, and discuss the Debye-Scherrer formula.
6. Derive Bragg's law for X-ray diffraction. Explain how XRD data is used to determine crystallite size and lattice parameter.
7. Outline the synthesis and characterization workflow of a nanomaterial using X-ray diffraction techniques.
8. What are miller indices? Explain how are they obtained?
9. Show that FCC unit cell packing factor is higher than BCC and SC unit cells?
10. Deduce and discuss 7 crystal systems and 14 Bravais lattices with neat diagrams?

Objective Questions

1. Crystallography is primarily concerned with the
 - A) Study of fluid flow
 - B) Arrangement of atoms in crystalline solids
 - C) Study of nuclear reactions
 - D) Design of electrical circuits
2. A crystal with a continuous ordered 3D arrangement is called a
 - A) Single crystal
 - B) Amorphous solid
 - C) Liquid crystal
 - D) Polycrystal
3. A polycrystalline material consists of many small crystals called
 - A) Quanta
 - B) Grains
 - C) Domains only
 - D) Cells
4. The smallest repeating unit of a crystal lattice is the
 - A) Basis
 - B) Unit cell
 - C) Grain boundary
 - D) Defect
5. A crystal structure is obtained by combining a lattice with a
 - A) Basis
 - B) Dislocation
 - C) Photon
 - D) Grain boundary
6. The three lengths and three angles defining a unit cell are called
 - A) Miller indices
 - B) Lattice parameters
 - C) Packing factors
 - D) Bragg parameters
7. The number of Bravais lattices in three dimensions is
 - A) 7
 - B) 10
 - C) 14
 - D) 16
8. The coordination number of simple cubic structure is
 - A) 4
 - B) 6
 - C) 8
 - D) 12
9. The coordination number of BCC structure is
 - A) 6
 - B) 8
 - C) 10
 - D) 12
10. The coordination number of FCC structure is
 - A) 6
 - B) 8
 - C) 10
 - D) 12
11. The number of effective atoms in a BCC unit cell is
 - A) 1
 - B) 2
 - C) 4
 - D) 6
12. The number of effective atoms in an FCC unit cell is
 - A) 1
 - B) 2
 - C) 4
 - D) 8
13. The approximate atomic packing factor of simple cubic is
 - A) 0.34
 - B) 0.52
 - C) 0.68
 - D) 0.74
14. The approximate atomic packing factor of FCC is
 - A) 0.34
 - B) 0.52
 - C) 0.68
 - D) 0.74
15. Miller indices are used to specify
 - A) Crystal planes
 - B) Atomic masses
 - C) Grain sizes
 - D) Defect energies
16. Miller indices of a plane are conventionally written as

- A) [hkl] B) (hkl) C) <hkl> D) {hkl} only
17. A missing atom from its normal lattice position produces a
A) Vacancy defect B) Line defect C) Surface defect D) Volume defect
18. An interstitial defect occurs when an atom occupies
A) A normal lattice site B) An interstitial site
C) A grain boundary only D) A vacuum chamber
19. A dislocation is a
A) Point defect B) Line defect C) Surface defect D) Volume defect
20. As particle size decreases to the nanoscale, the surface-to-volume ratio
A) Decreases B) Remains constant C) Increases D) Becomes zero
21. Bragg's law for X-ray diffraction is
A) $n\lambda = d \sin\theta$ B) $n\lambda = 2d \sin\theta$ C) $\lambda = nd \cos\theta$ D) $d = 2n\lambda \sin\theta$
22. In Bragg diffraction, θ represents the
A) Scattering angle B) Bragg angle C) Critical angle D) Acceptance angle
23. The powder method of X-ray diffraction is especially useful for
A) Polycrystalline materials B) Liquids only
C) Gases only D) Individual electrons
24. The Debye–Scherrer formula is used to estimate
A) Crystallite size B) Atomic number C) Magnetic moment D) Dielectric constant
25. Broadening of XRD peaks at nanoscale is associated with
A) Small crystallite size B) Large grain size only
C) Zero diffraction D) Absence of crystal planes

Fill in the blanks:

- The science dealing with the arrangement of atoms in crystalline solids is called _____.
- A regularly repeating three-dimensional array of points is called a _____ lattice.
- The atom or group of atoms attached to every lattice point is called the _____.
- The smallest repeating unit of a crystal is the _____ cell.
- The lengths and angles that define a unit cell are called _____ parameters.
- The number of nearest neighbors surrounding an atom is its _____ number.
- There are _____ Bravais lattices in three dimensions.
- The simple cubic structure contains _____ effective atom per unit cell.
- The BCC structure contains _____ effective atoms per unit cell.
- The FCC structure contains _____ effective atoms per unit cell.
- The packing fraction of FCC is approximately _____.
- The crystallographic planes are represented by _____ indices.
- A missing atom from a lattice site is called a _____ defect.
- An atom occupying a normally vacant space between lattice points produces an _____ defect.
- A dislocation is a _____ defect.
- Grain boundaries are examples of _____ defects.
- The surface-to-volume ratio generally increases when particle size _____.
- X-ray diffraction is used to study the _____ structure of materials.

19. Bragg's law is written as $n\lambda =$ _____.
20. The angle appearing in Bragg's law is called the _____ angle.
21. The powder XRD method is commonly used for _____ materials.
22. The Debye–Scherrer equation gives an estimate of average _____ size.
23. XRD peak broadening becomes significant for _____ crystallites.
24. A material made up of many grains is called a _____ material.
25. A solid lacking long-range periodic atomic order is called an _____ solid.

MRECW

MCQs: 1-B, 2-A, 3-B, 4-B, 5-A, 6-B, 7-C, 8-B, 9-B, 10-D, 11-B, 12-C, 13-B, 14-D, 15-A, 16-B, 17-A, 18-B, 19-B, 20-C, 21-B, 22-B, 23-A, 24-A, 25-A

Fill in the blanks: 1. crystallography; 2. space; 3. basis; 4. unit; 5. lattice; 6. coordination; 7. 14; 8. 1; 9. 2; 10. 4; 11. 0.74; 12. Miller; 13. vacancy; 14. interstitial; 15. line; 16. surface; 17. decreases; 18. crystal; 19. $2d \sin\theta$; 20. Bragg; 21. polycrystalline; 22. crystallite; 23. nanometer-scale; 24. polycrystalline; 25. amorphous

UNIT – 2: QUANTUM MECHANICS

Contents: - Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box, Bloch's theorem (qualitative), Kronig-Penney model (qualitative): E-k diagram, formation of energy bands, origin of bandgap, classification of solids, concept of discrete energy levels and quantum confinement in nanomaterials.

2.1. INTRODUCTION

In our everyday experience, we observe particles and waves without much ambiguity. Particles moving from one point to another point in space, carry momentum and energy as they possess definite mass and velocity at the given instant of time. Waves on the other hand also transfer momentum and energy but without anything transported physically. Consider the example of a stone dropped into a well and the ripples that spread out from the point from where the stone in contact with water in the well, apparently have in common only the ability to carry energy and momentum. In the first case, the stone, a physical object carries energy and momentum and in the later case, only a disturbance in a medium, a wave carries forward the energy and the momentum. Thus, in classical physics, particles and wave are treated as separate components. Newtonian mechanics of particles and wave optics are traditionally independent disciplines, each with its own chain of experiments and principles based on their result.

The story is entirely different when we enter into subatomic world. In the microscopic world of atoms and molecules, electrons and neutrons, the rules are different and as we shall see later, it is perfectly alright to treat these fast subatomic particles both as matter particles and matter waves. The physics that treat behavior of these subatomic particles appropriately is known as quantum mechanics. Particle diffraction experiments, photo emission process, Compton Effect, discovery of electron microscope and modern counterparts like SEM (scanning electron microscope) and TEM (tunneling electron microscope), quantum devices like SQUID (superconducting quantum interfering devices), MRI scanning and other semiconductor devices are all result of development of quantum mechanics.

Origin of Quantum Mechanics:

The origin of quantum mechanics lies in the explanation of thermal radiation from hot bodies of matter satisfactorily by Max Planck around the years 1900. Planck very radically and boldly formulated that the radiation emitted by bodies is not continuous (as required by classical continuous wave theory) but in discrete form in terms of packets of energy called 'quanta'. Followed by the Einstein's photoelectric experiment, Compton effect (scattering of x-ray photon by electron) and subsequent building up of electron microscope on experiment side, Bohr's postulates, Pauli's principle, de-Broglie hypothesis together with mathematical formulation of Schrödinger, Born, Pauli and others, today a strong foundation has been laid between 1900-1930 for the development of quantum mechanics. Prior to this period, the major stumbling block in the development of science is the failure of classic wave theory (which in fact explained the experiments by Hertz and young so well and which could be analyzed so

precisely with the Maxwell's equation) to explain the observed spectrum of thermal radiation.

Drawbacks of Classical Mechanics:

- Classical mechanics failed to explain the stability of the atom.
- The non-relativistic motion of atoms, electrons, protons, etc.
- Origin of discrete spectra of atoms.
- The spectrum of black body radiation.
- Variation of specific heat of metals and gases.
- Failure to explain the photoelectric effect, Compton effect, Zeeman effect, Raman effect, etc.

Classical vs. Quantum Mechanics

Classical Mechanics	Quantum Mechanics
1. It deals with macroscopic particles. 2. It is based on Newton's laws of motion and Maxwell's electromagnetic wave theory. 3. Any amount of energy may be emitted or absorbed continuously. 4. The state of a system is defined exactly by specifying their positions and velocities. 5. The future state can be predicted with certainty.	1. It deals with microscopic particles 2. It is based on the Schrödinger equation 3. In Planck's postulation, only discrete values of energy are emitted or absorbed – origin of "quantum" 4. Because of Heisenberg's uncertainty principle and de Broglie hypothesis dual nature of matter (both particle & wave), the state of a system cannot be specified exactly 5. It gives probabilities of finding particles at various locations in space

2.2. DE-BROGLIE HYPOTHESIS

The de-Broglie hypothesis states that all matter has wave-like properties and stemmed from the idea that "nature loves symmetry". Since energy possesses dual nature, matter must also exhibit dual nature. This implies that the elementary particles like electrons, protons, neutrons etc. must exhibit wave-like nature too.

According to de Broglie theory "every moving particle is associated with a wave (matter wave or de-Broglie wave) having wave length. The wavelength of the wave is given by,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where, h is the Plank's constant and $p = mv$ is the momentum of the particle of mass 'm' moving with the velocity 'v'.

To derive an expression for de-Broglie wavelength, we recall that energy of a photon is

$$E = h\nu = \frac{hc}{\lambda} \quad (\because c = \nu\lambda) \quad \text{----- (1)}$$

Again, in view of Einstein's mass-energy relation:

$$E = mc^2 \quad \text{----- (2)}$$

Now, from equations (1) and (2), we get

$$\Rightarrow \frac{hc}{\lambda} = mc^2 \Rightarrow \frac{h\nu}{\lambda} = mc^2 \therefore \lambda = \frac{h}{mc}$$

where mc is the momentum (p) of the photon.

$$\text{Therefore, the de-Broglie wavelength is } \lambda = \frac{h}{p} \quad \text{----- (3)}$$

This equation (Eq. (3)) gives the de-Broglie hypothesis of a particle in motion.

2) de Broglie wavelength in terms of K.E.

The kinetic energy of a moving particle $E = \frac{1}{2}mv^2 = \frac{1}{2m}p^2$

$$E = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mE}$$

Substituting the value of P in equation (3) we get, $\lambda = \frac{h}{\sqrt{2mE}}$

3) de-Broglie wavelength of electrons:

Let us consider that an electron of mass m and charge e is accelerated through a potential difference of V volts. The energy acquired by the electron is eV,

$$eV = \frac{1}{2}mv^2 \Rightarrow eV = \frac{p^2}{2m} \Rightarrow p = \sqrt{2meV}$$

Substituting the value of P in equation (3) we get,

De-Broglie wavelength of electron, $\lambda = \frac{h}{\sqrt{2meV}}$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times V}} \text{ m} \Rightarrow \lambda = \frac{12.28 \times 10^{-10}}{\sqrt{V}} \text{ m}$$

$$\Rightarrow \lambda = \frac{12.28}{\sqrt{V}} \text{ \AA}$$

2.3. PHYSICAL SIGNIFICANCE OF WAVE FUNCTION

In light waves, water waves and in sound waves the quantity varies periodically is electric and magnetic fields, water surface and pressure respectively. In matter waves the quantity varies periodically is called wave function, symbol ψ (psi). The value of the wave function associated with a moving particle at a point (x,y,z) and time t gives the probability of finding the particle at that time and at that point.

Properties of Matter Waves:

1) Matter-waves are associated with any moving body and their wavelength is de Broglie wave

length.

- 2) Wavelength of matter-waves depends on the mass of the body and decreases with increase in mass. Because of this, the wave-like behavior of heavier objects is not very evident whereas the wave nature of subatomic particles can be observed experimentally.
- 3) Matter waves are not electromagnetic. They are new kind of waves.
- 4) Unlike other waves, there is no physical quantity that varies periodically in the case of Matter waves. The matter waves are not physical phenomenon; it is just a wave of probability.
- 5) Matter waves are represented by a wave packet made up of a group of waves of slightly differing wavelengths. Hence, we talk of group velocity of matter waves rather than the phase velocity.
- 6) Amplitude of the matter-waves at a particular space and time depends on the probability of finding the particle at that space and time.
- 7) These waves can travel faster than the velocity of light. The velocity of matter waves is not constant it depends on the velocity of the material particle.
- 8) Matter-waves show similar properties as other waves such as interference and diffraction.

2.4. DAVISON – GERMER'S EXPERIMENT

If the de Broglie hypothesis is valid, then the associated with matter should exhibit interference and diffraction effects. Thus, a beam of electron diffraction by a crystal should show interference phenomena. Davisson and Germer, in 1927 the first experimental that gave convincing proof of the wave nature of matter.

Experimental setup:

Apparatus: 1). Electron gun, 2). Ni Target, 3). Electron detector and 4.) Galvanometer.

The experimental arrangement of Davisson-Germer consisted of an **electron gun**. The electron gun is used to produce a highly accelerated and collimated electron beam by applying high potential. An anode connected to variable voltage source accelerates the electrons. These electrons were scattered by **targeted nickel** crystal. The number of electrons scattered by the crystal in different directions was measured with the help of a **detector** known as *Faraday cylinder*, which can be move freely on a graduated circular scale. A **galvanometer** is connected to the detector, which is used to measure current.

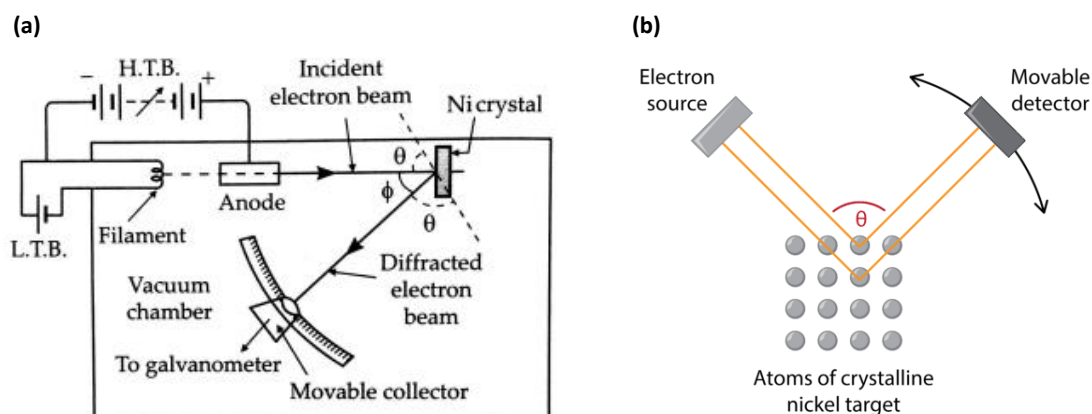


Figure 2.1: (a) Davisson-Germer experiment setup and (b) Lattice spacing of the Ni crystal.

Working:

Electrons produced by the electron gun pass through the accelerating anode, these electrons accelerate and falls normally on the nickel crystal N. The electrons are scattered by the atoms in the crystal as given by the Bragg's diffraction formula for X- rays as:

$$2d \sin\theta = n\lambda$$

For a given accelerating voltage V, the numbers of electrons are noted for different directions by the detector. A graph is plotted between the number of electrons scattered and the angle of scattering.

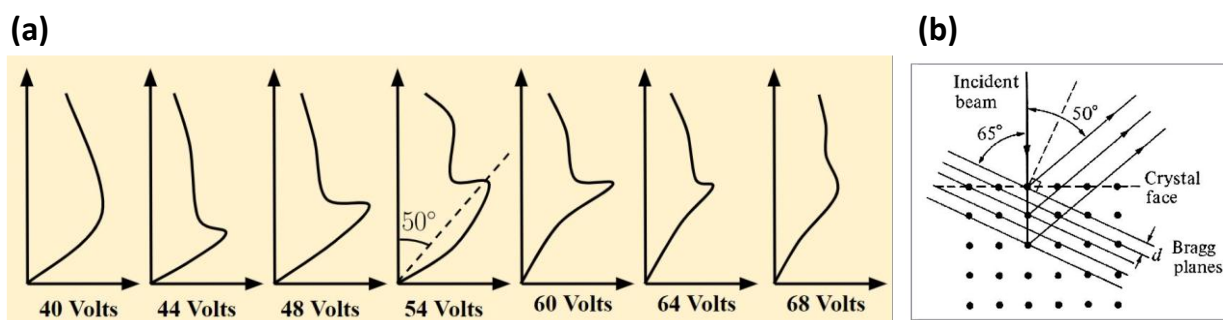


Figure 2.2: (a) Davisson-Germer experiment.

The peak at 50° indicates the most pronounced reflection occurs at a voltage of 54 V. According to the de-Broglie relation, a beam of 54 V had a wavelength of 0.165 nm. This matched the predictions of Bragg's law. For $n = 1$, $\theta = 50^\circ$, and for the spacing of the crystalline planes of nickel ($d = 0.091$ nm) obtained from previous X-ray scattering experiments on crystalline nickel.

$$2(0.091)\sin 65^\circ = (1)\lambda$$

$$\lambda = 2(0.091 \times 10^{-9}). \sin 65$$

$$\lambda = 1.65 \text{ \AA}$$

From de Broglie's equation,

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{12.28}{\sqrt{V}} \text{ \AA}$$

$$\lambda = \frac{12.28}{\sqrt{54}} \text{ \AA}$$

$$\lambda = 1.65 \text{ \AA}$$

Thus, the close agreement between Bragg's law, conforms de Broglie concept of matter waves. Therefore, Davison and Germer experiment calculate the wavelength of electron using both de-Broglie and Bragg's law.

2.5. HEISENBERG UNCERTAINTY PRINCIPLE

In quantum mechanical world, the Heisenberg's uncertainty principle (or simply the uncertainty principle) is one of a variety of mathematical inequalities asserting a fundamental

limit to the precision with which certain pairs of physical properties of a particle, known as complementary variables or canonically conjugate variables such as position x and momentum p , can be known. The concept was first introduced in 1927, by a German physicist Werner Heisenberg.

“The Heisenberg uncertainty principle states that it is impossible to find both the exact position (x) and the momentum ($p = mv$) simultaneously of a subatomic particle”.

More clearly, if one knows the precise momentum (p) of the particle, it is impossible to know the precise position (x), and vice versa. This relationship also applies to energy (E) and time (t), as well as angular momentum (L) and angular displacement (θ). Uncertainties in the products of “conjugate pairs” (momentum/position) and (energy/time) were defined by Heisenberg as having a minimum value corresponding to Planck’s constant divided by 4π . More clearly:

$$\Delta p \cdot \Delta x \geq \frac{h}{4\pi} \text{ (or) } \frac{\hbar}{2\pi}$$

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi} \text{ (or) } \frac{\hbar}{2\pi}$$

$$\Delta L \cdot \Delta \theta \geq \frac{h}{4\pi} \text{ (or) } \frac{\hbar}{2\pi}$$

Where Δ refers to the uncertainty in that variable, $\hbar$ is the reduced Planck's constant, which is obviously equal to the Planck's constant (h) divided by 2π .

2.6. POSTULATES OF QUANTUM MECHANICS:

Quantum mechanics is based on fundamental postulates:

- 1) State postulate: A system is described by a normalized wave function ψ .
- 2) Observable postulate: Every measurable quantity corresponds to a linear Hermitian operator.
- 3) Measurement postulate: Measurement yields one eigenvalue of the operator.
- 4) Probability postulate: Probability equals $|\psi|^2$.
- 5) Time evolution: Schrödinger equation governs evolution.

Postulate 1 (State of a system): The state of a quantum system is completely specified by a A quantum mechanical system is completely described by a normalized wave function $\psi(r, t)$, also called the state function or state vector. A quantum state is completely described by the wave function $\psi(r, t)$. The probability density is $|\psi|^2 = \psi * \psi$. Acceptable wave functions are single-valued, finite, continuous, differentiable and normalizable.

Normalization:
$$\int |\psi|^2 d\tau = 1$$

The normalization condition ensures that the total probability of finding the particle in all space is unity, making the wave function a physically meaningful and complete description of the quantum state. This postulate forms the foundation of quantum mechanics, as all measurable properties of a system are derived from its wave function.

Postulate 2 (Observables and Operators): Every measurable physical quantity (observable) of a quantum mechanical system is represented by a linear Hermitian operator acting on the wave function. The only possible results of measuring an observable are the eigenvalues of its corresponding operator. An observable is any physical quantity that can be measured experimentally.

Examples of Observables: Position (x), Momentum (p), Energy (E), Angular momentum (L), Spin (S), Kinetic energy, Potential energy. Every observable has a corresponding quantum mechanical operator.

Examples:

- Position $\hat{x}=x$
- Momentum $\hat{p}=-i\hbar\partial/\partial x$
- Hamiltonian $\hat{H}=-(\hbar^2/2m) \nabla^2+V$.

Postulate 3 (Measurement): When a measurement of an observable is performed on a quantum system, the only possible results are the eigenvalues of the corresponding Hermitian operator. Immediately after the measurement, the wave function collapses into the eigenstate associated with the measured eigenvalue.

A measurement of observable A gives one of the eigenvalues a satisfying,

$$\hat{A}\psi = a\psi$$

If the large number of measurements of a dynamic variable A are made on a system which is prepared to be in one and the same state before each measurement, the result of the measurement is distributed over an average value known as expectation value $\langle A \rangle$.

The expectation value is given by,

$$\langle A \rangle = \frac{\int \Psi^* \hat{A} \psi dt}{\int \psi \Psi^* dt}$$

Postulate 4 (Superposition): If $\psi_1, \psi_2, \psi_3 \dots$ are possible quantum states of a system, then any normalized linear combination of these states is also a valid quantum state of the system.

If ψ_1 and ψ_2 are valid states, $c_1, c_2, c_3 \dots$ are complex constants called probability amplitudes. then $c_1\psi_1+c_2\psi_2$ is also a valid state. This principle is known as the Superposition Principle, and it is one of the most fundamental concepts in quantum mechanics.

Postulate 5 (Time Evolution Postulate Schrodinger Equation): The time evolution of an isolated quantum system is governed by the Schrödinger equation. If the system is described by a wave function $\psi(r,t)$ at time t its future state is uniquely determined by the time-dependent Schrödinger equation. The wave function evolves according to the time-dependent Schrödinger equation:

$$i\hbar \partial\psi(x,t)/\partial t = \hat{H}\psi(x,t)$$

2.7. OPERATORS IN QUANTUM MECHANICS

An operator is a mathematical rule acting on a wave function. An operator acts on a wave function to produce another function.

Common operators:

Dynamical variable	Quantum Mechanical Operator	Result (multiplies by ψ)
Position	$\hat{x}$	x
Momentum	$\hat{p} = -i\hbar \partial/\partial x$	p
Kinetic energy	$\hat{T} = (\hbar^2/2m) \nabla^2$	T
Potential energy	$\hat{V} = V(x)$	V
Hamiltonian	$\hat{H} = \hat{T} + \hat{V} = (\hbar^2/2m) \nabla^2 + V$	H

Types of Operators:

- 1. Multiplication Operator:** Some operators simply multiply the wave function. Example: Position operator, $\hat{x} = x$ thus, $\hat{x}\Psi = x\Psi$.

Differential Operator: Many quantum mechanical operators involve differentiation.

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}, \text{ when it acts on the wave function, } \hat{p}_x\Psi = -i\hbar \frac{\partial\Psi}{\partial x}$$

- 2. Linear Operators:** Quantum mechanical operators are linear. An operator satisfies distributive law.

$$\hat{A}(a\Psi_1 + b\Psi_2) = a\hat{A}\Psi_1 + b\hat{A}\Psi_2$$

where a, b are constants and Ψ_1, Ψ_2 are wave functions.

- Commutator: $[A, B] = AB - BA$. If $[A, B] = 0$, the operators commute.
- Multiplication of a linear operator $\hat{A}$ by constant gives a linear operator ($a\hat{A}$)

$$a\hat{A}\psi = (a\hat{A})\psi$$

- 3. Hermitian Operators:** an operator 'A' is said to be Hermitian if it is equal to its adjoint $A^\dagger$

$$A^\dagger = A$$

Hermitian operators possess important physical properties:

- Their eigenvalues are always real, ensuring measured quantities are real numbers.
- Eigen functions corresponding to different eigenvalues are orthogonal.
- They form a complete set of basis functions, allowing any quantum state to be expressed as a combination of Eigen functions.

Properties:

- Linear: $\hat{A}(c_1\Psi_1 + c_2\Psi_2) = c_1\hat{A}\Psi_1 + c_2\hat{A}\Psi_2$.
- Hermitian operators have real eigenvalues and orthogonal eigenfunctions.
- Commutator: $[A, B] = AB - BA$. If $[A, B] = 0$, the operators commute. The important relation $[x, p] = i\hbar$ leads to the Heisenberg uncertainty principle.
- Example: If $D = \frac{d}{dx}$ and $\psi = x^2$, then $D\psi = 2x$.

2.8. EIGENVALUES AND EIGEN FUNCTIONS

- One of the most fundamental concepts in quantum mechanics is the idea of eigenvalues and Eigen functions. Every measurable physical quantity (such as position, momentum, energy, and angular momentum) is represented by a Hermitian operator. When an operator acts on

a special wave function called an Eigen function, the result is the same function multiplied by a constant known as the eigenvalue.

- If $\hat{A}\psi = a\psi$, then $\hat{A}$ is an Operator, ψ is an eigenfunction and a is its eigenvalue. This equation is called Eigen value equation.
- The eigenvalue is the measurable value of the observable, while the eigenfunction represents the corresponding quantum state.

A physically acceptable Eigen function must satisfy the following conditions:

- An Eigen function is a non-zero function that, when acted upon by a given linear operator, produce a scalar multiple of itself. This scalar multiple is known as an Eigen value.
- They should be single valued, continuous, finite, square integrable, differentiable and must satisfy appropriate boundary conditions.
- Eigen functions of Hermitian operators are orthogonal.
- Eigen functions have several important properties such as Linearity, Orthogonality, Completeness, Normalization.
- Degenerate eigenvalues correspond to multiple Eigen functions.

Examples:

- $d/dx(e^{2x}) = 2e^{2x}$. Therefore, the eigenvalue is 2.
- Momentum operator: $\hat{p}e^{ikx} = \hbar k e^{ikx}$. Hence $\hbar k$ is the eigenvalue.

Applications of Eigenvalue problems in different areas of physics:

- ⇒ Hydrogen atom – Energy eigenvalues determine the atomic energy levels.
- ⇒ Particle in a box – Gives quantized energy levels.
- ⇒ Harmonic oscillator – Produces equally spaced energy eigenvalues.
- ⇒ Angular momentum – Eigenvalues determine allowed angular momentum states.
- ⇒ Quantum computing – Measurement outcomes correspond to eigenvalues of quantum operators.

2.9. EXPECTATION VALUE

The expectation value is one of the most important concepts in quantum mechanics. It represents the average value of a physical quantity (observable) obtained from a large number of measurements performed on an ensemble of identically prepared quantum systems.

Unlike classical mechanics, where a physical quantity has a definite value at any instant, quantum mechanics predicts only the probability of obtaining different measurement outcomes. The expectation value provides the statistically averaged result of these measurements.

The average value of x is

$$\langle x \rangle = \frac{x_1 + x_2 + x_3 + \cdots + x_i + \cdots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

Thus, if a variable takes n discrete values, we add all these values of the variable and divide the sum by the total number of values to obtain the average value of the variable.

For example, if the position of an electron is measured repeatedly on many identical systems, the results may vary due to the probabilistic nature of quantum mechanics. However,

the average of all these measurements approaches the expectation value of the position.

The expectation value is the average value obtained from many measurements on identically prepared systems.

Expectation value in Operator notation: $\langle A \rangle = \frac{\int \Psi^* \hat{A} \Psi dt}{\int \Psi^* \Psi dt}$

General expression: $\langle A \rangle = \int \Psi^*(r,t) \hat{A} \Psi(r,t) d\tau$.

Special cases:

- ✓ $\langle x \rangle = \int \Psi^*(x) x \Psi(x) d\tau$ (Expectation value of position).
- ✓ $\langle p \rangle = \int \Psi^*(-i\hbar \partial/\partial x) \Psi d\tau$ (Expectation value of Momentum which gives the average momentum of a particle).
- ✓ $\langle E \rangle = \int \Psi^* \hat{H} \Psi d\tau$ (Expectation value of Energy, which gives average energy of the system).

Example:

For a particle in a one-dimensional box, $\Psi = \sqrt{2/L} \sin(n\pi x/L)$, the expectation value of position is $\langle x \rangle = L/2$.

2.10. SCHRÖDINGER'S TIME INDEPENDENT WAVE EQUATION

The Schrödinger equation is a linear partial differential equation that describes the wave function or state function of a quantum-mechanical system.

There are two types of Schrodinger's wave equations, one is Schrodinger's time independent wave equation and another one is Schrodinger's time dependent wave equation.

Schrodinger's time independent wave equation:

Consider a particle of mass “m” moving with velocity “v” in space. Suppose a system of stationary waves is associated with the particles at any point in space in the neighborhood of particle. The wave associated with this particle having frequency ‘v’ is given by

$$\psi = \psi_0 \sin 2\pi\vartheta t \text{ ----- (1)}$$

⇒ Differentiating with respect to the time on both sides,

$$\frac{\partial \psi}{\partial t} = \frac{\partial}{\partial t} (\psi_0 \sin 2\pi\vartheta t)$$

$$\frac{\partial \psi}{\partial t} = \psi_0 2\pi\vartheta \cos 2\pi\vartheta t$$

⇒ Again, differentiating with respect to the time,

$$\begin{aligned} \frac{\partial^2 \psi}{\partial t^2} &= \psi_0 2\pi\vartheta (-\sin 2\pi\vartheta t) 2\pi\vartheta \\ &= -4\pi^2\vartheta^2 \psi_0 (\sin 2\pi\vartheta t) \end{aligned}$$

$$\frac{\partial^2 \psi}{\partial t^2} = -4\pi^2\vartheta^2 \psi \text{ ----- (2)} \quad (\text{from eqn. 1, } \psi = \psi_0 \sin 2\pi\vartheta t)$$

The general wave equation in three dimensions is,

$$\frac{\partial^2 \psi}{\partial t^2} = v^2 \nabla^2 \psi \text{ ----- (3)}$$

By comparing the eqn (2) and (3), we get

$$v^2 \nabla^2 \psi = -4\pi^2 \vartheta^2 \psi$$

$$v^2 \nabla^2 \psi = -4\pi^2 \frac{v^2}{\lambda^2} \psi \quad (\text{frequency } \vartheta = \frac{v}{\lambda})$$

$$\nabla^2 \psi = \frac{-4\pi^2}{\lambda^2} \psi$$

According to the de-broglie hypothesis, $\lambda^2 = \frac{h^2}{m^2 v^2}$

$$\therefore \nabla^2 \psi = \frac{-4\pi^2}{\left(\frac{h^2}{m^2 v^2}\right)} \psi$$

$$\nabla^2 \psi = \frac{-4\pi^2 m^2 v^2}{h^2} \psi \text{ ----- (4)}$$

We know that, total energy (E) = KE + PE

$$E = \frac{1}{2} m v^2 + V$$

$$E - V = \frac{1}{2} m v^2$$

$$m^2 v^2 = 2m (E - V) \text{ ----- (5)}$$

By substituting eqn (5) in (4), we get,

$$\nabla^2 \psi = \frac{-4\pi^2 2m (E - V)}{h^2} \psi$$

$$\nabla^2 \psi = \frac{-8\pi^2 m (E - V)}{h^2} \psi$$

$$\nabla^2 \psi + \frac{8\pi^2 m (E - V)}{h^2} \psi = 0 \quad (\text{or}) \quad \nabla^2 \psi + \frac{2m (E - V)}{\hbar^2} \psi = 0 \quad (\text{where } \hbar = \frac{h}{2\pi})$$

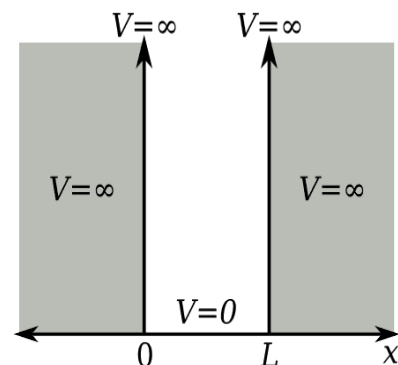
This is the Schrodinger time independent wave equation.

2.11. PARTICLE IN A 1-D BOX

Consider one-dimensional closed box of width L. A particle of mass 'm' is moving in a one-dimensional region along X-axis specified by the limits $x = 0$ and $x = L$ as shown in figure.

Boundary conditions: -

- The potential energy (V) of the particle inside the box is zero. $V(x) = 0, 0 \leq x \leq L$
- The potential energy (V) of the particle at the walls is infinity. $V(x) = \infty, 0 \geq x \geq L$
- The Schrödinger time independent wave equation for the particle is given by



$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m(E-V)}{h^2}\psi(x) = 0 \text{ ----- (1)}$$

But $V = 0$ between the walls

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m(E)}{h^2}\psi(x) = 0 \text{ ----- (2)}$$

let, $k^2 = \frac{8\pi^2m(E)}{h^2}$, Propagation constant

$$\frac{d^2\psi}{dx^2} + k^2\psi(x) = 0 \text{ ----- (3)}$$

The general solution of eq (3) is,

$$\psi(x) = A \sin kx + B \cos Kx \text{ ----- (4)}$$

where A and B are arbitrary conditions and these will be determined by the boundary conditions.

Boundary Conditions:

When $\psi(x) = 0$, $x = 1$.

$$0 = A \sin k(0) + B \cos K(0)$$

$$0 = A \sin(0) + B \cos(0)$$

$$0 = A(0) + B(1)$$

$$B = 0$$

Hence, the wave equation (4) becomes

$$\psi(x) = A \sin kx$$

When, $\psi(x) = 0$, $x = L$.

$$0 = A \sin k(L) + 0 \cos K(L)$$

$$0 = A \sin kL$$

$$0 = \sin kL$$

$$\sin n\pi = A \sin KL$$

$$KL = n\pi$$

$$K = \frac{n\pi}{L}, \text{ where } n = 1, 2, \dots$$

Hence, the wave equation becomes

$$\psi(x) = A \sin\left(\frac{n\pi}{L}x\right)$$

To find the 'A' value, by applying the normalization conditions,

$$\int_0^L |\Psi(x)|^2 dx = 1$$

$$\int_0^L \left(A \sin\left(\frac{n\pi x}{L}\right)\right)^2 dx = 1$$

$$A^2 \int_0^L \left(\sin\left(\frac{n\pi x}{L}\right)\right)^2 dx = 1$$

$$\frac{A^2}{2} \int_0^L (1 - \cos\left(\frac{2n\pi x}{L}\right)) dx = 1 \quad (\cos(2x) = 1 - \sin^2 x)$$

By integrating and applying the appropriate limits, finally we obtain the value of 'A' as

$$A^2 \left[\frac{L}{2}\right] = 1$$

$$\therefore A = \sqrt{\frac{2}{L}}$$

(i). Wave function of electron/particle is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right) \quad n = 1, 2, 3, \dots$$

(ii). Probability density function (P)

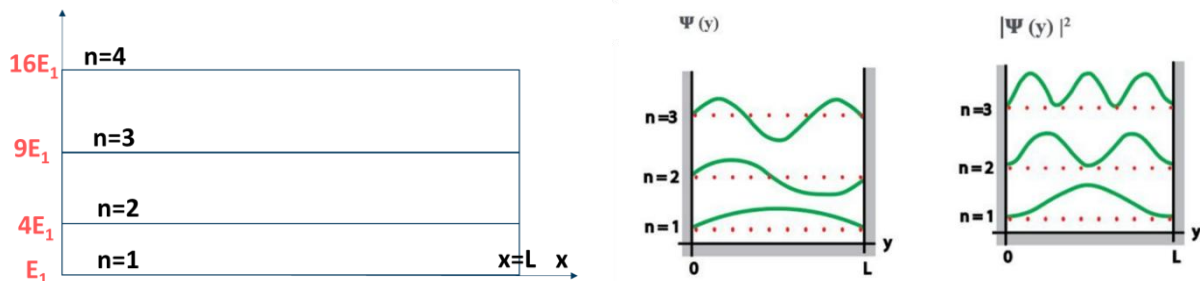
$$P = |\psi(x)|^2 = \frac{2}{L} \sin^2\left(\frac{n\pi}{L}x\right)$$

(iv). Energy function of the particle

$$E_n = \left(\frac{n^2 h^2}{8mL^2} \right)$$

Energy levels of particle (Electron):

$$E_1 = \left(\frac{h^2}{8mL^2} \right) \quad E_2 = \left(\frac{4h^2}{8mL^2} \right) \quad E_3 = \left(\frac{9h^2}{8mL^2} \right)$$



2.12. BLOCH'S THEOREM

In metals positive ions are regularly arranged and maintain periodicity. When an electron is moving inside the metal, the potential is minimum at the positive ions & it is maximum between two ions. Hence the potential of the electron varies periodically with a period equal to lattice periodicity. This type of periodical potential extends in all directions.

To study motion of electron & determine its energy values we have to apply Schrodinger time independent wave equation. The Schrodinger equation for an electron moving in one-dimensional periodic potential is

$$\frac{d^2\psi}{dx^2} + \frac{2m(E - V)}{\hbar^2} \psi = 0 \quad \text{where } \hbar = \frac{h}{4\pi}$$

V is potential at a distance X from the origin which is assumed to vary periodically in space with an interval equal to lattice constant a.

$$V(x) = V(x+a)$$

The solution of this equation is $\psi = e^{+iKX} \cdot U(X)$ ----- (1)

Where $U_K(X) = U_K(X + a)$

i.e. the solution are plane waves modulated by the function $U_K(X)$ which has the same periodicity as the lattice. This theorem is known as Bloch theorem. Functions of type (eq (1)) are called Bloch functions.

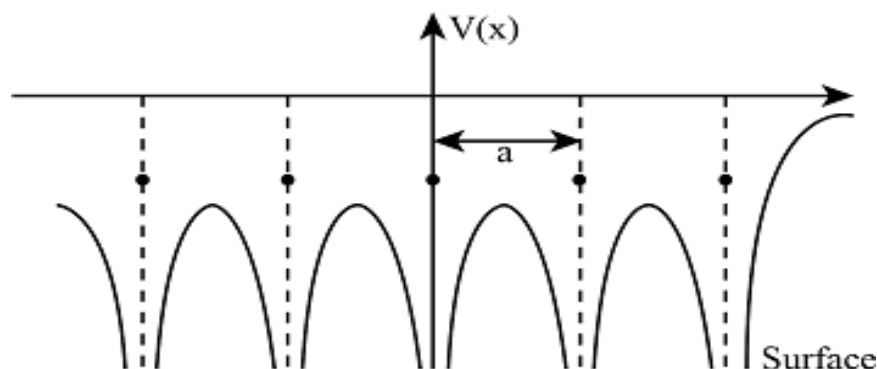


Figure 2.4: Bloch Theorem.

2.13. KRONIG-PENNEY MODEL

According to this model it is assumed that when electrons move in one-dimensional perfect crystal lattice, potential energy of an electron is represented in the form of rectangular walls as shown in fig.

In regions where $0 < X < a$, the potential energy is assumed to be V_0 and in the region $-b < X < 0$ the potential is zero.

The Schrodinger equations are

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V_0) \psi = 0 \quad \text{for } 0 < X < a \text{ ----- (1)}$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E) \psi = 0 \quad \text{for } -b < X < 0 \text{ ----- (2)}$$

Let energy of the electron is less than V_0 , then put

$$\alpha^2 = \frac{2m}{\hbar^2} (E - V_0) \text{ and } \beta^2 = \frac{2m}{\hbar^2} (E) \text{ ----- (3)}$$

Where α and β are real numbers.

Now, Eqn. (1) and (2) becomes,

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0$$

$$\frac{d^2\psi}{dx^2} + \beta^2\psi = 0$$

From Bloch theorem, the solution of wave equation will be of form of a plane wave e^{iKX} modulated with a periodic function $U_K(X)$

$$\psi = e^{iKX} \cdot U_K(X) \text{ ----- (4)}$$

where, $U_K(X)$ is the periodic function in X with period $a + b$. ($k = 2\pi/\lambda$; wave vector or propagation vector along x -axis)

To simplify this equation, Kronig and Penney suggested delta function such that $V_0 \rightarrow \infty$ and $a \rightarrow \infty$ but the product $(V_0 a)$ remains finite & is known as barrier strength.

Under these circumstances $\sinh\alpha a \rightarrow a$ and $\cosh\alpha a \rightarrow 1$ as $a \rightarrow 0$.

Now, we define the quantity, $P = \frac{mV_0ab}{\hbar^2}$, known as scattering power which is a measure of strength with which electrons are attracted to the ions in crystal.

$$P \frac{\sin \beta b}{\beta b} + \cos \beta b = \cos kb \text{ ----- (5)}$$

The R.H.S of the above equation can take values between -1 to +1 only. Therefore, the condition can be satisfied only those values of βb for which L.H.S lies between +1 & -1. i.e., some ranges of energies are allowed & some other ranges are not allowed. Hence, isolated particle energy levels change to energy bands.

If a graph, $P \frac{\sin \beta b}{\beta b} + \cos \beta b$ is plotted against βb curve is obtained as follows.

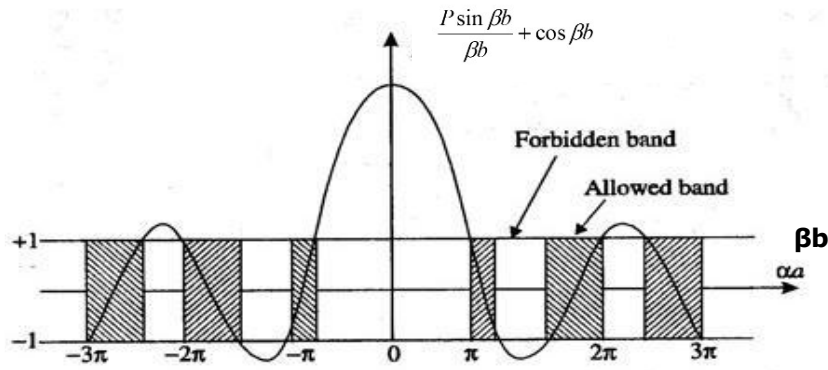


Fig. 2.5: Energy spectrum.

The following important conclusions are drawn from Fig. 2.5.

- 1) The energy spectrum of electrons consists of allowed energy bands separated by forbidden regions.
- 2) The motion of electrons confined only in allowed bands.
- 3) As βb increases width of allowed energy bands is increased and forbidden regions become narrow.
- 4) The width of allowed band decreases with increasing value of P i.e allowed band becomes narrower & forbidden gap becomes wider.
- 5) As, $P \rightarrow \infty$, eq (5) has solution only if $\sin \beta b = 0$.

$$b = \pm n\pi, \Rightarrow \beta = \pm \frac{n\pi}{b}, \Rightarrow \beta^2 = \frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{b^2}$$

$$E = \frac{n^2\pi^2\hbar^2}{2mb^2}$$

$$E = \frac{n^2h^2}{8mb^2}$$

This equation shows that allowed band reduces to discrete energy levels similar to particle present in a potential box. This leads to quantum free electron theory.

- 6) As, $P \rightarrow 0$, eq (5)

$$\cos \beta b = \cos kb$$

$$\beta b = kb$$

$$\beta^2 = \frac{2mE}{\hbar^2} = k^2$$

$$E = \frac{k^2\hbar^2}{2m} = \frac{k^2h^2}{8m}$$

As, $p = k\hbar$, momentum of the electrons, the kinetic energy,

$$E = \frac{p^2}{2m}$$

This indicates that particle is completely free & no energy level exist. This leads to classical free electron theory. Hence by varying P from 0 to ∞ we find that completely free electrons become completely bounded particle.

2.14. E-K DIAGRAM

Using the equation, $P \frac{\sin \beta b}{\beta b} + \cos \beta b = \cos kb$, it is possible to plot a curve showing the energy E as a function of K as shown in Fig. 2.6. It is clear from fig that energy of electron is continuously increasing from $K = 0$ to $\frac{\pi}{a}$ the right-hand side of above equation becomes +1 or -1 for values of $k = \pm \frac{n\pi}{a}$ and hence discontinuity appears in E-K graph at $k = \pm \frac{n\pi}{a}$ from Fig.2.7 it is seen that energy spectrum of electron is consisting allowed regions and forbidden regions. As described above, the Kronig-Penney model describes the electron behavior in a periodic potential like a crystal lattice which can be directly related to some specific regions in the E-K diagram called the Brillouin zones. Brillouin zones are specific regions in the E-K diagram that define the allowed energy bands for electrons moving within a crystal lattice.

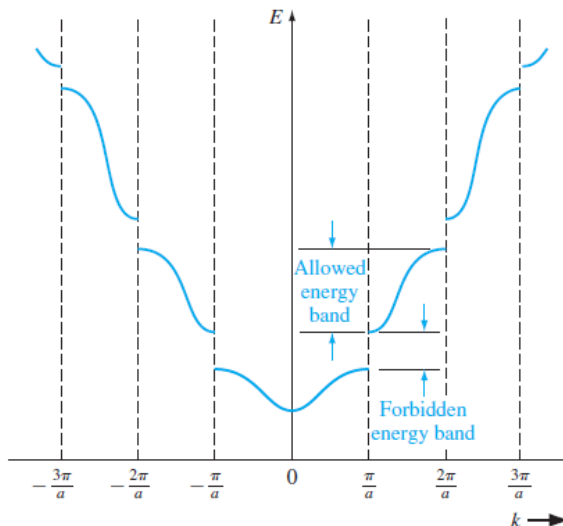


Figure 2.6: Brillouin Zones or E-K diagram.

The Kronig-Penney model helps us to understand how the periodic potential creates these band gaps and how electrons are confined to specific energy ranges within these zones. The allowed region which extends from $-\frac{\pi}{a}$ to $+\frac{\pi}{a}$ is known as the first Brillouin zone. After a discontinuity in energy band, called the **forbidden gap**, another allowed region extends from $-\frac{2\pi}{a}$ to $+\frac{2\pi}{a}$ and $\frac{\pi}{a}$ to $\frac{2\pi}{a}$ is called second Brillouin Zone.

2.15. ORIGIN OF BANDGAP AND FORMATION OF ENERGY BANDS

When isolated atoms come together to form a solid, their discrete atomic energy levels interact and split into a very large number of closely spaced levels, which effectively form continuous ranges of allowed energies called **energy bands**. This band formation underpins the electrical, optical, and thermal properties of crystalline solids.

From Atomic Levels to Bands:

- Isolated atom: Electrons occupy sharp, discrete energy levels (e.g., 1s, 2s, 2p, ...).
- Two atoms brought close: Their outer electron wavefunctions overlap. Because of the Pauli exclusion principle (no two electrons in an interacting system can have the same quantum state), each original level splits into two slightly different levels.
- N atoms in a crystal: For N identical atoms, each atomic level splits into N closely spaced levels. As N becomes macroscopic ($\sim 10^{23}$), these levels are so dense that they form a quasi-continuous energy band.
- Interatomic spacing effect: As the lattice spacing decreases toward the equilibrium value, different bands (e.g., derived from 2s and 2p orbitals) can broaden, overlap, or

separate, and an energy gap may open between some bands.

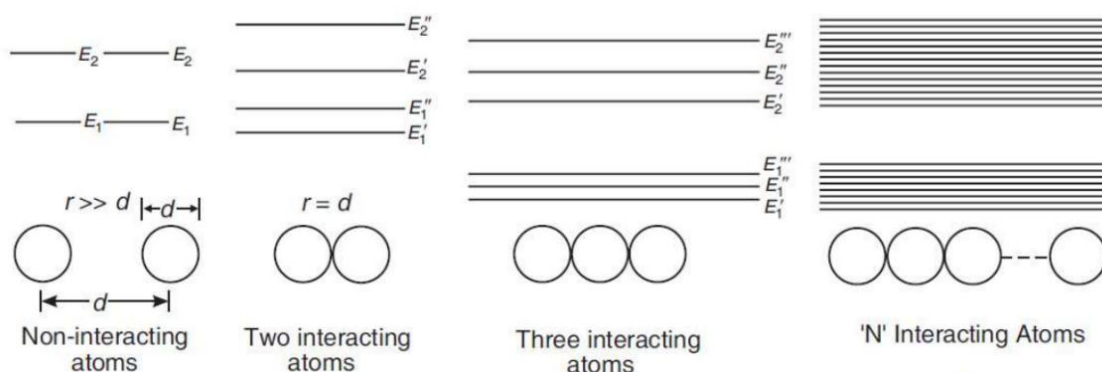


Figure 2.7: Formation of energy bands.

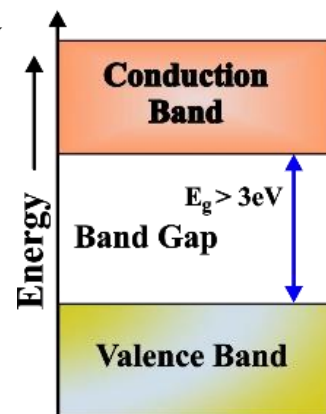
- The width of energy band depends on the degree of over lapping of electrons of adjacent atoms & is largest for outermost electrons.
 - The band corresponding to outer most orbit is called *conduction band*, which is empty or partially filled by the electrons. Electrons in this band can move freely under an electric field and contribute to conduction.
 - The highest energy band that is *valence band*, which is completely (or nearly completely) filled with electrons at 0 K. It is formed mainly from the outer (valence) atomic orbitals.
 - The gap between these two allowed bands is called *forbidden energy gap* or *band gap*, where no allowed electronic states exist.
- $$E_g = E_C - E_V$$
- The size of E_g determines whether a material behaves as a conductor, semiconductor, or insulator.

2.16. CLASSIFICATION OF SOLIDS

The electrical properties of solid depends upon its energy band structure & the way in which energy bands are occupied by the electrons. Basing on the width of forbidden band, solids are classified into insulators, Semi-Conductors and conductors. They are described as follows.

1. **Insulators:** In insulators, the conduction band is completely empty & valence band is completely filled. The forbidden energy band is very wide ($E_g > 3 \text{ eV}$) as shown in Fig. 2.11. Due to this fact electrons cannot jump from valance band to conduction band under normal conditions. In insulators the valence electrons are bound very tightly to their parent atoms. Increase in temperature enables some electrons to go to the conduction band so insulators become conductors at very high temperatures.

Eg: Glass, Fiber, Plastic, Rubber, Wood, Paper, Diamond etc.,



2. **Semi-Conductors:** In semiconductors forbidden band is very small ($E_g < 3 \text{ eV}$) as shown in Fig. 2.11 Germanium and silicon are best examples. In semi-conductor the electrical properties lie between insulators and conductors. At OK there are no electrons in conduction band and valence band is completely filled & semiconductor behaves as an insulator.

When a small amount of energy supplied the electrons can easily jump from valence band to conduction band. These electrons leave holes in valence band which behaves as a positive charge & also participate in the conduction. As temperature increases conductivity increases & it behaves as a conductor. In semi-conductors, the conductivities are of order of $10^2 \Omega$. Semiconductors are having negative temperature coefficient of resistance.

Eg: Germanium (Ge), Silicon (Si) etc.

3. **Conductors:** In case of conductors, there is no forbidden band ($E_g = 0$) and both valence and conduction bands overlap each other as shown in Fig. 2.11. In conduction band, plenty of free electrons are available at room temperature for the process of electric conduction. The electrons from valence band freely enter into conduction band. The resistivity increases with increase of temperature as mobility of electrons reduced due to collisions. Hence these are having positive temperature coefficient of resistance. The total current in conductors is simply a flow of electrons.

Eg: Metals; Silver (Ag), Copper (Cu), Iron (Fe) etc.

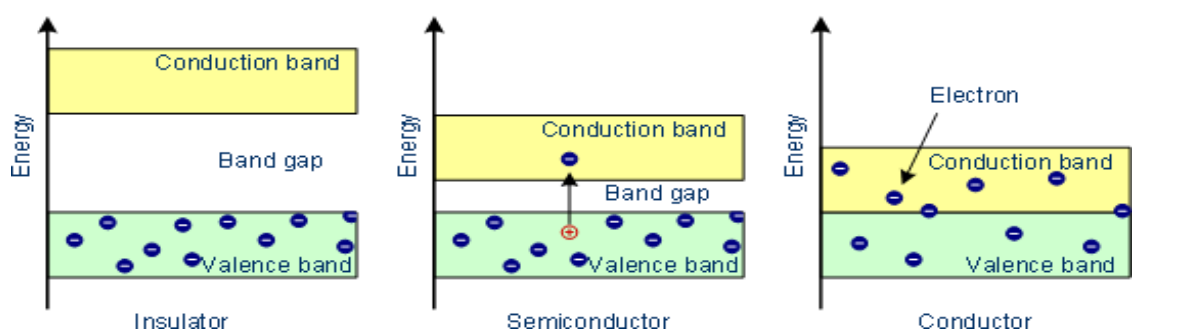


Figure 2.8: Formation of energy bands in solids.

2.17. CONCEPT OF DISCRETE ENERGY LEVELS AND QUANTUM CONFINEMENT IN NANOMATERIALS

Definition: In quantum mechanics, electrons and other bound particles are restricted to specific, allowed values of energy or in other words, in quantum mechanics, a microscopic particle confined to a finite region cannot possess arbitrary values of energy. It can exist only in certain permitted energy states called discrete or quantized energy levels. These energy levels arise due to the wave nature of matter and the boundary conditions imposed on the particle's wave function.

Think of a staircase in which only Step 1, Step 2, or Step 3 etc. are possible, but they cannot exist in between the steps (e.g., there is no "Step 1.5").

Why these states exist?

These states exist due to wave nature of matter. At the atomic scale, electrons behave like waves (wave-particle duality). An electron bound to a nucleus forms a standing wave around that nucleus. For the wave to be stable and not cancel itself out destructively, its wavelength must fit neatly into the circular orbit.

Only specific circumferences allow whole numbers of wavelengths to fit. Because only specific wavelengths are allowed, only specific momentum values are allowed. And because energy depends on momentum, only specific energy levels are allowed.

Example: Particle in 1D Box

The Particle in a 1D Box (an electron trapped inside a rigid container of length L). By solving the Schrödinger Equation and applying boundary conditions (meaning the wave probability goes to zero at the walls), we get the allowed energy values:

$$E_n = \frac{n^2 h^2}{8mL^2}$$

where: $n = 1, 2, 3, \dots$ principal quantum number or our "step" number

h is Planck's constant

m is the mass of the particle

L is the length of the box

Applications

Discrete energy levels explain many phenomena that classical physics cannot explain properly,

- 1. Atomic spectra:** Atoms emit or absorb photons of specific frequencies because electrons can occupy only specific energy levels.
- 2. Lasers:** Laser operation depends on transitions between discrete energy states.
- 3. Semiconductors:** The electronic energy structure of solids is fundamental to semiconductor devices.
- 4. LEDs:** Light emission occurs when electrons and holes undergo energy transitions.
- 5. Quantum computing:** Quantum systems use discrete quantum states as the basis for information processing.
- 6. Nanotechnology:** When structures become extremely small, quantum confinement produces discrete energy states.

Quantum Confinement in Nanomaterials:

Quantum confinement occurs when the physical dimensions of a material become smaller than (or comparable to) the de Broglie wavelength of its electrons (usually a few nanometers). It is the spatial confinement of electron-hole pairs (excitons) in one or more

dimensions within a material and also its electronic energy levels are discrete.

When the movement of electrons in one, two, or three spatial dimensions is restricted, they can no longer act like they are in a vast, open space. Instead, they are squeezed into a tiny "potential well."

Bulk Material: Electrons can move freely in all three dimensions. Energy levels are closely spaced, forming continuous energy bands.

Nanomaterial: Electrons motion is restricted. The continuous energy bands begin to break back down into discrete, well-separated energy levels, much like the individual energy levels of an isolated atom.

For example: Particle in 1-D box.

$$E_n = \frac{n^2 h^2}{8mL^2}$$

In a bulk material the Length of box L will be large. This makes the spacing between energy levels so small that they blur together into a continuous band. While in case of nanomaterials L is extremely small which increases the spacing between energy levels.

Types of Quantum Confinement

Quantum confinement is classified according to the number of dimensions in which the electron is confined.

1. Bulk Materials (Zero Dimensions of Confinement/3D Systems)

Confinement: None. The electrons are not confined in any dimension. Electrons can move freely in all three spatial dimensions.

Electronic Property: Continuous energy bands with a standard, fixed band gap.

2. Quantum Wells (One Dimension of Confinement / 2D Systems)

Confinement: Confined in 1 direction (perpendicular to the layer), but free to move in 2 dimensions along the plane.

Electronic Property: Energy levels split into sub-bands in the confinement direction.

Applications: Semiconductor lasers, high-electron-mobility transistors (HEMTs), and photodetectors.

3. Quantum Wires (Two Dimensions of Confinement / 1D Systems)

Confinement: Confined in 2 directions, with freedom of movement restricted to 1 direction (along the length of the wire).

Electronic Property: Sharp spikes appear in the density of states, restricting electron scattering and making them exceptional for carrying current efficiently.

Applications: Carbon nanotubes, silicon nanowires, and nanoscale interconnects for next-generation microelectronics.

4. Quantum Dots (Three Dimensions of Confinement / 0D Systems)

Confinement: Confined in all 3 dimensions.

Electronic Property: No free movement remains. Electrons occupy fully discrete, atom-like energy levels. As we discussed earlier, tuning the size directly alters the band gap and emission color.

Applications: QLED displays, biological fluorescence labeling, and qubits for quantum computing.

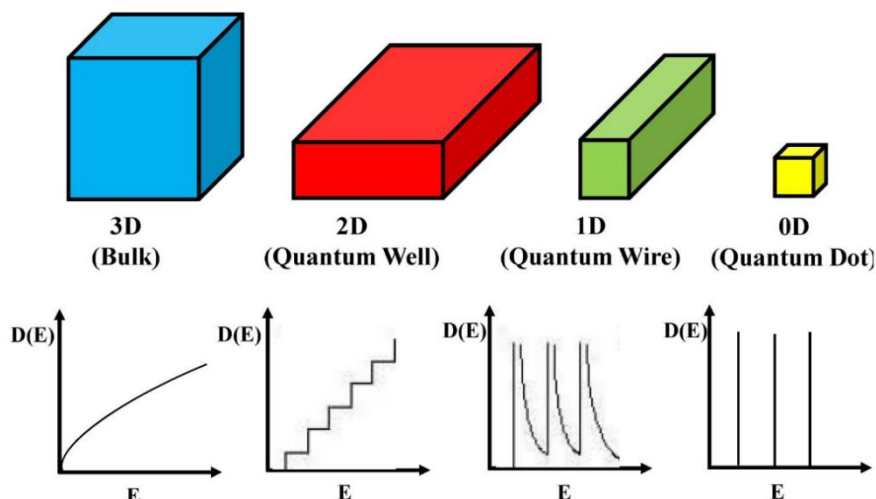


Figure 2.9: Types of Quantum Confinement.

Problems

Problem1: An electron moves with the velocity 1.0×10^6 m/s. Find its de Broglie wavelength.

Solution: $\lambda = h/mv = 6.626 \times 10^{-34} / (9.11 \times 10^{-31} \times 10^6) = 7.27 \times 10^{-10}$ m.

Answer: 0.727 nm.

Problem2: Find the de Broglie wavelength of an electron accelerated through 100 V.

Solution: $\lambda = h/\sqrt{2meV} = 1.227/\sqrt{V}$ Å ≈ 0.123 nm.

Answer: ≈ 1.23 Å.

Problem3: A proton moves with the velocity 2.0×10^5 m/s. Find its wavelength λ .

Solution: $\lambda = h/mv = 6.626 \times 10^{-34} / (1.67 \times 10^{-27} \times 2 \times 10^5) = 1.98 \times 10^{-12}$ m.

Answer: 1.98 pm.

Problem4: Uncertainty in position $\Delta x = 2.0 \times 10^{-10}$ m. Find minimum Δp .

Solution: $\Delta p \geq \hbar/(2\Delta x) = 1.055 \times 10^{-34} / (4 \times 10^{-10}) = 2.64 \times 10^{-25}$ kg m/s.

Answer: 2.64×10^{-25} kg m/s.

Problem5: If uncertainty in momentum is $\Delta p = 1.0 \times 10^{-24}$ kg m/s, find minimum uncertainty in Δx .

Solution: $\Delta x \geq \hbar/(2\Delta p) = 1.055 \times 10^{-34} / (2 \times 10^{-24}) = 5.28 \times 10^{-11}$ m.

Answer: 0.0528 nm.

Problem6: An electron is confined to a box of length $L = 1.0$ nm. Find the energy of 1st, 2nd and 3rd excited energy levels?

Solution: $E_1 = h^2/(8mL^2) = (6.626 \times 10^{-34})^2 / [8(9.11 \times 10^{-31})(10^{-9})^2] = 6.02 \times 10^{-20}$ J. In eV = 0.376 eV.

Thus $E_1 = 0.376$ eV, $E_2 = 1.504$ eV, $E_3 = 3.384$ eV.

Answer: 0.376, 1.50 and 3.38 eV.

Problem8: For a particle in a 1Dimensional box, find the ratio E_3/E_1 .

Solution: $E_n \propto n^2$. Therefore $E_3/E_1 = 3^2/1^2 = 9$.

Answer: 9.

Problem9: An electron in a 1.0 nm box falls from $n=3$ to $n=1$. Find emitted photon energy.

Solution: $\Delta E = (9-1)E_1 = 8(0.376) = 3.008 \text{ eV}$. In joules $= 3.008 \times 1.602 \times 10^{-19} = 4.82 \times 10^{-19} \text{ J}$.

Answer: $\approx 3.01 \text{ eV}$.

Problem10: For the transition $n=4$ to $n=2$ in a 1.0 nm box, find photon wavelength.

Solution: $\Delta E = (16-4)E_1 = 12(0.376) = 4.51 \text{ eV}$. $\lambda = hc/\Delta E = 1240/4.51 \text{ nm} \approx 275 \text{ nm}$.

Answer: $\approx 275 \text{ nm}$.

Problem11: An electron has $\lambda = 0.50 \text{ nm}$. Find momentum and kinetic energy.

Solution: $p = h/\lambda = 6.626 \times 10^{-34} / (5 \times 10^{-10}) = 1.325 \times 10^{-24} \text{ kg m/s}$. $KE = p^2/(2m) = 9.64 \times 10^{-19} \text{ J} \approx 6.02 \text{ eV}$.

Answer: $p \approx 1.33 \times 10^{-24} \text{ kg m/s}$, $KE \approx 6.02 \text{ eV}$.

Problem12: For an electron in a 2.0 nm box, find E_1 .

Solution: Since $E_1 \propto 1/L^2$ and E_1 at 1 nm is 0.376 eV, at 2 nm $E_1 = 0.376/4 = 0.094 \text{ eV}$.

Answer: 0.094 eV.

Problem13: For the ground state in a 1D box from 0 to L , calculate $\langle x \rangle$.

Solution: $\psi_1 = \sqrt{(2/L)} \sin(\pi x/L)$. $\langle x \rangle = \int_0^L \psi^* x \psi dx = L/2$.

Answer: $\langle x \rangle = L/2$.

Problem14: An electron is confined to $L = 0.5 \text{ nm}$. Find E_1 and compare it with the 1.0 nm case.

Solution: $E \propto 1/L^2$. Thus $E_1(0.5 \text{ nm}) = 4E_1(1 \text{ nm}) = 4(0.376) = 1.504 \text{ eV}$.

Answer: 1.50 eV; four times larger.

Problem15: For an electron in a 0.5 nm box, calculate the photon wavelength for transition $n=3 \rightarrow n=1$.

Solution: $E_1 = 1.504 \text{ eV}$. $\Delta E = 8E_1 = 12.03 \text{ eV}$. $\lambda = 1240/12.03 = 103 \text{ nm}$.

Answer: $\approx 103 \text{ nm}$.

Problem 16: A microscope using photons is employed to locate an electron in an atom with in a distance of $0.1 \text{ \AA}$. Figure out the uncertainty in the momentum of the electron located in this way? Assess the uncertainty in velocity? Rest mass of an electron $= 9.1 \times 10^{-31} \text{ kg}$.

Solution: Distance $= 0.1 \text{ \AA}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$

$$\Delta X. \Delta P = \hbar/2 \text{ or } \hbar/2\pi, \quad \Delta P = \hbar/(2 \times \Delta X) = 6.626 \times 10^{-34} / 0.1 \times 10^{-10} \\ = 6.63 \times 10^{-33} \text{ kg m/s}$$

$$\text{Uncertainty in the velocity} = \Delta P = m_0. \Delta V$$

$$\Delta V = m_0 / \Delta P = 7.28 \times 10^7 \text{ m/s}$$

Problem 17: Compute the energy difference between the ground state and the first excited state for an electron in a one-dimensional box of length 10^{-8} cm .

Solution: $E_n = \frac{n^2 h^2}{8ml^2}$

$$= 1 \times (6.626 \times 10^{-34})^2 / 8 \times 9.1 \times 10^{-31} \times (1 \times 10^{-10})^2 \\ = 6.030 \times 10^{-18} \text{ J}$$

$$E_2 = 4 \times (6.626 \times 10^{-34})^2 / 8 \times 9.1 \times 10^{-31} \times (1 \times 10^{-10})^2 = 2.412 \times 10^{-17} \text{ J}$$

$$E_2 - E_1 = 2.412 \times 10^{-17} \text{ J} - 6.030 \times 10^{-18} \text{ J} = 1.8089 \times 10^{-17} \text{ J} \\ = 112.91 \text{ eV}$$

Problem 17: Compute the De-Broglie wavelength of an electron whose kinetic energy is 10 eV.

Solution: The de-Broglie wavelength of an electron $\lambda = \frac{h}{\sqrt{2m(K.E)}}$

$$= \frac{6.626 \times 10^{-34} \text{ J/s}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 10 \times 1.602 \times 10^{-19}}} = 3.88 \times 10^{-10} \text{ m}$$

Problem 17: Calculate the energy in eV required to pump an electron from ground state to excited state in metal of length 10^{-10} m .

Solution: $E_n = \frac{n^2 h^2}{8ml^2}$

$$\text{Energy of ground state, } = E_1 = \frac{1 \times (6.62 \times 10^{-34})^2}{8 \times (9.1 \times 10^{-31})(10^{-10})^2} = 0.631 \times 10^{-17} \text{ J}$$

$$\text{Energy of first excited state, } E_2 = \frac{2^2 \times (6.62 \times 10^{-34})^2}{8 \times (9.1 \times 10^{-31})(10^{-10})^2} = 2.412 \times 10^{-17} \text{ J}$$

$$\begin{aligned} \text{Energy required is, } E_2 - E_1 &= 2.412 \times 10^{-17} - 0.631 \times 10^{-17} = 1.8089 \times 10^{-17} \text{ J} \\ &= \frac{1.8089 \times 10^{-17} \text{ J}}{1.602 \times 10^{-19} \text{ C}} = 113.05 \text{ eV} \end{aligned}$$

Review Questions

Short Answer Questions

1. State the de-Broglie hypothesis for matter waves.
2. What is Heisenberg's uncertainty principle?
3. Give the physical significance of the wave function Ψ .
4. What is an operator in quantum mechanics?
5. Define eigenvalues and eigenfunctions.
6. Write the 1D time-independent Schrödinger wave equation.
7. State Bloch's theorem.
8. What is the physical origin of energy band gaps in solids?
9. What is quantum confinement in nanomaterials?
10. Define expectation value of a physical quantity.

Long Answer Questions

1. Setup and solve the 1D time-independent Schrödinger wave equation for a particle in a 1D potential box with infinite potential barriers. Obtain the normalized wave functions and energy eigenvalues.
2. Explain the postulates of quantum mechanics in detail and derive expressions for operators corresponding to position, momentum, and energy.
3. Describe the Kronig-Penney model qualitatively and explain how it leads to the formation of allowed and forbidden energy bands in solids using the E-k diagram.
4. State and explain the Heisenberg Uncertainty Principle. Describe an experiment (e.g., gamma-ray microscope or single-slit diffraction) demonstrating this principle.
5. Classify solids into conductors, semiconductors, and insulators based on band theory. Explain how the E-k diagram helps in understanding effective mass.
6. Explain the concept of quantum confinement in 0D, 1D, and 2D nanoscale structures, and compare their density of states with 3D bulk materials.
7. Derive the de-Broglie wavelength for a particle accelerated through a potential difference V.
8. Describe Davisson and Germer's experiment verifying matter waves. Explain the physical significance and normalization condition of a wave function φ .
9. Derive the probability density expression and continuity equation. Solve the Schrödinger equation for a step potential barrier and discuss the concept of quantum

tunneling.

10. Discuss the transition from continuous energy levels in bulk materials to discrete energy levels in nanostructures due to quantum mechanics principles.

Objective Type Questions

Multiple Choice Questions

- The wave nature of matter was proposed by
 - Planck
 - de Broglie
 - Rutherford
 - Maxwell
- Planck proposed that electromagnetic radiation is emitted or absorbed in
 - Continuous waves only
 - Discrete energy quanta
 - Mechanical pulses
 - Sound waves
- The energy of a photon is given by
 - $E = h\nu$
 - $E = h/\nu$
 - $E = \nu/h$
 - $E = hc^2/\nu$
- The de Broglie wavelength of a particle is
 - h/p
 - p/h
 - hp
 - hmv
- The Davisson–Germer experiment demonstrated
 - Electron diffraction
 - Nuclear fission
 - Photon polarization
 - Magnetic resonance
- Heisenberg's uncertainty principle relates uncertainty in
 - Charge and mass
 - Position and momentum
 - Temperature and volume
 - Frequency and intensity
- The probability density associated with a wave function ψ is
 - ψ
 - $|\psi|^2$
 - $1/\psi$
 - $d\psi/dx$
- A physically acceptable wave function should be
 - Infinite everywhere
 - Single-valued and finite
 - Discontinuous everywhere
 - Zero everywhere
- The operator corresponding to momentum in one dimension is proportional to
 - d/dx
 - $-i\hbar d/dx$
 - x^2
 - d^2/dt^2
- When an operator acts on its eigenfunction, the result is
 - A random function
 - Eigenvalue times the eigenfunction
 - Always zero
 - Always infinite
- The average measured value of an observable is called its
 - Eigenvalue
 - Expectation value
 - Wave number
 - Probability amplitude
- Schrödinger's time-independent equation describes
 - Stationary states
 - Classical trajectories
 - Thermal conduction
 - Circuit transients
- The energy of a particle in a one-dimensional infinite box is
 - Continuous
 - Quantized
 - Always negative
 - Always zero
- For a particle in a 1D box, the energy varies as

- A) n
 B) n^2
 C) $1/n$
 D) $\sqrt{n}$
15. The lowest energy state of a particle in a box corresponds to
 A) $n = 0$
 B) $n = 1$
 C) $n = -1$
 D) $n = 2$ only
16. Bloch's theorem applies to particles moving in a
 A) Periodic potential
 B) Random potential only
 C) Vacuum only
 D) Uniform magnetic field only
17. The Kronig–Penney model explains
 A) Energy bands in solids
 B) Nuclear decay
 C) Optical pumping
 D) Magnetic hysteresis
18. The E-k diagram represents the relationship between
 A) Energy and wave vector
 B) Energy and temperature only
 C) Electric field and charge
 D) Entropy and pressure
19. The energy interval in which no allowed electron states exist is the
 A) Valence band
 B) Forbidden energy gap
 C) Fermi sphere
 D) Conduction path
20. In a conductor, the valence and conduction bands may
 A) Be widely separated
 B) Overlap
 C) Disappear
 D) Become quantized only
21. A semiconductor generally has a
 A) Small forbidden energy gap
 B) Zero band gap
 C) Very large nuclear gap
 D) Negative lattice constant
22. An insulator generally has a
 A) Large forbidden energy gap
 B) Zero band gap
 C) Band overlap
 D) Filled conduction band
23. Quantum confinement in nanomaterials produces
 A) More discrete energy levels
 B) No energy levels
 C) Classical continuous states only
 D) Zero surface effects
24. The principle that quantum states can be described probabilistically is associated with the interpretation of
 A) Wave function
 B) Lattice constant
 C) Packing factor
 D) Numerical aperture
25. The dual nature of matter means matter exhibits both
 A) Heat and sound
 B) Particle and wave properties
 C) Electric and magnetic charge
 D) Mass and temperature

Multiple Choice Questions

- The quantum of electromagnetic radiation is called a _____.
- Planck's relation is $E =$ _____.
- The matter-wave hypothesis was proposed by _____.
- The de Broglie wavelength is $\lambda =$ _____.
- The Davisson–Germer experiment demonstrated the wave nature of _____.
- Heisenberg's principle involves uncertainty in position and _____.

7. The symbol commonly used for a quantum wave function is _____.
8. The quantity $|\psi|^2$ represents _____ density.
9. A physically acceptable wave function should be single-valued, finite and _____.
10. The momentum operator in one dimension is $-i\hbar$ _____.
11. The result of an operator acting on an eigenfunction contains an _____.
12. The average value of an observable is called its _____ value.
13. Schrödinger's time-independent equation describes _____ states.
14. The allowed energies of a particle in a box are _____.
15. For a 1D box, energy is proportional to the square of the quantum number _____.
16. The ground state of a particle in a box has quantum number _____.
17. Bloch's theorem applies to a _____ potential.
18. The Kronig–Penney model explains the formation of energy _____.
19. The E-k diagram gives the relation between energy and _____ vector.
20. The region between allowed bands where no states exist is called the _____ gap.
21. In conductors, valence and conduction bands may _____.
22. Semiconductors have a relatively _____ forbidden energy gap.
23. Insulators generally have a relatively _____ forbidden energy gap.
24. Quantum confinement produces more _____ energy levels.
25. Matter exhibiting both wave and particle properties is said to possess _____ nature.

MCQs: 1-B, 2-B, 3-A, 4-A, 5-A, 6-B, 7-B, 8-B, 9-B, 10-B, 11-B, 12-A, 13-B, 14-B, 15-B, 16-A, 17-A, 18-A, 19-B, 20-B, 21-A, 22-A, 23-A, 24-A, 25-B

Fill in the blanks: 1. photon; 2. $h\nu$; 3. Louis de Broglie; 4. h/p ; 5. electrons; 6. momentum; 7. ψ ; 8. probability; 9. continuous; 10. d/dx ; 11. eigenvalue; 12. expectation; 13. stationary; 14. quantized; 15. n ; 16. 1; 17. periodic; 18. bands; 19. wave; 20. forbidden; 21. overlap; 22. small; 23. large; 24. discrete; 25. Dual.

UNIT – 3: QUANTUM COMPUTING

Contents: - Introduction, linear algebra for quantum computation, Dirac's Bra and Ket notation and their properties, Hilbert space, Bloch's sphere, concept of quantum computer, classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, evolution of quantum systems, quantum measurements, entanglement, quantum gates, challenges and advantages of quantum computing over classical computation, quantum algorithms: Deutsch-Jozsa, Shor, Grover (qualitative).

3.1. INTRODUCTION: Brief History of Quantum Computers

Quantum computing is an emerging field of science and technology that combines the principles of quantum mechanics with computer science to develop a new generation of computational systems. Quantum Computing is the area of study focused on developing computing methods based on the principles of quantum theory. Quantum Physics explains the nature and behavior of energy and matter on the quantum (atomic and subatomic) scale. Elementary particles such as protons, neutrons, and electrons can exist in two or more states at a time. This fundamental behavior is utilized in designing the quantum computation processing units. Quantum computing uses a combination of bits of 1's, 0's, and both 1 and 0 at a time to perform computational tasks with greater efficiency.

- Quantum computing is based on the fundamental principles of quantum mechanics, such as superposition, entanglement, interference, and quantum measurement.
- Richard Feynman proposed quantum computing in the 1980. Quantum computers can simulate more efficiently than classical computers.
- In Quantum computing, the information is encoded in a quantum system such as atoms, ions, or quantum dots.
- Quantum computers are based on quantum bits (qubits) and use quantum effects like superposition and entanglement to their benefit, hence overcoming the problems of classical computing.

Classical Computer: A computer that uses voltages flowing through circuits and gates, which can be controlled and manipulated entirely by classical mechanics.

Quantum Computer: A quantum computer is a machine that performs calculations based on the laws of quantum mechanics, which is the behavior of particles at the sub-atomic level.

"I think I can safely say that nobody understands quantum mechanics" – Feynman

Differences between classical and quantum computing

Essential features	Classical computing	Quantum computing
Basis of computing	Large-scale multipurpose computer based on classical physics.	A high-speed computer based on quantum mechanics.
Typical	One to several	One

number of processors		
Information storage	Bit-based information storage using voltage/charge.	Quantum bit-based information storage using electron spin or polarization.
Bit values	Bits having a value of either 0 or 1 can have a single value at any instant.	Qubits have a value of 0, 1, or sometimes a linear combination of both (a property known as superposition).
Number of possible states	The number of possible states is 2, which is either 0 or 1.	The number of possible states is infinite since it can hold combinations of 0 or 1 along with some complex information.
Output	Deterministic (repetition of Computation on the same input gives the same output).	Probabilistic (repetition of computation on superposed states gives a probabilistic answer).
Gates used for Processing	Logic gates (AND, OR, NOT, etc.)	Quantum gates (X, Y, Z, H, CNOT, etc.)
Operations	Operations use Boolean Algebra.	Operations use linear algebra and are represented with unitary matrices.
Circuit implementation	Circuit implemented in macroscopic technologies	Circuits are implemented in the microscopic technologies.
Data processing	Data processing is carried out by logic and in sequential order.	Data processing is carried out by quantum logic at parallel instances.
Speed	Limited by classical physics	Can perform certain calculations much faster than classical computers
Applications	Software, Gaming, Simulations, etc.	Cryptography, Chemistry, Machine learning, etc.

3.2. LINEAR ALGEBRA FOR QUANTUM COMPUTING

Dirac Representation and Matrix Operations:

Dirac notation is used in quantum computing to describe quantum states, quantum operations, and quantum measurements. It was named after the physicist Paul Dirac, who developed the notation in 1939.

Vector Space:

A vector space is a set of objects called vectors that can be added together and multiplied by scalars (numbers) and still stay within the space.

In Quantum mechanics, the Bra-Ket notation is a standard way to describe quantum states. The notation $|\psi\rangle$ is known as ‘ket’ notation, and $\langle\psi|$ is known as ‘bra’ notation. Both are together called the Dirac notation.

3.3. DIRAC'S BRA AND KET NOTATION AND THEIR PROPERTIES

Basic definitions:

- **Ket:** $|\psi\rangle$ represents a quantum state vector.
- **Bra:** $\langle\psi|$ is the conjugate transpose of the ket $|\psi\rangle$.
- **Inner product:** The bracket $\langle\phi|\psi\rangle$ is the inner product of two states, yielding a complex scalar.

Linearity Property of Bra-Ket Notation:

- **Linearity of bras:** If any ket vector, $|\phi\rangle$, the linearity property is

$$(\langle\psi_1|\alpha+\beta\langle\psi_2|\beta)\,|\phi\rangle = \alpha\langle\psi_1|\phi\rangle+\beta\langle\psi_2|\phi\rangle.$$

- **Linearity of kets:** If any Bra vector, $\langle\phi|$, the linearity property is

$$\langle\phi|(\alpha|\psi_1\rangle+\beta|\psi_2\rangle) = \alpha\langle\phi|\psi_1\rangle+\beta\langle\phi|\psi_2\rangle$$

Inner product and its properties:

The inner product of two vectors $|\psi\rangle$ and $|\phi\rangle$ (also called the dot product) measures their overlap. It represents the $\langle\phi|\psi\rangle$.

- **Non-negativity:** If any ket vector, $|\psi\rangle$, the inner product is $\langle\psi|\psi\rangle\geq 0$, with $\langle\psi|\psi\rangle=0$, if and only if $|\psi\rangle$ is the zero vector.
- **Conjugate symmetry:** The inner product follows conjugate symmetry, i.e, $\langle\phi|\psi\rangle=\langle\psi|\phi\rangle^*$.
- **Orthonormality:** For a set of orthonormal basis vectors $|\psi_i\rangle$, the inner product is $\langle\psi_i|\psi_j\rangle=\delta_{ij}$ (the Kronecker delta), where it is 1 if $i=j$, is called the orthonormal property, and 0 if $i\neq j$, is called the orthogonal property.

Outer Product and its properties:

The **outer product** creates a matrix from two vectors: $|\psi\rangle\langle\phi|$.

- **Completeness condition (Resolution of identity):** For a complete basis, the sum of the outer products of the basis vectors with themselves is the identity operator: $\sum_i |\psi_i\rangle\langle\psi_i| = \mathbf{I}$.
- **Projection operator:** The outer product $|\psi\rangle\langle\psi|$ is a projection operator that projects any state onto the state $|\psi\rangle$.

Operator action and its properties:

- **Operator action:** An operator $\hat{A}$ acting on a ket is written as $\hat{A}|\psi\rangle$.
 - **Linearity of Operators:** If $\hat{A}$ is an operator,
- $$\hat{A}(\alpha|\psi\rangle+\beta|\phi\rangle) = \alpha\hat{A}|\psi\rangle + \beta\hat{A}|\phi\rangle.$$
- **Hermitian Conjugation (Adjoint):** The adjoint of an operator $\hat{A}$ is denoted by $\hat{A}^\dagger$ and is related by $\langle\phi|\hat{A}|\psi\rangle^*=\langle\psi|\hat{A}^\dagger|\phi\rangle$.

- **Hermitian (Self-adjoint) Operators:** The operator $\hat{A}$ is and the Hermitian operator $\hat{A}^\dagger$ is follows, $\hat{A}^\dagger = \hat{A}$.

Properties of the Hermitian operator are:

- Eigenvalues are real
- Eigenstates corresponding to different eigenvalues are orthogonal
- **Expectation Values:** The operator $\hat{A}$ and any ket vector $|\psi\rangle$, the expectation value is $\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle$.
- **Matrix elements:** The matrix elements of an operator $\hat{A}$ between two states $|\psi\rangle$ and $|\phi\rangle$ are given by the bra-ket expression $\langle \phi | \hat{A} | \psi \rangle$.

Matrix representation of 0 and 1 states:

Consider a quantum state ψ in a vector space represented by $|\psi\rangle$. The notation $|\rangle$ indicates that the object is a vector, called a ket vector. The examples of such ket vectors are $|\psi\rangle$, $|\phi\rangle$, and $|u\rangle$, etc.

The wave function could be expressed in ket notation as $|\psi\rangle$ (ket Vector), where ψ is the wave function.

Hence, any arbitrary state can be represented as

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \text{ or } |\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

The ‘ket’ vector is typically represented as a column vector, and the ‘bra’ vector is typically represented as a row vector. The matrix for of the states $|0\rangle$ and $|1\rangle$ as follows,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\langle 0| = [1 \quad 0] \text{ and } \langle 1| = [0 \quad 1]$$

Operators and matrices:

An operator is a mathematical rule that transforms a given function into another function. Consider an operator ‘A’ transforms the vector $|a\rangle$ to another vector $|b\rangle$, it can be written as

$$\hat{A} |a\rangle = |b\rangle$$

There are different types of operators like Linear operator, Identity operator, Null operator, Inverse operator, Singular & non-singular operator, etc.

Identity operator (I):

The identity operator is an operator that, operating on a function, leaves the function unchanged, i.e., $I |a\rangle = |a\rangle$. It is given in matrix form by

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This is also called the identity matrix. There will be no change when I operate on either $|0\rangle$ state or $|1\rangle$ state. It is explained as follows.

$$I|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$I|0\rangle = |0\rangle$$

Similarly,

$$I|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$I|1\rangle = |1\rangle$$

The identity matrix acts as number 1. It is always a square matrix.

Conjugate matrices:

If the elements in a matrix A are complex numbers, then the matrix obtained by the corresponding conjugate complex elements is called the conjugate of A and is denoted by A^* .

For example,

$$\text{If, } A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}, \text{ then } A^* = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

$$\text{If, } A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}, \text{ then } A^* = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

Transpose matrices:

If columns and rows of a matrix A are interchanged, then the resultant matrix is the transpose of A and represented as A^T .

For example,

$$\text{If, } A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}, \text{ then } A^T = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\text{If, } A = \begin{bmatrix} 1 & 2i \\ 4i + 1 & 0 \end{bmatrix}, \text{ then } A^T = \begin{bmatrix} 1 & 4i + 1 \\ 2i & 0 \end{bmatrix}$$

Hermitian matrices:

The transpose of the complex conjugate of a matrix is known as a Hermitian operator, and the resultant matrix is known as a Hermitian matrix. It is represented by $A^\dagger$.

Let A be a matrix, A^* be its complex conjugate, and $(A^*)^T$ be its transpose; then its Hermitian matrix is $A^\dagger = (A^*)^T$.

$$\text{If, } A = \begin{bmatrix} 1 & 2i \\ 4i + 1 & 0 \end{bmatrix}, \text{ then } A^* = \begin{bmatrix} 1 & -2i \\ -4i + 1 & 0 \end{bmatrix},$$

$$\text{then } A^\dagger = \begin{bmatrix} 1 & -4i + 1 \\ -2i & 0 \end{bmatrix}$$

Unitary matrices:

Matrix A is said to be unitary if it produces an identity matrix I when multiplied by its conjugate transpose, $AA^\dagger = I$.

In other words, A is a unitary matrix if its conjugate transpose is equal to its reciprocal,

i.e.
$$A^\dagger = \frac{1}{A} = A^{-1}$$

Column and Row Matrices and their inner product:

The Column Vectors are called ket Vectors, denoted by $|\psi\rangle$, and are represented by Column Matrices.

The Row Vectors are called Bra Vectors, denoted by $\langle\phi|$, and are represented by Row Matrices.

Let us consider a ket vector represented in the form of a column matrix,

$$|\psi\rangle = \begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix},$$

The Row Matrix is represented as $\langle\psi| = [\alpha^*_1 \quad \beta^*_1]$

Here,
$$\begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix}^\dagger = [\alpha^*_1 \quad \beta^*_1]$$

Thus, the Bra is the complex conjugate of the ket and vice versa.

Inner product:

The inner product of two vectors U and V in the complex space is a function that takes U and V as inputs and produces a complex number as output.

In terms of Dirac notation, the inner product is given as $\langle U|V\rangle = C$ In matrix form, U and V are written as a

$$\langle U| = \begin{bmatrix} x_1 & y_1 \end{bmatrix} \text{ and } |V\rangle = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

Their inner product is written as $\langle U|V\rangle$, but $\langle U|$ is equal to conjugate transpose of $|U\rangle$

i.e.,
$$\langle U| = |U^*\rangle^{-1} = |U\rangle^\dagger = [x^* \ y^*]$$

$$\langle U|V\rangle = [x_1 \quad y_1] \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = x_1^* x_2 + y_1^* y_2$$

The square root of the inner product of a vector with itself is also called the norm or the length of the vector. It is given by $|U| = \sqrt{\langle U|U\rangle}$.

Ex: Find the inner product of $|U\rangle = \begin{bmatrix} 3+i \\ 4-i \end{bmatrix}$ and $|V\rangle = \begin{bmatrix} 3i \\ 4i \end{bmatrix}$

First, we shall find the conjugate transpose of $|U\rangle$, $|U^*\rangle = \begin{bmatrix} 3-i \\ 4+i \end{bmatrix}$

$$|U\rangle^\dagger = [3-i \quad 4+i]$$

$$|U\rangle = |U\rangle^\dagger = [3-i \quad 4+i]$$

$$\langle U|V\rangle = [3-i \quad 4+i] \begin{bmatrix} 3i \\ 4 \end{bmatrix}$$

$$\langle U|V\rangle = [(3-i) * (3i) + (4+i) * 4]$$

$$\langle U|V\rangle = [9i + 3 + 16 + 4i]$$

$$\langle U|V \rangle == [13i + 19]$$

Orthogonality:

If the inner product of two vectors is 0, then they are said to be orthogonal (or perpendicular) to each other.

If $\langle U|V \rangle = 0$, then $|U\rangle$ and $|V\rangle$ are perpendicular.

Consider, $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\langle 0|1 \rangle = [1 \quad 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

Hence, $|0\rangle$ is perpendicular to $|1\rangle$.

The most important property of the inner product of a vector with itself is equal to one.

$$\text{ie, } \langle \psi|\psi \rangle = 1.$$

This is known as the normalization condition.

The physical significance of normalization is that the "probability amplitude" of the quantum system.

Orthonormality:

If each element of a set of vectors is normalized and the elements are orthogonal with respect to each other, we say the set is orthonormal.

Consider the set: $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\langle 0|0 \rangle = [1 \quad 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \quad \text{“normalized”}$$

$$\langle 0|1 \rangle = [1 \quad 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \quad \text{“orthogonal”}$$

$$\langle 1|0 \rangle = [0 \quad 1] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \quad \text{“orthogonal”}$$

$$\langle 1|1 \rangle = [0 \quad 1] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1 \quad \text{“normalized”}$$

Hence, the set of $|0\rangle$ and $|1\rangle$ is orthogonal.

From the above relations, if states $|\psi\rangle$ and $|\phi\rangle$ are said to be orthonormal. if

1. $|\psi\rangle$ and $|\phi\rangle$ are normalized.
2. $|\psi\rangle$ and $|\phi\rangle$ are orthogonal to each other.

Probability:

Let us consider a Quantum State, $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$

$$|\psi\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

The inner product $\langle \psi|\psi \rangle$ is given by,

$$\langle \psi | \psi \rangle = [\alpha^* \quad \beta^*] \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha^* \alpha + \beta^* \beta = |\alpha|^2 + |\beta|^2$$

This could also be written as $|\psi|^2 = \psi \psi^*$

Thus, the above equation represents Probability Density. As per the principle of Normalization

$$|\psi|^2 = \psi \psi^* = \langle \psi | \psi \rangle = 1 = |\alpha|^2 + |\beta|^2$$

Thus, it implies $|\psi\rangle$ is normalized.

3.4. HILBERT SPACE IN QUANTUM COMPUTING

A Hilbert space is a complete vector space with an inner product, serving as the mathematical framework for quantum mechanics and quantum computation. In this space, quantum states are represented as vectors, and quantum operations are represented as linear operators.

- **Inner product:** Defines lengths and angles between vectors, allowing calculation of probabilities through $\langle \phi | \psi \rangle$.
- **Normalization:** Quantum states must satisfy $\langle \psi | \psi \rangle = 1$.
- **Orthogonality:** Two states $|\psi\rangle$ and $|\phi\rangle$ are orthogonal if $\langle \phi | \psi \rangle = 0$.
- **Tensor product:** Multi-qubit systems are described by tensor products of Hilbert spaces, expanding the dimensionality exponentially.
- **Completeness:** Every convergent sequence of vectors in Hilbert space has a well-defined limit, ensuring mathematical consistency.

Thus, Hilbert space provides the foundation to describe superposition, entanglement, and evolution of quantum systems.

Representation of Qubits by Bloch Sphere: (single qubit state):

The Bloch sphere is an imaginary sphere used to represent pure single-qubit states as points on its surface. It has a unit radius. Its North Pole and South Pole are selected to represent the basis states, namely $|0\rangle$ and $|1\rangle$. The North Pole represents $|0\rangle$ (say spin up $\uparrow$) and the South Pole represents $|1\rangle$ (say spin down $\downarrow$). All other points on the sphere represent superposed states (i.e., state space). The Bloch sphere allows the state of a qubit to be represented in spherical coordinates (i.e., r , θ and ϕ). It is as follows

The state qubit $|\psi\rangle$ on the Bloch sphere makes an angle ' θ ' with the z-axis, and its projection (azimuth) makes an angle ' ϕ ' with the x-axis as shown. It is clear from the figure that $0 < \theta < \pi$ and $0 < \phi < 2\pi$.

$|\psi\rangle$ is represented as, $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$.

Where α and β are complex probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$

The Bloch sphere is a powerful tool for understanding how quantum gates transform qubit states through rotations on the sphere.

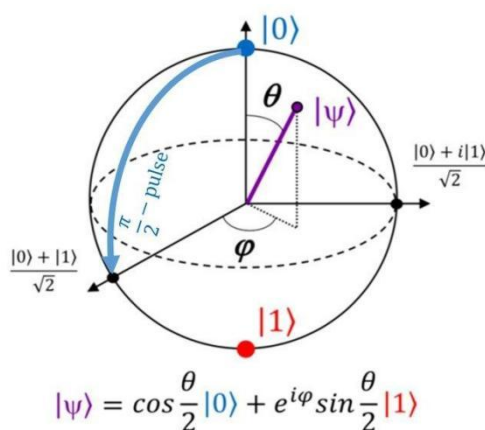


Figure 3.1: Bloch sphere model.

Significance of Bloch Sphere:

The Bloch sphere is not only a visualization, but also it is a tool for:

- ✓ Understanding Superposition: As a qubit can exist in both $|0\rangle$ and $|1\rangle$ states simultaneously, the Bloch sphere helps to visualize this as a point between the poles
- ✓ Explaining Quantum Gates: As Quantum gates like X, Y, and Z correspond to rotations around the sphere's axes, the Bloch sphere makes quantum gate effects intuitive.
- ✓ Visualizing Phase and Interference: The relative phase of a quantum state is shown as a rotation around the vertical axis.
- ✓ Correcting Quantum Algorithms: Bloch vectors help to understand quantum behavior.

3.6. CONCEPT OF QUANTUM COMPUTERS

A quantum computer is a type of computer or computing system that uses quantum mechanics to perform calculations and process information in fundamentally new ways. It utilizes quantum bits (qubits), which can represent 0, 1, or both simultaneously through a phenomenon known as superposition. Qubits can also become entangled, meaning the state of one qubit is directly related to the state of another, allowing quantum computers to perform complex calculations more efficiently, with potential applications in drug discovery, financial modelling, and materials science.

3.6.1. CLASSICAL BITS

Bit: A classical bit is the fundamental unit of information in traditional (classical) computing. It can exist in one of two distinct states, represented as 0 or 1. In modern computing and communications, bits are represented by the absence or presence of an electrical signal, encoding '0' and '1' respectively.

- Bits are usually implemented physically using two-level systems such as voltage (high/low), current (on/off), or magnetic polarization (up/down).
- Operations on bits are carried out using logic gates (AND, OR, NOT, etc.), which form the building blocks of classical circuits.

- A sequence of bits forms larger data units such as bytes, kilobytes, and gigabytes, enabling representation of numbers, text, and multimedia.
- Bits follow the principles of classical physics, meaning their states are deterministic; each bit is always either 0 or 1 at any given time.
- Computation with classical bits underpins all digital devices, from calculators and smartphones to supercomputers.

3.6.2. QUBITS

A Qubit (quantum bit) is the basic unit of quantum information. It is the quantum version of a bit, and its quantum state can be written in terms of two levels, labelled $|0\rangle$ and $|1\rangle$. The notation ' $| \rangle$ ' is known as 'ket' notation, and $\langle |$ is known as 'bra' notation. Both are together called as Dirac notations, 'Ket' is analogous to a column vector. They are also called basis vectors and represented by two-dimensional column vectors as follows:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The qubit can be in any one of the two states as well as in the superposed state, simultaneously. In quantum computation, two distinguishable states of a system are needed to represent a bit of data.

For example, two states of an electron orbiting a single atom. Spin up is taken as $|1\rangle$, and spin down is taken as $|0\rangle$. Similarly, the ground state energy level is $|0\rangle$, and the excited state level is $|1\rangle$.

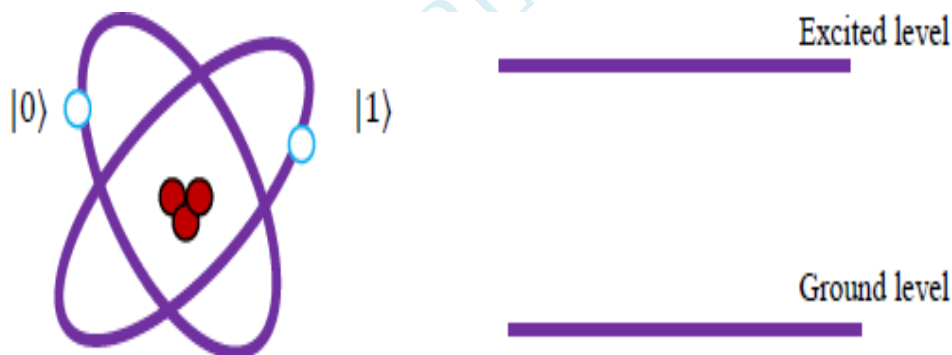


Figure 3.2: Qubit represented by two electronic levels in the atom.

Properties of qubits:

1. Qubits make use of discrete energy state particles such as electrons and photons.
2. Qubits exist in two quantum state $|0\rangle$ and $|1\rangle$ or in a linear combination of both states. This is known as superposition.
3. Unlike classical bits, a qubit can work with the overlap of both 0 & 1 states.
For example, a 4-bit register can store one number from 0 to 15 (because $2n = 24 = 16$). But a 4-qubit register can store all 16 numbers.
4. When the qubit is measured, it collapses to one of the two basis states $|0\rangle$ or $|1\rangle$
5. Quantum entanglement and quantum tunnelling are two exclusive properties of a qubit.

6. The state of the qubits is represented using the Bloch sphere.

3.7. MULTIPLE QUBIT SYSTEM

A multiple-qubit system arises when two or more qubits are combined, forming a higher-dimensional Hilbert space. Unlike classical bits, the state space of qubits grows exponentially with the number of qubits.

- State representation:
- One qubit $\rightarrow$ 2 basis states ($|0\rangle$, $|1\rangle$)
- Two qubits $\rightarrow$ 4 basis states ($|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$)
- Three qubits $\rightarrow$ 8 basis states and so on. In general, an n -qubit system has 2^n basis states.

Tensor product: Multi-qubit states are formed using the tensor product of single-qubit states, and they are represented as $|\psi\rangle \otimes |\phi\rangle$.

Key Principles of Quantum Computing (Characteristics of Qubits):

Quantum Superposition:

In quantum computing, quantum superposition is the fundamental principle that allows a quantum bit (qubit) to exist in multiple states (both 0 and 1) simultaneously until it is measured.

The qubit is a linear combination (superposition) of the two states $|0\rangle$ and $|1\rangle$. Superposition enables a qubit to hold multiple states at once, exponentially increasing the computational possibilities.

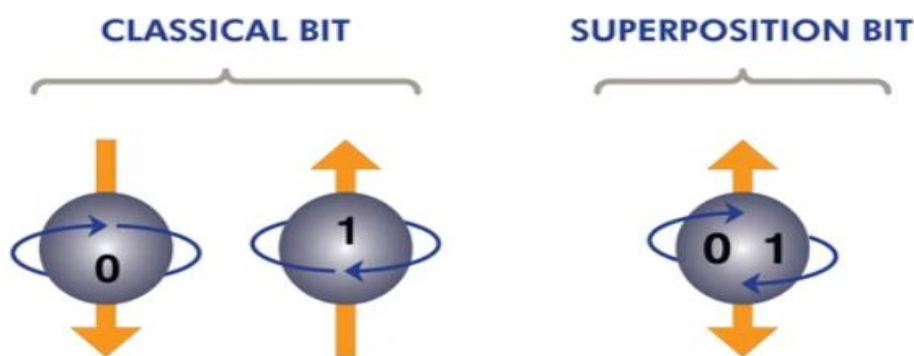


Figure 3.3: Illustration of the classical bit and superposition bit.

For example, if α and β are the probability amplitudes of an electron in the ground state (i.e., in $|0\rangle$ state) and in the excited state (i.e., in $|1\rangle$ state), then the linear combination of the two states is

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

The numbers α and β are complex, but due to normalization conditions.

$$|\alpha|^2 + |\beta|^2 = 1$$

Here, $|\alpha|^2$ is the probability of finding $|\psi\rangle$ in state $|0\rangle$ And $|\beta|^2$ is the probability of finding $|\psi\rangle$ in state $|1\rangle$. So, when a qubit is measured, it only gives either '0' or '1' as the measurement

result – probabilistically.

Consider the following example of qubit representation,

$$|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

$$\therefore \alpha = \frac{1}{\sqrt{2}} \text{ and } \beta = \frac{1}{\sqrt{2}}$$

$$|\alpha|^2 = |\beta|^2 = \frac{1}{2}$$

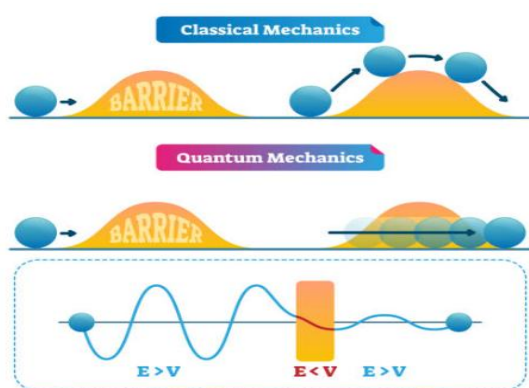
This means that with 50% probability, the qubit will be found in $|0\rangle$ state as well as in $|1\rangle$ state. The superposed states are also called space states, whereas $|0\rangle$ and $|1\rangle$ are called basis states.

- A classical bit can only be in the state corresponding to 0 or the state corresponding to 1.
- Superposition is an ambiguous state in which a particle can be both a 0 and 1. That is a linear combination of the two.
- Like flipping a coin, and the coin is always rotating in the air.
- The probability of measuring 0 or 1 for a qubit is, in general, neither 0% nor 100%.

QUANTUM TUNNELING

In quantum computing, quantum tunneling is the phenomenon where a quantum particle, like an electron, can pass through a potential energy barrier that it classically would not have enough energy to overcome. (or)

Quantum tunneling is a quantum mechanical phenomenon where a particle can pass through a potential energy barrier even if it doesn't have enough energy to overcome it, according to classical physics.



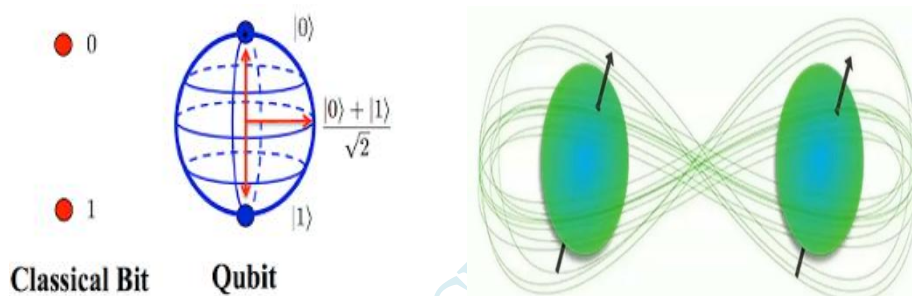
3.8. QUANTUM MEASUREMENT

The measurement process in quantum computing is probabilistic. When a qubit is measured, it collapses from its superposition state to one of the basis states, $|0\rangle$ or $|1\rangle$, with probabilities determined by the coefficients α and β . This probabilistic nature introduces new quantum algorithm design. Uncertainty principle arises from the nature of quantum measurement.

3.9. QUANTUM ENTANGLEMENT

Quantum entanglement is when two particles link together in a certain way, no matter how far apart they are in space. Their state remains the same. It occurs when two subatomic particles become linked in such a way that their properties remain connected, no matter how far apart they are. A change to one particle seems to instantaneously influence the other, even if billions of light-years separate them.

- Entanglement allows qubits to share information instantaneously, enabling speedups in algorithms, quantum teleportation, and quantum cryptography.
- Measurement effect: Measuring one qubit of an entangled pair collapses the entire system, showing the deep interdependence of the states.
- Resource: Entanglement is considered a key resource that gives quantum computers their computational advantage over classical machines.



In the above figure, have two entangled electrons such that their total spin is known to be zero (but both are in a state of superposition).

Applications of Quantum Entanglement:

Quantum Computing: Entanglement is fundamental to how quantum computers work. It allows qubits to be correlated, which is essential for developing new and faster algorithms to solve complex problems.

Quantum Cryptography: This technology uses entanglement to create perfectly secure communication channels. Information can be encrypted in a way that is virtually impossible to hack without detection.

Quantum Teleportation: This isn't like in science fiction. It's a process where the exact quantum state of one particle can be transferred to another distant particle using entanglement, which is crucial for building quantum communication networks.

Quantum decoherence: Quantum decoherence is the loss of quantum properties like superposition due to a system's interaction with its environment, causing it to behave classically.

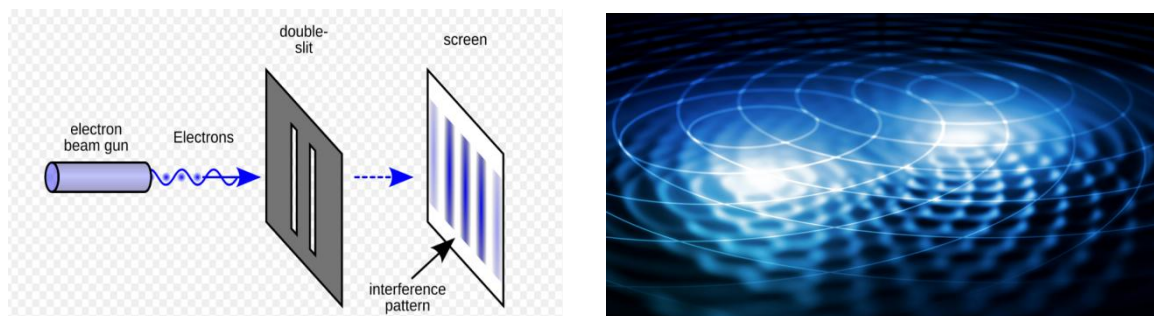
This interaction is a form of quantum noise, which includes factors like electromagnetic signals, thermal fluctuations, magnetic and electric fields, cross-talk between qubits, Thermal fluctuations, and is a major challenge for quantum computing.

E.g.: A qubit losing its superposition due to environmental noise like thermal fluctuations.

Scalability: With more qubits, quantum computers gain exponentially greater computational

power, enabling them to handle problems that classical computers cannot.

Quantum Interference: Quantum interference arises from the wave-like nature of the quantum particles such as electrons or photons. When particle is in a superposition of multiple states, these states can interfere with each other leads to constructive or destructive interference. Quantum interference has been experimentally verified through the double-slit experiment.



Key aspects:

- Due to wave-particle duality, at the subatomic level, particles have wavelike properties.
- In quantum computing interference is used to affect probability amplitudes.
- Each qubit corresponds to a wave, each wave to a specific path to solve a problem.
- Due to superposition, all the paths are available at the same time which results in qubit to exhibit constructive and destructive interference, which is used to find the probability of finding the best and wrong solutions.

3.10. QUANTUM GATES

A quantum gate is a very simple computing device that performs a quantum operation on qubits. Quantum gates are essential components of a quantum computer and serve as the building blocks of all quantum algorithms. Quantum gates are mathematically represented as transformation matrices that operate on inputs to give outputs.

There are different types of quantum gates: single-qubit gates and multiple-qubit gates. These gates can flip a qubit from 0 to 1, as well as allow superposition states to be created.

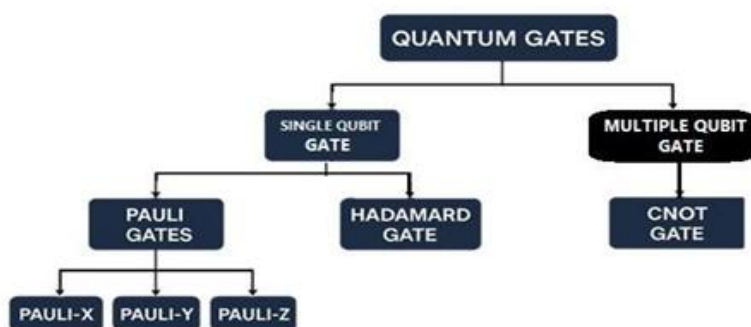


Figure 3.4: Types of quantum gates.

Single-Qubit Gates:

Single qubit inputs are $|0\rangle$ to $|1\rangle$ and can be represented by matrix forms as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Single Qubit Gates are X- gate, Y- gate, Z- gate, H –gate, S- gate, T- gate.

1. Pauli-X Gate or Quantum Not Gate:

X- gate is the single-qubit input gate and is also called the Pauli X–gate or quantum X- gate.

NOT gate.

The Matrix form of X is given by $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Action of the X- gate on inputs: When X- gate operates on inputs $|0\rangle$, $|1\rangle$, $|\psi\rangle$;

When X operates on $|0\rangle$ and $|1\rangle$, the output will be inverted (i.e., $|0\rangle$ becomes $|1\rangle$ and $|1\rangle$ becomes $|0\rangle$).

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$X|1\rangle = |0\rangle$$

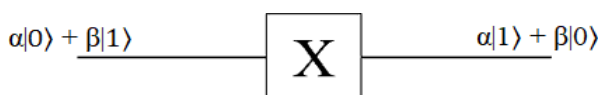
Since X inverts each input, it is also called a bit-flip gate.

If a superposed qubit goes through the X gate, the result will be

$$X|\psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$$

$$X|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

Gate representation:



Truth table:

X -gate	
Input	Output
$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 1\rangle + \beta 0\rangle$

2. Pauli-Y Gate:

Y- gate is the single-qubit input gate. This is also called the Pauli Y–gate. The matrix form of the Y gate is, $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

Action of Y gate on inputs: When Y operates on $|0\rangle$ and $|1\rangle$,

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 + 0 \\ i + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$Y|0\rangle = i|1\rangle$$

$$Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ 0 & 0 \end{bmatrix} = - \begin{bmatrix} i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = i|0\rangle$$

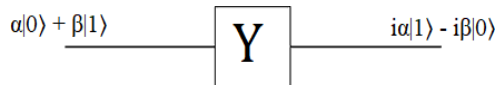
$$Y|1\rangle = -i|0\rangle$$

If a superposed qubit goes through a Y gate, the result will be

$$Y|\psi\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix} = -i\beta|0\rangle + i\alpha|1\rangle$$

$$Y|\psi\rangle = -i\beta|0\rangle + i\alpha|1\rangle$$

Gate representation:



Truth Table:

Y -gate	
Input	Output
$ 0\rangle$	$i 1\rangle$
$ 1\rangle$	$-i 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$i\alpha 1\rangle - i\beta 0\rangle$

3. Pauli-Z Gate:

Z-gate is the single-qubit input gate. This is also called the Pauli Z-gate. The matrix form of the Z gate is $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Action of Z gate on inputs: When Z operates on $|0\rangle$ and $|1\rangle$, the phase will change. Hence, this is also called a phase-flip gate.

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 + 0 \\ 0 + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 + 0 \\ 0 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -|1\rangle$$

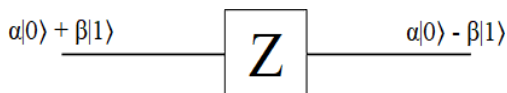
$$Z|1\rangle = -|1\rangle$$

If a superposed qubit goes through a Z gate, the result will be

$$Z|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

$$Z|\psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

Gate representation:



Truth Table:

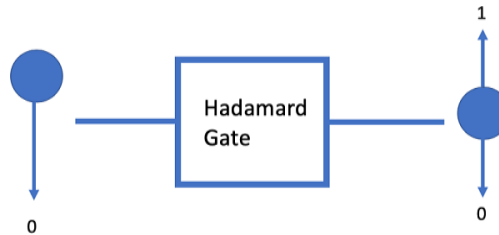
Z -gate	
Input	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$- 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle - \beta 1\rangle$

4. Hadamard Gate (H-gate):

It is a single-qubit gate and is also a gate superposition. The H gate acts on a single qubit input and produces a superposition state output. Matrix form of H gate is,

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

and its symbol is,



Action of the H gate on inputs:

Let us find out what happens when the Hadamard gate operates on a qubit that is in the $|0\rangle$ state

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} |0\rangle + |1\rangle$$

$$H|0\rangle = \frac{1}{\sqrt{2}} |0\rangle + |1\rangle$$

Let us find out what happens when the Hadamard gate operates on a qubit that is in the $|1\rangle$ state.

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} |0\rangle - |1\rangle$$

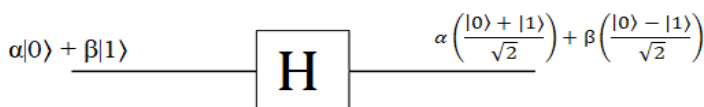
$$H|1\rangle = \frac{1}{\sqrt{2}} |0\rangle - |1\rangle$$

If a superposed qubit goes through the H gate, the result will be

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix} = \left(\frac{\alpha + \beta}{\sqrt{2}}\right) |0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}}\right) |1\rangle$$

$$H|\psi\rangle = \left(\frac{\alpha + \beta}{\sqrt{2}}\right) |0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}}\right) |1\rangle$$

Gate representation:



Truth Table:

H -gate	
Input	Output
$ 0\rangle$	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$
$ 1\rangle$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha\left(\frac{ 0\rangle + 1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{ 0\rangle - 1\rangle}{\sqrt{2}}\right)$

Note: The Difference between X, Y, Z, and H gates is that in X, Y, and Z gates, the output is in a single state, whereas in the H gate output is a superposed state.

5. Phase Gate (S Gate):

It is a single-qubit gate. The Phase gate or S gate is a gate that transfers $|0\rangle$ into $|0\rangle$ and $|1\rangle$ into $i|1\rangle$. The matrix form of the S gate is

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Action of S gate on inputs: Consider the S gate applied to a state $|0\rangle$; it will remain the same.

$$S|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$S|0\rangle = |0\rangle$$

If the S gate applies to a state $|1\rangle$, it will be transformed into $i|1\rangle$,

$$S|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

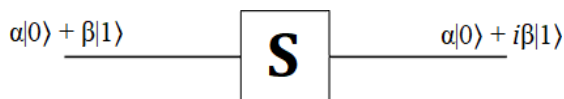
$$S|1\rangle = i|1\rangle$$

If S gate applies to a state $\alpha|0\rangle + \beta|1\rangle$, it will be transformed into $\alpha|0\rangle + i\beta|1\rangle$,

$$S|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ i\beta \end{bmatrix} = \alpha|0\rangle + i\beta|1\rangle$$

$$S|\psi\rangle = \alpha|0\rangle + i\beta|1\rangle$$

Gate representation:



Truth table:

S -gate	
Input	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$i 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle + i\beta 1\rangle$

6. T-Gate:

It is a single-qubit gate, and it is also called the $\pi/8$ gate. Its matrix form is

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Action of T gate on inputs:

If the T gate operates on an input $|0\rangle$, then the output is also $|0\rangle$

$$T|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$T|0\rangle = |0\rangle$$

If T gate operates on the input $|1\rangle$, then the output is $e^{i\pi/4} |1\rangle$

$$T|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = e^{i\pi/4} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = e^{i\pi/4} |1\rangle$$

$$T|1\rangle = e^{i\pi/4} |1\rangle$$

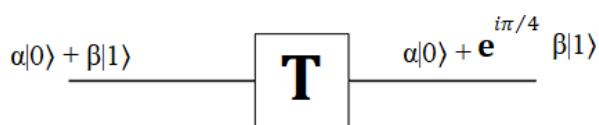
If T gate operates on superposed state $\alpha|0\rangle + \beta|1\rangle$, It transforms to $\alpha|0\rangle + e^{i\pi/4}\beta|1\rangle$

$$T|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\pi/4}\beta \end{bmatrix} = \alpha|0\rangle + e^{i\pi/4}\beta|1\rangle$$

$$T|\psi\rangle = \alpha|0\rangle + e^{i\pi/4}\beta|1\rangle$$

Gate representation:

Truth table:



T -gate	
Input	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$e^{i\pi/4} 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle + e^{i\pi/4}\beta 1\rangle$

Multiple Qubit gates:

Quantum gates operating on multiple qubits are called multiple-qubit gates. Multiple Qubit Gates operate on two or More Input Qubits. A multiple-qubit gate consists of a control gate and a target gate. The action of the gate is as follows;

- The Target qubit is altered only when the control qubit is $|1\rangle$, and
- The control qubit remains unaltered during the transformations.

For two qubits, the input qubits are $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.

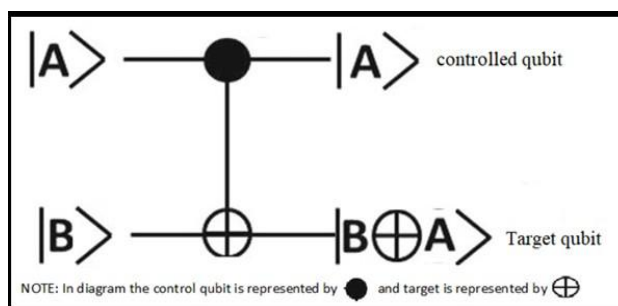
For three qubits, inputs qubits are $|000\rangle$, $|001\rangle$, $|010\rangle$, $|011\rangle$, $|100\rangle$, $|101\rangle$, $|110\rangle$ and $|111\rangle$

In a multiple-qubit gate, the input qubit is applied in the form $|AB\rangle$, the first term $|A\rangle$ goes to the control qubit, and the second term $|B\rangle$ goes to the target qubit.

Some of the multiple-qubit gates are as follows: Controlled gate (CNOT gate), Swap gate, Controlled Z gate, and Toffoli gate (CCNOT gate).

- Controlled Gate (CNOT):** The CNOT gate is a two-qubit operation, where the first qubit is referred to as the control qubit $|A\rangle$ and the second qubit as the target qubit $|B\rangle$. If the control qubit is $|1\rangle$, then it will flip the target qubit state from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$. When the control qubit is in state $|0\rangle$, then the target qubit remains unchanged.

The symbolic representation is as follows. The upper line represents the control qubit, and the bottom line represents the target qubit.



In the combined qubit, the first term is the control qubit, and the second term is the target qubit. For example, in $|AB\rangle$, A is the control qubit, and B is the target qubit.

The matrix form of the CNOT Gate is given by

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Gate	Input to the gate	Output of the gate
1	$ a, b\rangle$	$ a, a \oplus b\rangle$
2	$ a, a \oplus b\rangle$	$ b, a \oplus b\rangle$
3	$ b, a \oplus b\rangle$	$ b, a\rangle$

The Transformation could be expressed as $|A, B\rangle \rightarrow |A, B \oplus A\rangle$

Action of the gate:

Consider the operations of the CNOT gate on the four inputs $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.

- i. Operation of CNOT Gate for input $|00\rangle$: $|00\rangle \rightarrow |00\rangle$.

Here, the control qubit is $|0\rangle$. Hence, no change in the state of Target qubit $|0\rangle$.

- ii. Operation of CNOT Gate for input $|01\rangle$: $|01\rangle \rightarrow |01\rangle$.

Here, the control qubit is $|0\rangle$. Hence, no change in the state of Target qubit $|1\rangle$.

- iii. Operation of CNOT Gate for input $|10\rangle$: $|10\rangle \rightarrow |11\rangle$.

Here, the control qubit is $|1\rangle$. Hence, the state of the Target qubit flips from $|0\rangle$ to $|1\rangle$.

- iv. Operation of CNOT Gate for input $|11\rangle$: $|11\rangle \rightarrow |10\rangle$.

Here, the control qubit is $|1\rangle$. Hence, the state of the Target qubit flips from $|1\rangle$ to $|0\rangle$.

Input		Output	
$ A\rangle$	$ B\rangle$	$ A\rangle$	$ B \oplus A\rangle$
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$

The Truth Table of the operation of the CNOT gate is as follows.

CNOT -gate	
Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

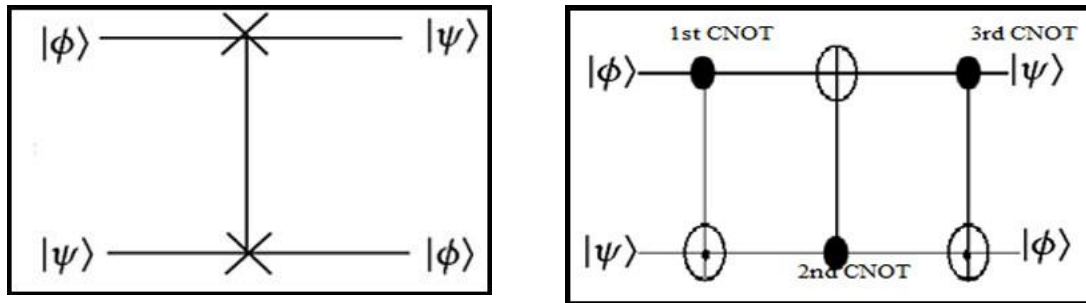
2. Swap Gate:

The SWAP gate is a two-qubit operation gate that swaps the state of the two qubits involved in the operation. It contains 3 CNOT gates.

The Matrix representation of the Swap Gate is as follows.

$$\text{SWAP Gate} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The schematic symbol of the swap gate circuit and its equivalent are shown in the figure below.



The swap gate is a combined circuit of 3 CNOT gates (1st CNOT gate, 2nd CNOT gate, and 3rd CNOT gate), and the overall effect is that two input qubits are swapped at the output. Gate representation is,

The Action of the swap gate is as follows.

- i. Operation of SWAP Gate for input $|00\rangle$: $|00\rangle \rightarrow |00\rangle$.
- ii. Operation of SWAP Gate for input $|01\rangle$: $|01\rangle \rightarrow |10\rangle$.
- iii. Operation of SWAP Gate for input $|10\rangle$: $|10\rangle \rightarrow |01\rangle$.
- iv. Operation of SWAP Gate for input $|11\rangle$: $|11\rangle \rightarrow |11\rangle$.

Truth table of the swap gate:

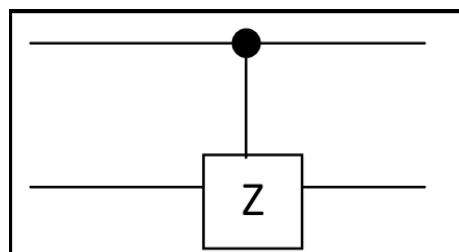
Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 10\rangle$
$ 10\rangle$	$ 01\rangle$
$ 11\rangle$	$ 11\rangle$

3. Controlled Z Gate:

In Controlled Z Gate, the operation of the Z Gate is controlled by a Control Qubit. If the control Qubit is $|A\rangle = |1\rangle$, then only the Z gate transforms the Target Qubit $|B\rangle$ as per the Pauli-Z operation. The action of Controlled Z-Gate could be specified by a matrix as follows.

$$\text{Controlled Z Gate} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

The controlled Z gate is represented as follows.



Action of controlled Z gate is

- i. Operation of the Gate for input $|00\rangle$: $|00\rangle \rightarrow |00\rangle$. Control qubit is $|0\rangle$, Z gate remains unchanged.
- ii. Operation of the Gate for input $|01\rangle$: $|01\rangle \rightarrow |01\rangle$. Control qubit is $|0\rangle$, Z gate remains unchanged.
- iii. Operation of the Gate for input $|10\rangle$: $|10\rangle \rightarrow |10\rangle$. Control qubit is $|1\rangle$, Z gate remains unchanged.
- iv. Operation of the Gate for input $|11\rangle$: $|11\rangle \rightarrow -|11\rangle$. The control qubit is $|1\rangle$, Z gate flips the state from $|1\rangle$ to $-|1\rangle$.

Truth table of the controlled Z gate:

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 10\rangle$
$ 11\rangle$	$- 11\rangle$

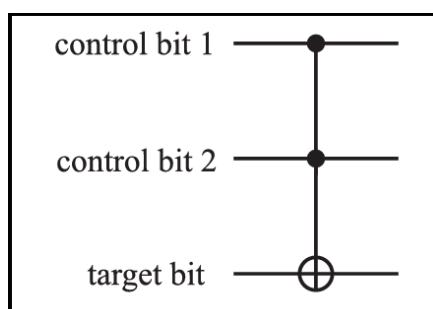
4. Toffoli Gate:

The Toffoli Gate is also known as the CCNOT Gate (Controlled-Controlled-Not). It has three inputs, out of which two are Control Qubits, and one is the Target Qubit. The Target Qubit flips only when both the Control Qubits are $|1\rangle$. The two Control Qubits are not altered during the operation.

The matrix representation of the Toffoli Gate is,

$$\text{Toffoli Gate} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Gate representation:



Truth Table of Toffoli Gate:

Input to the gate $ abc\rangle$	Output of the gate $ a'b'c'\rangle$
$ 100\rangle$	$ 100\rangle$
$ 001\rangle$	$ 001\rangle$
$ 010\rangle$	$ 010\rangle$
$ 011\rangle$	$ 011\rangle$
$ 101\rangle$	$ 101\rangle$
$ 110\rangle$	$ 111\rangle$
$ 111\rangle$	$ 110\rangle$

3.11. QUANTUM COMPUTING SYSTEM FOR INFORMATION PROCESSING

A quantum computing system processes information by exploiting the principles of quantum mechanics, offering capabilities far beyond classical systems.

- **Representation of Information:** Information is encoded in qubits, which can exist in superposition, allowing storage and manipulation of many states simultaneously.
- **Information Transformation:** Quantum operations are carried out using quantum gates (unitary transformations), which rotate and entangle qubits rather than just flipping states.
- **Parallelism:** Superposition enables quantum computers to evaluate multiple possibilities at once, providing massive parallelism.
- **Entanglement:** Strong correlations between qubits allow highly efficient information sharing and processing.
- **Measurement:** Extracting results involves measuring qubits, which collapses their states into definite outcomes, with probabilities linked to amplitudes.
- **Algorithms:** Quantum algorithms (e.g., Shor's for factoring, Grover's for searching) demonstrate clear speedups over classical methods for specific tasks.

Thus, quantum computing systems represent a new paradigm of information processing, leveraging the unique properties of quantum mechanics to solve problems in cryptography, optimization, machine learning, and beyond.

3.12. EVOLUTION OF QUANTUM SYSTEMS

The evolution of quantum systems describes how the state of a quantum system changes over time according to the principles of quantum mechanics.

- **Schrödinger Equation:** The time evolution of a quantum state $|\psi(t)\rangle$ is governed by the time-dependent Schrodinger equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

where H is the Hamiltonian (energy operator) of the system.

- **Unitary Evolution:** Solutions to the Schrodinger equation are expressed using a unitary operator $U(t)$: $|\psi(t)\rangle = U(t) |\psi(0)\rangle$, $U(t) = e^{-iHt/\hbar}$. Ensuring that the total probability remains conserved.
- **Quantum Gates:** In quantum computing, time evolution is simulated by unitary gates, which transform qubit states without loss of information.
- **Reversibility:** Since evolution is unitary, it is reversible until a measurement is performed.
- **Measurement Interruption:** When measured, the system undergoes a non-unitary collapse into one of its basis states, ending continuous evolution.

Thus, the evolution of quantum systems ensures predictable transformations under quantum operations, forming the backbone of quantum computing algorithms.

3.13. CHALLENGES AND ADVANTAGES OF QUANTUM COMPUTING OVER CLASSICAL COMPUTATION

Advantages:

- Parallelism through Superposition – Qubits can represent many states simultaneously, enabling massive parallel computation.
- Speedups in Algorithms – Algorithms like Shor's (factoring large numbers) and Grover's (database search) outperform their best-known classical counterparts.
- Cryptography and Security – Quantum key distribution (QKD) provides theoretically unbreakable communication security.
- Complex Simulations – Quantum computers can efficiently simulate quantum systems (molecules, materials) that are intractable for classical computers.
- Optimization and AI – Potential applications in optimization problems, machine learning, and big data analysis.

Challenges:

- Decoherence – Quantum states are fragile and easily disturbed by the environment, causing loss of information.
- Error Rates – Quantum operations have higher error probabilities compared to classical logic gates, requiring complex quantum error correction.

3.14. QUANTUM ALGORITHMS

- Quantum algorithms use superposition, entanglement, and interference to achieve faster solutions than classical methods. They operate on qubits through quantum gates, allowing parallel exploration of many states.
- The Deutsch–Jozsa Algorithm first showed quantum advantage in decision problems.
- Shor's Algorithm enables efficient integer factorization, challenging modern cryptography.
- Grover's Algorithm speeds up database search and optimization with quadratic efficiency.
- Overall, quantum algorithms promise breakthroughs in cryptography, AI, optimization, and scientific simulations.

3.15. DEUTSCH–JOZSA ALGORITHM:

The Deutsch-Jozsa algorithm efficiently determines if a "black box" function $f(x)$ is constant (always 0 or always 1) or balanced (1 for half inputs, 0 for the other half) using a single quantum query, unlike classical methods needing many.

Deutsch-Jozsa Algorithm: Step-by-Step:

The Problem: Given a function $f: \{0, 1\}^n \rightarrow \{0, 1\}$, determine if f is constant or balanced using a single query to f (as a quantum oracle U_f).

1. Initialization:

- Start with n input qubits in the state $|0\rangle^{\otimes n}$ and one ancillary qubit in $|1\rangle$.
- Initial State: $|\psi_0\rangle = |0\rangle^{\otimes n}|1\rangle$.

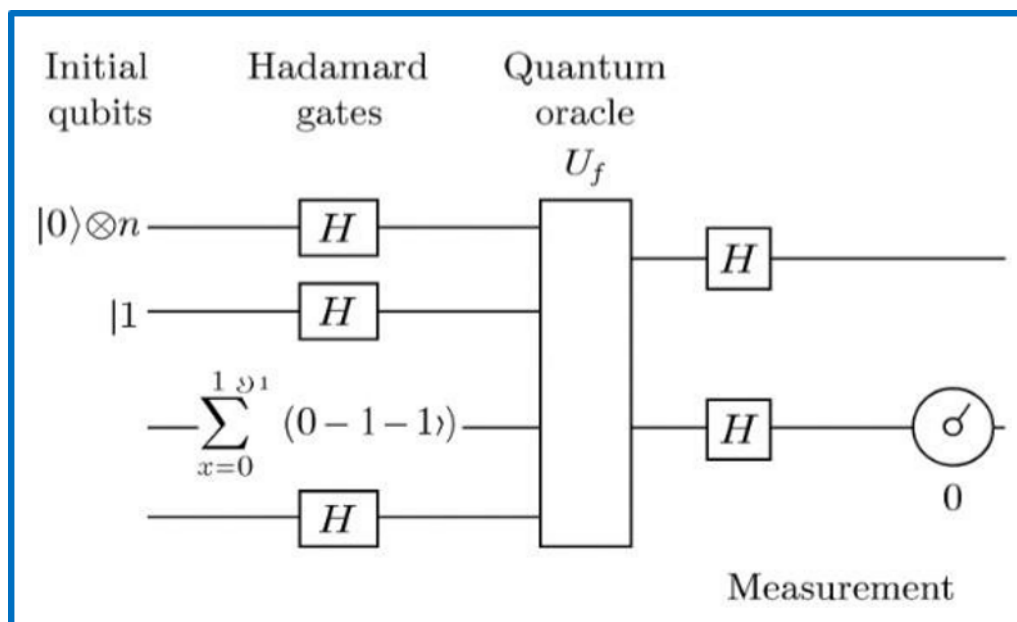
2. Superposition (Hadamard Gates):

- Apply Hadamard (H) gates to all n input qubits and the ancillary qubit.
- This creates an equal superposition of all possible inputs for the n qubits and puts the ancillary qubit in a special state.

- State after Hadamard: $|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

3. Oracle Query (U_f):

- Apply the quantum oracle U_f , which acts as $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ (XOR operation).
- This effectively applies a phase shift of $(-1)^{f(x)}$ to the input state $|x\rangle$.
- State after Oracle: $|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$.

**1. Interference (More Hadamards):**

- Apply Hadamard gates to the input register again.
- This causes destructive/constructive interference, amplifying one outcome and canceling others.

2. Measurement:

- Measure the n input qubits.
- **Result:** If the function f was constant, the measurement outcome would be $|0\rangle^{\otimes n}$ (all zeros).
- **Result:** If the function f was balanced, the measurement outcome will be any other state (at least one '1').

- This single query provides the answer, demonstrating a quantum advantage.

Quantum Circuit for Deutsch – Jozsa Algorithm

Significance:

The Deutsch–Jozsa algorithm is significant because it was the first quantum algorithm to demonstrate an exponential speedup over classical computation, requiring only a single query to determine whether a function is constant or balanced, whereas a classical computer may need up to $2^{n-1} + 1$ queries.

This marked a milestone in quantum computing by proving that quantum parallelism and interference can solve specific problems much more efficiently than classical approaches, laying the foundation for more advanced quantum algorithms such as Simon's and Shor's, and highlighting the true potential of quantum computers in outperforming classical systems.

3.16. SHOR'S ALGORITHM

Shor's Algorithm uses a quantum computer to find the period of a modular exponential function, which classically helps factor large numbers efficiently; its steps involve choosing a random number, using Quantum Phase Estimation (QPE) via Quantum Fourier Transform (QFT) to find the period r of $ax \pmod{N}$, and then using classical methods (continued fractions, GCD) to derive factors p, q from r and a , repeating if necessary.

Simplified Step-by-Step Breakdown:

- 1. Classical Setup:** Choose a composite number N to factor. Pick a random integer a such that $1 < a < N$. Calculate $g = \gcd(a, N)$ using the Euclidean algorithm. If $g > 1$, you've found a factor; if $g=1$, proceed to the quantum part.
- 2. Quantum Period Finding (QPE):**
 - ⇒ Initialize: Create two quantum registers: the first for input ($|x\rangle$) and the second for output ($|f(x)\rangle = |a^x \pmod{N}\rangle$).
 - ⇒ Apply Oracle: Apply the modular exponentiation function $Ua|x\rangle|0\rangle = |x\rangle|a^x \pmod{N}\rangle$ to get a state like $\sum x |x\rangle|a^x \pmod{N}\rangle$.
 - ⇒ Measure Output: Measure the second register to get a value c . The state collapses to $\sum k |x_0 + k_s\rangle|c\rangle$, where s is the period.
 - ⇒ Apply QFT: Apply the Quantum Fourier Transform to the first register to find the period s .
 - ⇒ Measure Input: Measure the first register to get a value m , which is related to s (specifically, $m \approx k \cdot \frac{2^t}{s}$ for some k).
- 3. Classical post-processing:**
 - ⇒ Use the measured m and the number of qubits t (from step 2) to estimate the fraction k/s .
 - ⇒ Employ the Continued Fractions algorithm to find potential values for the period s .

⇒ Check if $a^{s^2} \equiv -1 \pmod{N}$. If not, and s is even, calculate $\text{ged}(a^{s^2} - 1, N)$. This often reveals a non-trivial factor of N .

4. Repeat: If factors aren't found, choose a new a and restart.

This algorithm not only demonstrates the immense computational power of quantum systems but also motivates the development of post-quantum cryptography to secure future digital communication.

3.17. GROVER'S ALGORITHM

- Grover's Algorithm is a quantum search algorithm that provides a quadratic speedup over classical search methods.
- This algorithm is significant for applications in database searching, cryptanalysis, and optimization problems, demonstrating the practical computational advantage of quantum mechanics.
- In a classical system, searching for a particular item in an unsorted database of N entries requires, on average, $O(N)$ steps.
- Grover's algorithm, however, can find the desired item in about $O(\sqrt{N})$ steps using quantum parallelism and amplitude amplification.
- The algorithm works by initializing the system into a uniform superposition of all possible states, applying an oracle function that marks the correct solution by inverting its phase, and then amplifying its probability amplitude through the Grover iteration (involving inversion about the mean).
- After repeating this process approximately $\pi/4\sqrt{N}$ times, measuring the quantum state yields the correct solution with high probability.

Step-by-Step working flow of Grover's Algorithm:

5. Initialization:

- Start with n qubits in state $|0\rangle$.
- Apply Hadamard gates to create a uniform superposition of all $N = 2^n$ possible states, $|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$

6. Oracle Application (Marking the Solution):

- An oracle function identifies the correct solution $|w\rangle$.
- It flips the phase of the solution state:

$$O|x\rangle = \begin{cases} -|x\rangle & x = w \\ +|x\rangle & x \neq w \end{cases}$$

7. Amplitude Amplification (Grover Diffusion Operator):

- Apply the inversion about the mean step, which increases the probability amplitude of the correct solution while decreasing others.
- Mathematically: $D = 2|\psi\rangle\langle\psi| - I$

8. Grover Iteration:

- One Grover iteration = Oracle + Diffusion Operator.
- Repeat this about $\pi/4\sqrt{N}$ times.

9. Measurement:

- Measure the system.
- The state collapses with high probability to the correct solution $|w\rangle$.

The significance of Grover's Algorithm lies in its ability to provide a quadratic speedup for searching unsorted data. In classical computation, searching an unsorted database of N entries takes $O(N)$ steps on average, but Grover's Algorithm can accomplish the same task in only $O(\sqrt{N})$ steps using quantum parallelism and amplitude amplification.

Review Questions

Short Answer Questions

1. What is a Qubit?
2. Differentiate between a classical bit and a quantum bit.
3. Define Hilbert space in the context of quantum mechanics.
4. What is the Dirac Bra and Ket notation?
5. What does the Bloch sphere represent?
6. Define quantum entanglement.
7. What is a Hadamard gate?
8. Mention two key advantages of quantum computing over classical computing.
9. What is the main objective of Grover's algorithm?
10. State the primary application of Shor's algorithm.
11. Discuss the postulates of quantum mechanics?

Long Answer Questions

1. Explain the representation of qubit using Bloch sphere.
2. Compare classical bits and Qubits. Explain the state representation of single and multi-Qubit systems using superposition principles.
3. Discuss the fundamental single-Qubit and multi-Qubit quantum gates (Pauli-X, Y, Z, Hadamard, CNOT, Phase shift) along with their matrix representations and truth tables.
4. Explain the key phenomena of quantum dynamics: superposition, quantum measurement, state collapse, and quantum entanglement with appropriate mathematical representations.
5. Describe the Deutsch-Jozsa algorithm in detail. Explain how it demonstrates quantum speedup over classical algorithms.
6. Elucidate the difference between classical computing and quantum computing?
7. Discuss the primary challenges in physical realization of quantum computers, such as quantum decoherence and error correction.
8. Derive the matrix representation of a two-Qubit system state and explain how controlled operations (like CNOT) generate entangled states like Bell states.

9. Explain how quantum measurement alters the state of a quantum system, including projective measurements and expectation values in matrix form.
10. Provide a detailed comparative analysis between classical computing architectures and quantum computing architectures.
11. Write a note on evolution of quantum computing?
12. State Pauli matrices. Apply Pauli matrices on the states $|0\rangle$ and $|1\rangle$
13. Explain Hadamard gate. Apply Hadamard gate on the states $|0\rangle$ and $|1\rangle$

Objective Type Questions

Multiple Choice Questions

1. Quantum computing is based mainly on
 - A) Newtonian mechanics
 - B) Quantum mechanics
 - C) Classical thermodynamics
 - D) Fluid mechanics
2. The basic unit of quantum information is the
 - A) Bit
 - B) Byte
 - C) Qubit
 - D) Register
3. A classical bit can take
 - A) 0 or 1
 - B) Any complex value
 - C) Both 0 and 1 simultaneously
 - D) Any real number
4. A qubit can be in a superposition of
 - A) Only one state
 - B) $|0\rangle$ and $|1\rangle$
 - C) Three classical states
 - D) Infinite classical bits
5. Dirac notation consists of
 - A) Bragg and Miller notation
 - B) Bra and ket notation
 - C) Scalar and vector notation only
 - D) Boolean notation
6. The ket representing a quantum state is written as
 - A) $\langle\psi|$
 - B) $|\psi\rangle$
 - C) ψ^2
 - D) $[\psi]$
7. The bra corresponding to $|\psi\rangle$ is
 - A) $\psi|$
 - B) $\langle\psi|$
 - C) $|\psi\rangle$
 - D) ψ^2
8. The inner product of two quantum states gives a
 - A) Complex scalar
 - B) Crystal lattice
 - C) Classical bit
 - D) Photon frequency only
9. The state space of a quantum system is described mathematically by a
 - A) Hilbert space
 - B) Bravais lattice
 - C) Fermi surface
 - D) Crystal cell
10. The Bloch sphere is a geometrical representation of a
 - A) Single qubit
 - B) Crystal plane
 - C) Laser cavity
 - D) Fibre mode
11. Entanglement is a
 - A) Classical mechanical effect
 - B) Quantum correlation
 - C) Thermal effect
 - D) Crystal defect
12. Measurement of a qubit generally causes
 - A) State collapse to an outcome
 - B) Infinite amplification
 - C) No change ever
 - D) Creation of a crystal
13. Quantum gates are represented mathematically by
 - A) Unitary operations
 - B) Non-reversible scalar operations only
 - C) Bragg equations
 - D) Boolean truth tables only
14. The Hadamard gate is commonly used to create
 - A) Superposition
 - B) Decoherence
 - C) Magnetic saturation
 - D) Optical loss
15. The Pauli-X gate is analogous to a classical
 - A) NOT gate
 - B) AND gate

- C) OR gate
D) NAND gate
16. The Pauli-Z gate changes the
A) Phase of $|1\rangle$ relative to $|0\rangle$
B) Number of qubits
C) Crystal size
D) Laser wavelength only
17. CNOT is a
A) Controlled two-qubit gate
B) Single-qubit measurement
C) Classical memory
D) Fibre connector
18. Quantum evolution before measurement is generally
A) Unitary and reversible
B) Irreversible and random
C) Thermal only
D) Classical only
19. Decoherence is caused by
A) Interaction with the environment
B) Perfect isolation
C) Increasing crystal packing
D) Laser coherence
20. Shor's algorithm is important for
A) Integer factorization
B) Fibre attenuation
C) Crystal growth
D) Magnetic hysteresis
21. Grover's algorithm provides a speedup for
A) Unstructured search
B) X-ray diffraction
C) Laser pumping
D) Dielectric polarization
22. Deutsch–Jozsa determines whether a function is
A) Constant or balanced
B) Magnetic or dielectric
C) Direct or indirect
D) Linear or circular only
23. Quantum parallelism arises mainly from
A) Superposition
B) Hysteresis
C) Polarization
D) Refraction
24. A multiple-qubit system can represent a state in a space whose dimension grows as
A) n
B) 2^n
C) n^2 only
D) $\log n$
25. Quantum computing has potential applications in
A) Cryptography and simulation
B) Only mechanical engineering
C) Only crystal polishing
D) Only sound recording

Multiple Choice Questions

- The basic unit of quantum information is the _____.
- A classical bit can be either _____ or _____.
- A quantum bit is called a _____.
- A linear combination of $|0\rangle$ and $|1\rangle$ is called _____.
- The notation $|\psi\rangle$ is called a _____.
- The notation $\langle\psi|$ is called a _____.
- Bra and ket together form _____ notation.
- The mathematical vector space used for quantum states is called _____ space.
- The geometrical representation of a single qubit is the _____ sphere.
- A strong quantum correlation between particles is called _____.
- Measurement of a quantum state generally causes state _____.
- Quantum gates are represented by _____ matrices.
- The Hadamard gate is denoted by the letter _____.
- The Pauli-X gate is analogous to a classical _____ gate.
- The Pauli-Z gate changes the relative _____ of a qubit state.
- The controlled-NOT gate is abbreviated as _____.

17. Quantum evolution before measurement is _____.
18. Loss of quantum information through environmental interaction is called _____.
19. Shor's algorithm is used for integer _____.
20. Grover's algorithm is used for unstructured _____.
21. Deutsch–Jozsa distinguishes between constant and _____ functions.
22. Quantum parallelism is enabled by _____.
23. A system containing n qubits has a computational basis dimension of _____.
24. Quantum algorithms use superposition, entanglement and _____.
25. Quantum key distribution is commonly abbreviated as _____.

MRECW

MCQs: 1-B, 2-C, 3-A, 4-B, 5-B, 6-B, 7-B, 8-A, 9-A, 10-A, 11-B, 12-A, 13-A, 14-A, 15-A, 16-A, 17-A, 18-A, 19-A, 20-A, 21-A, 22-A, 23-A, 24-B, 25-A

Fill in the blanks: 1. qubit; 2. 0; 3. 1; 4. qubit; 5. superposition; 6. ket; 7. bra; 8. Dirac; 9. Hilbert; 10. Bloch; 11. entanglement; 12. collapse; 13. unitary; 14. H; 15. NOT; 16. phase; 17. CNOT; 18. unitary; 19. decoherence; 20. factoring; 21. search; 22. balanced; 23. superposition; 24. 2^n ; 25. interference; 26. QKD.

UNIT – 4 MAGNETIC & DIELECTRIC MATERIALS

Contents: - Magnetic Materials: Introduction to magnetic materials, origin of magnetic moment-classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, synthesis of ferrimagnetic materials using sol-gel method, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

Dielectric Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

4.1. INTRODUCTION TO MAGNETIC MATERIALS

Magnetism arises from the magnetic moment or magnetic dipole of the magnetic materials. Any material that can be magnetized by the application of external magnetic field is called magnetic material. Magnetic substances are the substances having the property of attracting the magnetic materials.

Basic Definitions:

Magnet: A piece of matter, which when suspended freely rests in a particular direction (North South) and which possesses a net magnetic moment and attracts ferrous bodies towards it, is called a magnet. This property of a magnet is called, magnetism.

Magnetic poles: The two points at the ends of a magnet at which magnetism is maximum, are called as magnetic poles. They are of two types, i) North pole ii) South pole.

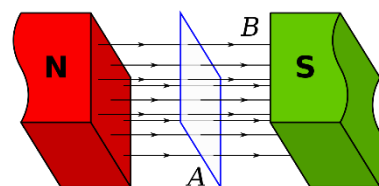
Magnetic Dipole: Two equal and opposite magnetic poles (north and south) separated by a small distance is called a magnetic dipole.

Magnetic dipole Moment (μ_m): It is defined as the product of the pole strength (m) and the distance between the two poles ($2l$) of the magnet.

$$\mu_m = m \times 2l, [\text{Ampere-metre}^2]$$

Magnetic induction or magnetic flux density (B): It is defined as the number of magnetic lines of force passing perpendicularly through a unit area.

$$B = \frac{\text{Magnetic flux}}{\text{area}} = \frac{\Phi}{A} [\text{Weber/metre}^2 \text{ or Tesla}]$$



Magnetic Field Intensity (H): Magnetic field intensity (H) at any point in the magnetic field is defined as the force experienced by a unit North pole placed at that point. The SI unit of magnetic field intensity is Ampere/meter (A/m).

The magnetic induction B due to magnetic field intensity H applied in vacuum is related by

$$B = \mu_0 H$$

where μ_0 is permeability of free space $= 4\pi \times 10^{-7} \text{ H/m}$

If the field is applied in a medium, the magnetic induction in the solid is given by

$$B = \mu H$$

where μ is permeability of the material in the medium.

$$\mu = \frac{B}{H}$$

Hence magnetic Permeability μ of any material is the ratio of the magnetic induction to the applied magnetic field intensity. The ratio of $\frac{\mu}{\mu_0}$ is called the relative permeability (μ_r).

$$\mu_r = \frac{\mu}{\mu_0}$$

Therefore,

$$B = \mu_0 \mu_r H$$

Magnetization (M): The magnetic dipole moment per unit volume of the material is called intensity of magnetization (I) or simply magnetization (M). The intensity of magnetization is directly related to the applied field H.

$$\text{Magnetization} = \frac{\text{Magnetic dipole moment}}{\text{volume of the material}} \text{ [Ampere / meter]}$$

Magnetic Susceptibility (χ): The magnetic susceptibility of a material is the ratio of the intensity of magnetization produced to the magnetic field intensity which produces the magnetization.

$$\chi = \frac{M}{H}$$

Relation between B, H and M:

We know,

$$B = \mu H$$

$$B = \mu_0 \mu_r H$$

$$B = \mu_0 \mu_r H + \mu_0 H - \mu_0 H$$

$$B = \mu_0 H + (\mu_r - 1) \mu_0 H$$

$$B = \mu_0 H + \mu_0 M \quad \text{where } M \text{ is magnetization} = H(\mu_r - 1)$$

$$B = \mu_0 (H + M)$$

The first term on the right side of above equation is due to external field. The second term is due to the magnetization.

4.2. ORIGIN OF MAGNETIC MOMENT

The magnetic properties of solids arise due to electrons undergoing different motions in the atoms, which give rise to magnetic dipole moments. These magnetic dipole moments are responsible for the magnetic properties of materials. In general, the magnetic dipole moment of the atom arises from **three** sources:

1. **The orbital motion of electrons:** The atom of any material consists of a central nucleus and the electrons move around the nucleus in specific orbits. Each electron orbit is equivalent to a tiny current loop and behaves as an elementary magnet having a magnetic dipole

moment. The total orbital magnetic moment of an atom is the sum of orbital magnetic moments of individual electrons.

2. **The electron spin:** Each electron is spinning about an axis through itself and this spin also gives rise to a magnetic dipole moment.
3. **The nuclear spin:** In addition to electronic contribution, nuclear spin also contributes to magnetic moment of atoms. The magnetic moment of the nucleus is about $1/2000$ of the magnetic moment of electron. Therefore, in studying magnetic properties of solids, the magnetic moment due to nuclear spin is neglected.

In general, the resulting magnetic moment of an atom is the sum of the orbital and spin magnetic moments of its electrons. The major contribution to atomic magnetic moment comes from the spin of unpaired valence electrons.

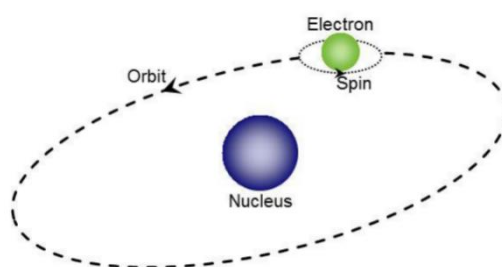


Figure 4.1: The orbit of a spinning electron about the nucleus of an atom.

4.3. CLASSIFICATION OF MAGNETIC MATERIALS

Materials can be classified as diamagnetic materials and a second group consists of paramagnetic, Ferro magnetic, anti-ferromagnetic and ferrimagnetic materials.

1. Diamagnetic Materials:

The materials with no permanent dipole moment are called diamagnetic materials. In these materials atoms consist of an even number of electrons. Thus, its electron orbits are closed or full, having two electrons or eight electrons. So, no permanent dipole moment exists in such materials.

Properties:

- P1 No permanent dipole moment.
- P2 No spontaneous magnetization.
- P3 Weakly magnetized in opposite direction to the applied field.
- P4 It repels the magnetic materials weakly. They do not allow magnetic lines of force to pass through it.
- P5 Susceptibility (χ) is small and negative. It is practically independent of temperature.
- P6 Relative permeability (μ_r) is small and less than one.

Eg: Examples: copper, gold, bismuth, mercury antimony etc.

2. Paramagnetic Materials:

Paramagnetic materials are substances that are weakly attracted to a magnetic field. This attraction occurs because they have unpaired electrons, which create tiny magnetic dipoles that

align with the applied magnetic field. However, this alignment is weak and temporary, meaning the material only exhibits magnetism while in the presence of an external magnetic field.

Properties:

- P1 Does not possess permanent dipole moment.
- P2 Net magnetization in any direction is zero in the absence of external field.
- P3 It attracts magnetic materials weakly. These materials allow magnetic lines of force.
- P4 Gets strongly magnetized.
- P5 Susceptibility is positive and small and depends on temperature.
- P6 Spin vectors are randomly oriented.

Eg: Oxygen, copper chloride, platinum,

3. Ferro magnetic materials:

Ferromagnetic materials are a class of materials which when placed in an external magnetic field become strongly magnetized in the direction of the field. Ferromagnetic materials exhibit spontaneous magnetization even in the absence of applied magnetic field. (Spontaneous magnetization in ferromagnetic materials refers to the inherent ability of these materials to develop a net magnetic moment even in the absence of an external magnetic field).

Properties:

- P1 Even in the absence of an external field, the dipoles are aligned in certain directions resulting in net magnet moment and hence they possess permanent dipole moment.
- P2 It strongly attracts the magnetic materials. Magnetic lines of force strongly prefer to go through these materials.
- P3 Above certain temperature known as Curie temperature, the ferromagnetic material gets converted to paramagnetic material.
- P4 Spin directions are parallel with equal magnitudes.
- P5 Exhibits hysteresis.
- P6 In ferromagnetic materials, dipoles are aligned uniformly in small regions known as domains.
- P7 These materials have high permeability, high susceptibility and strongly depends on temperature.

Eg: Iron, Cobalt, Nickel.

4. Anti-ferromagnetic Materials:

In these materials dipoles are aligned in parallel and anti-parallel directions in different domains. So, there is no net magnetic moment. In an external field they can be oriented in accordance with the field direction. Also, spin directions are anti-parallel. When the material is

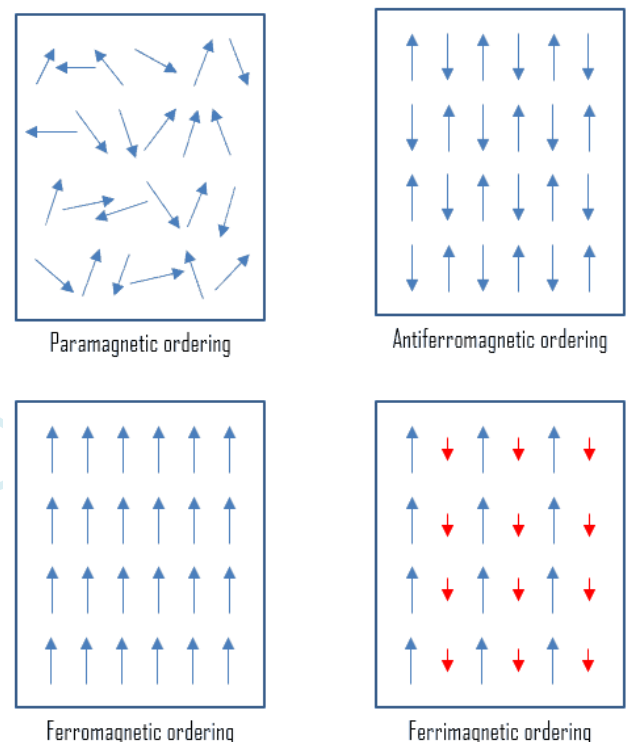


Figure 4.2: Magnetic moment alignment in different magnetic materials

placed in an external magnetic field and the temperature is 0 K, the magnetization is still zero. As the temperature is increased in the presence of an external field, the antiparallel arrangement of spins is slowly disturbed and magnetization (and hence the susceptibility) increases. The susceptibility reaches a maximum at Neel temperature, T_N .

Neel temperature: It is the critical temperature at which an antiferromagnetic material changes to a paramagnetic material.

Properties:

- P1 Low magnetic susceptibility.
- P2 These materials exhibit unique property like 'Neel temperature'.
- P3 No permanent magnetization when external field is removed.
- P4 Adjacent electron spins are anti-parallel with equal magnitudes, therefore there is no net magnetization.
- P5 Spin ordering means the magnetic moments are arranged in a specific pattern.
Examples: checker board or spiral patterns.

Eg: Chromium and metal oxides like nickel oxide, iron oxide etc.

5. Ferri Magnetic Materials or Ferrites

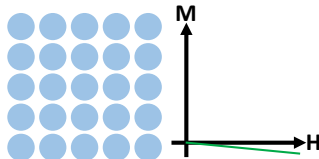
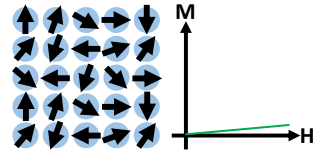
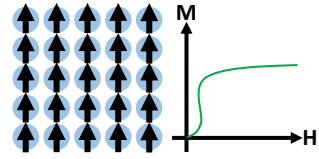
In ferrimagnetic materials, the electron spins are aligned in the same direction with unequal magnitudes, therefore possesses spontaneous magnetization like ferromagnetic materials. The susceptibility and permeability are > 1 and large.

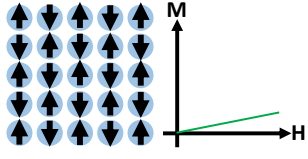
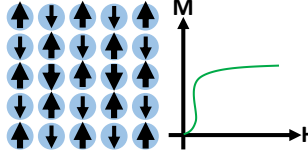
Properties:

- P1 Antiparallel alignment with unequal magnetic moments.
- P2 Spontaneous magnetization below Curie temperature due to unequal magnetic moments.
- P3 They possess high resistivity which is crucial for reducing energy loss.

Eg: Ferrites, Magnetite, Spinel ferrites, Garnets.

Classification of Magnetic materials (Summary)

Type of Magnetism	Susceptibility (χ)	Atomic Behavior	Magnetic Behavior	Examples
Diamagnetic	Negative and Small	Atoms possess no magnetic moment		Copper, gold, and water
Paramagnetic	Positive and Small	Atomic magnetic moments point in random directions		Aluminum, platinum, and titanium
Ferromagnetic	Positive and large	Parallel alignment and Same magnitudes (<i>Large net magnetization</i>)		Iron, nickel, cobalt, and steel.

Anti-Ferromagnetic	Positive and Small	Antiparallel alignment and Equal magnitudes (<i>Zero net magnetization</i>)		Manganese Oxide (MnO), Chromium (Cr), Nickel Oxide (NiO).
Ferrimagnetic	Positive and large	Antiparallel alignment and unequal magnitudes (<i>non-zero net magnetization</i>)		Magnetite (Fe_3O_4), Nickel Ferrite (NiFe_2O_4), Barium Ferrite ($\text{BaFe}_{12}\text{O}_{19}$)

4.4. HYSTERESIS LOOP

Hysteresis is a property of a ferromagnetic material defined as the lag in the changes of magnetization behind variations of the magnetic field. Usually, below the Ferro magnetic Curie temperature, the ferromagnetic material exhibit hysteresis, a phenomenon in which residual magnetization exists in the material even after the complete removal of the applied field. In simple we can define hysteresis as the lagging of magnetic induction B behind magnetizing field H .

Hysteresis Curve: The magnetic field induction B varies as a function of H along a closed loop, called the hysteresis loop. The curve (Fig. 4.3) begins at the origin O . As H is increased, the field B begins to increase slowly, then more rapidly and finally attaining a saturation value and becoming independent of H . The maximum value of B is the saturation flux density B_s and the corresponding magnetization is the saturation magnetization M_s . If H is now decreased, B also decreases but following a path AC instead of the original path AO (Fig. 4.3). Thus, B lags behind H . When H becomes zero, B does not become zero but has a value equal to OC (B_r). This magnetic flux density remaining in the materials is called the residual magnetism. It indicates that the material remains magnetized even in the absence of an external applied field H . The power of retaining the magnetism is called the retentivity or remanence of the material. Thus, the retentivity of a material is a measure of the magnetic flux density remaining in the material when the magnetizing field is removed.

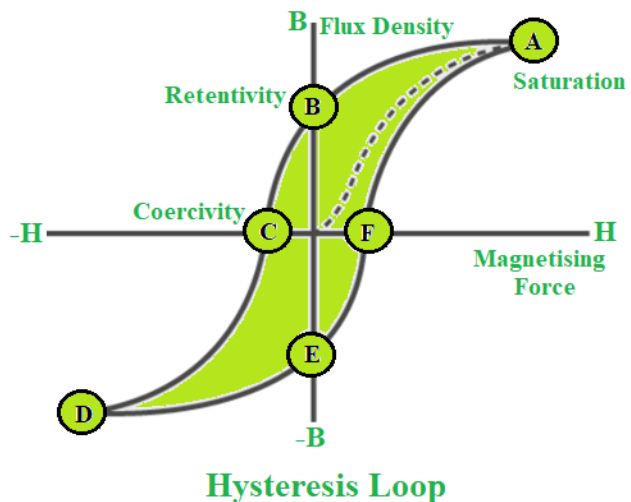


Figure 4.3: Hysteresis Curve / Hysteresis loop.

If the magnetic field H is now increased in the reverse direction, the value of B decreases along the path CD . It becomes zero, when H attains a value equal to OD . It means that to reduce magnetic induction within the material to zero, a field of magnitude H_c must be externally applied in a direction opposite to that of the original magnetizing field. H_c is called the

coercivity or the coercive force. Thus, coercivity is a measure of the magnetic field strength required to destroy the residual magnetism in the material. As the applied field H is increased further in the negative direction, saturation is ultimately reached in the reverse direction (point E on Figure) This creates a loop and this loop is known as hysteresis loop. Area of the hysteresis loop gives the energy loss per cycle of magnetization.

4.5. WEISS DOMAIN THEORY OF FERROMAGNETISM

Domain theory was proposed and explained by Weiss in 1907, it explains the hysteresis and the properties of ferromagnetic materials.

Postulates of domain theory:

- A ferromagnetic material is divided into a large number of small regions called domains; each domain size is $\sim 10^{-6}$ m.
- In each domain, the magnetic moments are aligned in the same direction,
- In the absence of an external magnetic field, these domains are randomly oriented, resulting in a net magnetization of zero.
- However, when an external magnetic field is applied, the domains tend to align with the field, leading to a strong magnetization or spontaneous magnetization of the material because in ferromagnetic materials there exist a special type of energy known as exchange energy¹ between the neighboring atoms.
- When a magnetic field is applied there are two processes that take place, 1) By the motion of domain walls and 2) By the rotation of domains.

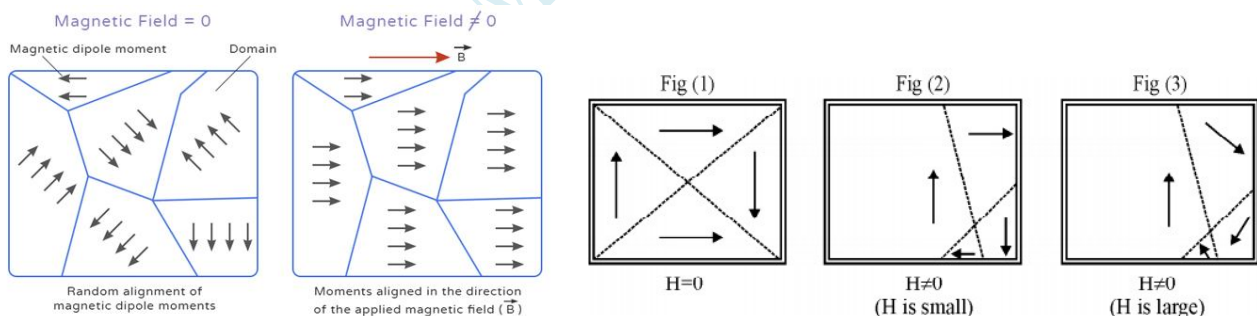


Figure 4.4: Domain orientation in ferromagnetic materials.

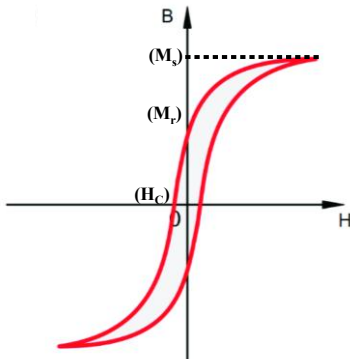
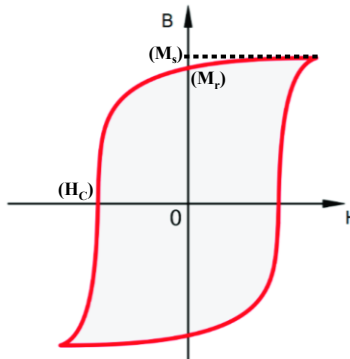
1. By motion of domain walls: The volume of the domain in which dipoles are favorably oriented with the field direction increases at the cost of other domains having dipole orientation not in the direction of the field.
2. By rotation of the domains: If the applied field is strong enough rotation of the direction of magnetization occurs in the direction of the field.

4.6. SOFT AND HARD MAGNETIC MATERIALS

The magnetic materials are classified into *soft* and *hard* categories based on how easily

¹Exchange energy is the energy required in assembling the atomic dipoles in to one domain and this is stored as potential energy in the system.

they magnetize and demagnetize. The key differences lie in coercivity, hysteresis loop area, and remanence.

S. No.	Soft Magnetic Materials	Hard Magnetic Materials
1	Soft magnetic materials are those materials which have a smaller area enclosed by their Hysteresis loop.	Hard magnetic materials are those materials which have a larger area enclosed by their Hysteresis loop
2	They have low remnant magnetization (M_r)	They have high remnant magnetization (M_r)
3	They have low coercivity (H_C)	They have high coercivity (H_C)
4	They have high initial permeability	They have low initial permeability
5	Hysteresis loss is less	Hysteresis loss is higher
6	Used in Transformer cores, motor armatures, inductors, and temporary electromagnets.	Used in Permanent magnets for electric motors, loudspeakers, refrigerator magnets, and magnetic data storage.
7	<u>Eg:</u> Silicon steel, nickel-iron alloys (permalloy), and soft ferrites.	<u>Eg:</u> Alnico, carbon steel, and rare-earth magnets like neodymium-iron-boron.
Hysteresis Loop		

4.7. SYNTHESIS OF FERRIMAGNETIC MATERIALS (SOL-GEL METHOD)

Ferrimagnetic materials are magnetic solids in which the magnetic moments of neighboring ions are aligned antiparallel but have unequal magnitudes. Consequently, the opposing magnetic moments do not completely cancel, producing a net spontaneous magnetization ($M \neq 0$).

Ferrites are the most widely studied ferrimagnetic materials. They are generally ceramic compounds containing iron oxide and one or more additional metallic elements. A common class is the spinel ferrite, represented by:



where M may be manganese (Mn), nickel (Ni), cobalt (Co), zinc (Zn), magnesium (Mg) etc.

Examples include:

- ⇒ MnFe_2O_4 : Manganese ferrite
- ⇒ NiFe_2O_4 : Nickel ferrite
- ⇒ CoFe_2O_4 : Cobalt ferrite
- ⇒ ZnFe_2O_4 : Zinc ferrite
- ⇒ MgFe_2O_4 : Magnesium ferrite

THE SOL–GEL METHOD:

The sol–gel method is a wet-chemical synthesis technique in which molecular or ionic precursors are converted into a solid oxide through the formation of a colloidal solution called a sol (solution) and a three-dimensional network called a gel (gelatin).

A sol is a stable suspension containing very small particles or polymeric species. During gelation, these species link together to form a continuous three-dimensional network. Drying removes the solvent, and subsequent heat treatment converts the dried gel into a crystalline ferrimagnetic oxide.

The general sequence is:

Metal precursors → Sol → Gel → Dry gel → Calcined ferrite

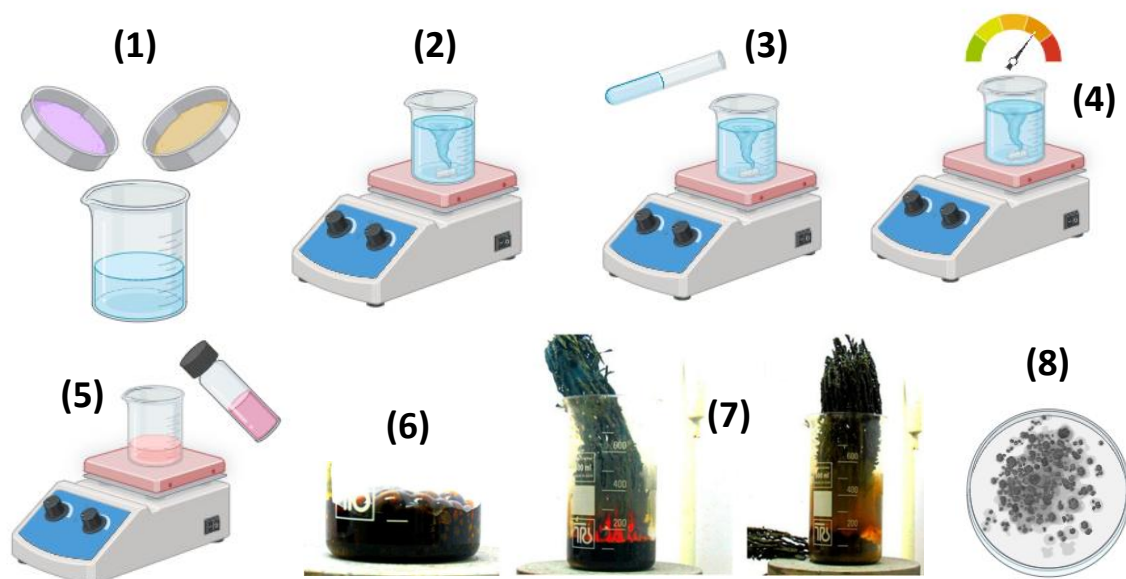


Figure 4.5: Solid state route method. Step 1: The nitrate precursors were dissolved in deionized water, creating a solution. Step 2: The solution beaker was placed on a magnetic stirrer with a hot plate, using a magnetic bead for stirring. Step 3: Citric acid was added as a chelating agent to achieve a homogeneous mixture at 80 °C. Step 4: The pH was adjusted to 6-7 with ammonia hydroxide. Step 5: Ethylene glycol was incorporated as a gelatin agent to aid gel formation. Step 6: The viscous gel began combustion, and Step 7: transformed into a gray porous mass in the beaker. Step 7: Black colored ferrite nanoparticles formed.

Key Steps of the Process:

Sol Preparation: Dissolve a metal alkoxide or salt precursor in water or alcohol to form a stable liquid suspension.

Hydrolysis and Condensation: Water replaces alkoxy groups and links molecules together to form metal-oxygen-metal bonds.

Gelation: The liquid solution turns into a sticky, semi-rigid 3D network or gel.

Aging and Drying: The gel sits to strengthen its network, then undergoes drying to remove trapped liquid (forming a xerogel or aerogel).

Calcination: High-heat treatment burns off remaining organics and locks in the final crystal structure.

The method is particularly useful for ferrite nanoparticles because it provides good control over chemical composition, homogeneity, particle size, and microstructure. It can also produce ferrite powders at lower processing temperatures than many conventional ceramic routes.

4.8. APPLICATIONS OF MAGNETIC MATERIALS

- ❖ Magnetic materials are used to operate such things as electrical motors, generators, and transformers.
- ❖ Much of the data storage technology (computer hard disc etc.) is based on magnetic properties.
- ❖ Magnetic properties are also used in loudspeakers, telephones, CD players, TV and Video recorders can also be viewed as magnetic materials.
- ❖ Magnetic materials are such as iron oxide are used to bind DNA molecules, cells and proteins.
- ✚ Diamagnetic materials: Which are used for a magnetic levitation where an object will be made to float in air above a strong magnet.

✚ Ferromagnetic materials: Materials are used in magnetic recording devices such as for cassette tapes, floppy discs for computers and the magnetic stripe on the back of credit cards.

⇒ Soft magnetic materials:

- Soft magnetic materials are mainly used in electromagnetic machinery and transformer cores in AC appliances.
- They are also used in switching circuits microwave isolators and matrix storage of computers.

⇒ Hard magnetic materials:

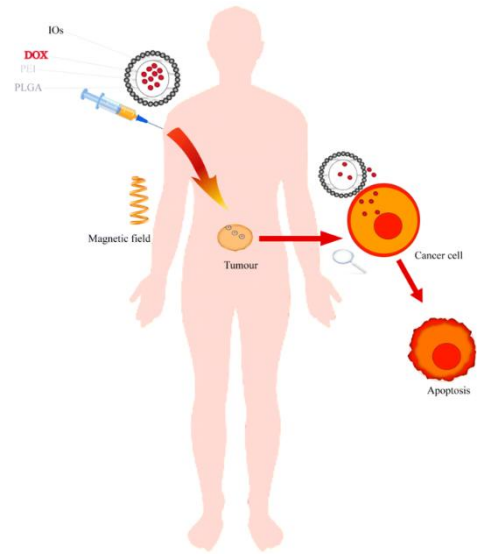
- Used in the production of magnets which are used in magnetic detectors, microphones, flux meters, voltage regulators, damping devices, magnetic separators and in making permanent magnets.
- Hard ferrites.

4.8.1. Magnetic hyperthermia for cancer treatment:

Magnetic hyperthermia is an experimental cancer treatment where magnetic nanoparticles (usually iron oxide) are delivered to a tumor and exposed to an alternating magnetic field. The particles absorb the electromagnetic energy and convert it into localized heat, raising tissue temperatures to 40–45°C to kill cancer cells or sensitize them to standard

therapies like radiotherapy and chemotherapy.

- Magnetic hyperthermia therapy (MHT) relies entirely on the unique physics of magnetic materials at the nanoscale.
- The heating arises from magnetic relaxation processes at the nanoscale. In an AMF, the magnetic moments of the nanoparticles repeatedly reorient and the particles physically rotate in the surrounding fluid. These processes dissipate energy as heat.
- Therefore, the injected ferromagnetic nanoparticles convert the magnetic energy into thermal energy (heat).
- The local temperature of the surrounding microenvironment rises rapidly, leading to the targeted destruction of localized cells or materials.



4.8.2. Magnets for EV:

Magnets are the foundational powerhouse of the modern electric vehicle (EV) revolution, primarily driving the electric motors that propel the vehicle. In a standard EV drivetrain, the traction motor relies heavily on permanent magnets embedded in the rotor. When an alternating electrical current passes through the stator coils, it generates a rotating electromagnetic field that interacts with these fixed magnets, producing the torque needed to rotate the wheels.

Key Types of Magnets Used in EVs: Engineers balance magnetic strength, heat tolerance, and weight when selecting materials:

Neodymium-Iron-Boron (NdFeB): Known as "Neo" magnets, these dominate over 75% of the global EV motor market. They offer the highest known magnetic strength-to-weight ratio (BH_{max}), allowing for incredibly small, lightweight, and hyper-efficient motors.

Samarium-Cobalt (SmCo): While slightly weaker than neodymium, SmCo possesses exceptional thermal stability. It is used selectively in performance vehicles where motors operate under extreme temperatures.

Ferrite Magnets: Made from iron oxide, these are significantly cheaper and highly abundant but have much lower energy density. They are reserved for lower-cost EVs or minor vehicle components like secondary actuators.

4.8.3. Giant Magneto Resistance (GMR) device:

Giant Magnetoresistance (GMR) is a quantum mechanical effect observed in thin-film structures composed of alternating ferromagnetic and non-magnetic conductive layers. When a magnetic field is applied to these structures, their electrical resistance changes drastically. Discovered independently by Albert Fert and Peter Grünberg in 1988 (earning them the 2007 Nobel Prize in Physics), GMR marked the birth of spintronics, a field of electronics that utilizes

the intrinsic spin of electrons in addition to their fundamental electrical charge.

Construction of a GMR Device:

A practical GMR device is typically constructed using a specialized multilayer geometry known as a Spin Valve. It is engineered by depositing ultra-thin atomic layers (ranging from 1 to a few nanometers thick) using high-precision techniques like sputtering or Molecular Beam Epitaxy (MBE).

The basic stack structure comprises four distinct layers:

Pinned (Fixed) Ferromagnetic Layer:

A layer made of magnetic material (e.g., Cobalt, Iron, or Permalloy) whose magnetic orientation is structurally locked or "pinned" in one single direction.

Antiferromagnetic Layer (Pinning Layer): A layer placed directly adjacent to the pinned layer (often made of Iron-Manganese or Platinum-Manganese). It locks the fixed layer's magnetic direction through a phenomenon known as exchange bias.

Non-Magnetic Spacer Layer: A highly conductive, non-magnetic metallic layer (typically Copper, Cu) that physically separates the two magnetic layers. Its thickness must be precisely tuned to prevent direct magnetic coupling between them.

Free Ferromagnetic Layer: A second magnetic layer whose magnetization orientation is free to rotate and align itself with the direction of any external, ambient magnetic field.'

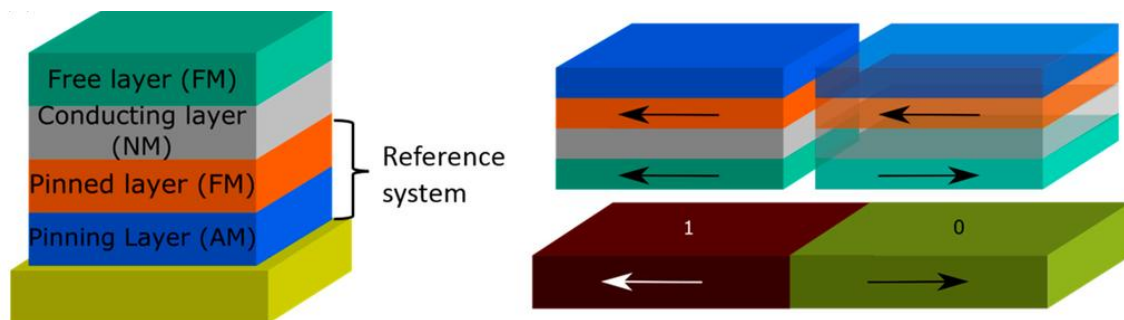


Figure 4.6: Schematic setup of the stack configuration of a GMR spin-valve sensor.

Working Principle

The fundamental physics governing a GMR device relies on spin-dependent electron scattering. Electrons possess an intrinsic quantum property called spin, which can be oriented either "spin-up" or "spin-down". When an electron passes through a ferromagnetic material, it moves with minimal resistance if its spin matches the material's magnetization direction; if its spin opposes the magnetization direction, it experiences heavy scattering, creating high electrical resistance.

Applications: Hard Disk Drive (HDD) Read Heads, Magnetoresistive Random Access Memory (MRAM), Automotive Sensors (ABS), Bio-Sensors (targeted medicine).

4.9. INTRODUCTION TO DIELECTRIC MATERIALS

A dielectric material is an electrical insulator that does not allow current to flow, but it can store electrical energy when exposed to an electric field. When you place a dielectric in an electric field, its internal charges shift slightly, a process known as polarization.

Eg: Paper, wood, plastic, glass, rubber, mica, air, asbestos, porcelain, ceramics, etc.

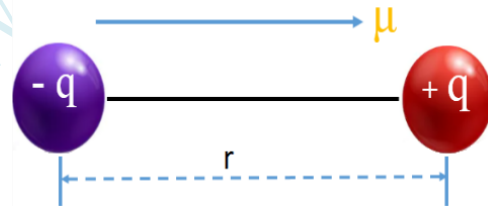
Permittivity (ϵ): It is a measure of how easily a material can be polarized by an electric field.

Dielectric constant: It is the ratio between the permittivity of the medium and the permittivity of the free space ($\epsilon_r = \epsilon/\epsilon_0$). It is a measure of polarization in the dielectric material, where $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

Polarization: The process of losing the neutrality of an atom or molecule and the formation of positive and negative charge centers due to the application of an external field is called polarization.

Dipole: When two equal and opposite charges are separated by a small distance, it constitutes a dipole.

Dipole moment (μ): It is the product of the charge (q) and the distance between the charge centers (r). It is a vector quantity pointing from a negative charge towards positive charge. The dipole moment is defined as, (μ) = $q \times r$. Its unit is Coulomb – meter (C-m) or debye (D). $1 \text{ D} = 3.3 \times 10^{-30} \text{ C-m}$.



Polarizability (α): Induced dipole moment ' μ ' is directly proportional to the strength of the applied field ' E ' ($\mu = \alpha E$), where the constant of proportionality ' α ' is known as the Polarizability. So, Polarizability ' α ' can be defined as the induced dipole moment per unit electric field.

Polarization vector (P): It is defined as the dipole moment per unit volume of the dielectric material ($P = N\mu$). The unit of polarization is Cm^{-2} . It is a vector quantity and pointing from negative charge to positive charge.

Electric displacement (D): It is also known as the electric flux density and defined as, $D = \epsilon E = \epsilon_r \epsilon_0 E$; D and P are related as: $D = \epsilon_0 E + P$ and $D = \epsilon_r \epsilon_0 E$.

Thus, $\epsilon_r \epsilon_0 E = \epsilon_0 E + P$ or $P = \epsilon_0 (\epsilon_r - 1) E$.

Electric susceptibility (χ_e): It is a dimensionless number that shows how easily a material gets polarized when an external electric field is applied. Polarisation is directly proportional to the direction of applied electric field. Hence, the polarization vector can be written as

$$P = \epsilon_0 \chi_e E$$

where χ_e is the electric susceptibility. Therefore,

$$\chi_e = \frac{P}{\epsilon_0 E} = \frac{\epsilon_0 (\epsilon_r - 1) E}{\epsilon_0 E} = \epsilon_r - 1$$

4.10. TYPES OF POLARIZATION

Dielectric polarization happens when the tiny charges inside a material shift (realign) because of an electric field. This alignment reduces the overall internal electric field, enabling the capacitor to store more electrical charge at the same voltage, effectively increasing its capacitance. Polarization in dielectrics mainly categorized into four types: 1. electronic, 2. ionic, 3. orientational, and 4. space charge.

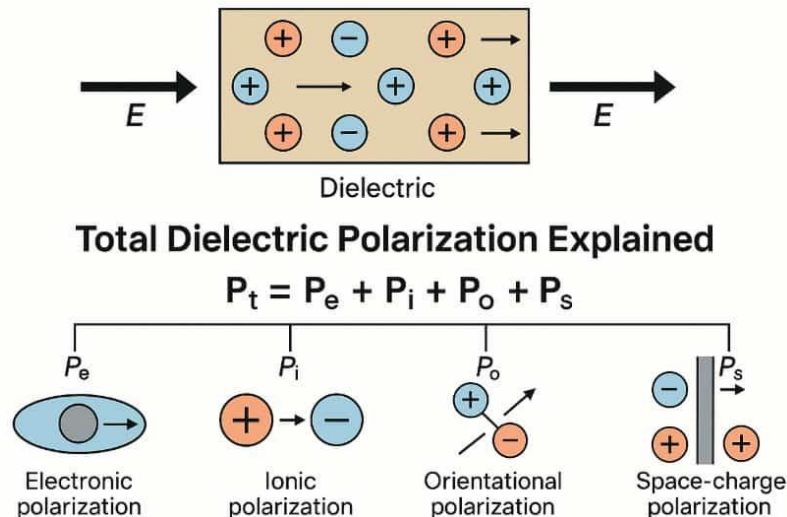


Figure 4.8: Total Polarization Diagram.

1. Electronic Polarization:

- This effect occurs in all atoms under the application of an electric field.
- The nucleus of the atom and the center of its electron cloud shift away from each other, creating a tiny dipole with very small polarization effect.
- This kind of shift will produce an electric dipole within the atom i.e. dipole moment is proportional to the magnitude of field strength.
- Electronic polarization occurs in monoatomic gases at optical frequency of 10^{15} Hz, and it is independent of temperature.
- Electronic polarization takes place in time period of 10^{-14} to 10^{-15} s

⇒ Electronic polarizability $(\alpha_e) = 4\pi\epsilon_0 R^3$

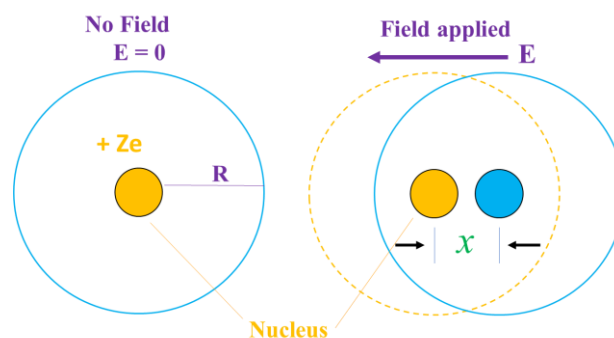


Figure 4.9: Displacement of nucleus in an atom when placed in an electric field E.

2. Ionic Polarization:

- Ionic Polarization occurs due to relative displacements between positive and negative ions in an ionic crystal.
- In ionic solids such as ceramic materials, the ions are symmetrically arranged in a crystal lattice with a net zero polarization.
- Once an electric field is applied, the cations and anions are attracted to opposite directions. This creates a relatively large ionic displacement (compared to electronic displacement), which can give rise to high dielectric constants in ceramics popularly used in capacitors.
- The displacement is independent of temperature.
- Ionic polarization takes place in time period of 10^{-11} to 10^{-14} s.
- Electronic polarizability $(\alpha_i) = \frac{e^2}{\omega_0^2} \left[\frac{1}{M} + \frac{1}{m} \right]$, where ω_0 be the angular frequency, M and m be the masses of cation and anion.

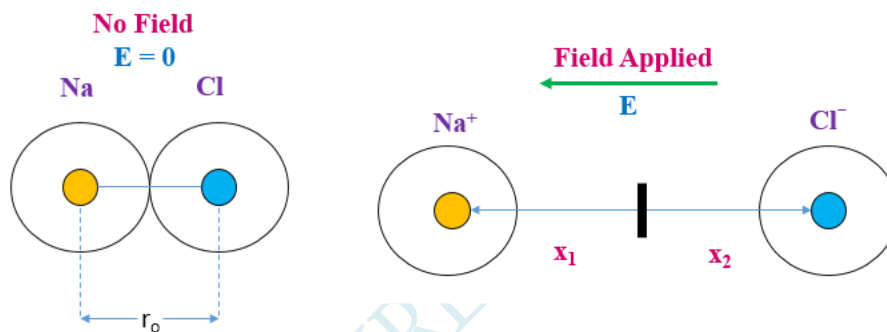


Figure 4.10: Displacement of ions when placed in an electric field E.

3. Orientational Polarization:

- Certain solids have permanent molecular dipoles that, under an electric field, rotate themselves in the direction of the applied field, creating a net average dipole moment per molecule.
- The process of orientation of permanent dipole moments along the axis of applied electric field is called *orientational polarization*.
- Orientation polarization built-up time is of the order of 10^{-10} s or more.

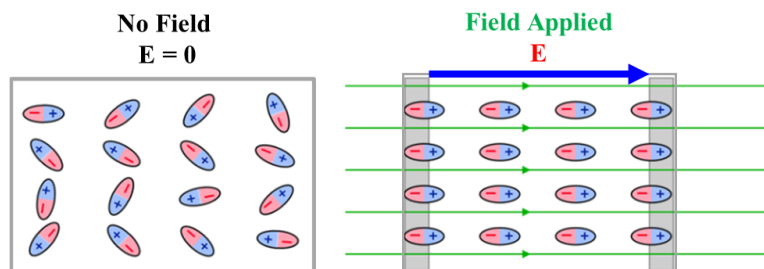


Figure 4.11: Orientation polarization.

- Orientational polarizability, $\alpha_o = \frac{\mu^2}{3KT}$, where T is absolute temperature and μ is

dipole moment.

4. Space Charge (Interfacial) Polarization:

- Space charge polarization arises due to the accumulation of charges at the electrodes or at the interphase in the multiphase material.
- The ions diffuse over appreciable distance in response to the applied field, gives rise to a redistribution of charges in the dielectric medium. This type is known as Space-charge polarization.
- The space-charge polarization is not a significant factor in most common dielectrics. Hence, we take the space-charge polarizability as $\alpha_s = 0$.

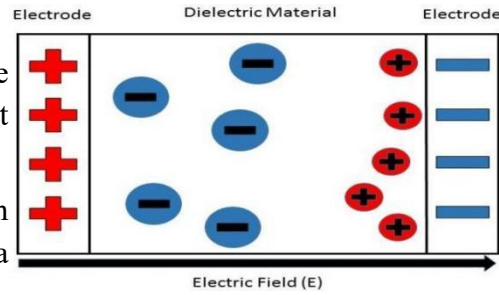


Figure 4.12: Space charge polarization

Total Polarizability:

The total polarizability (α) is the sum of four main types: electronic, ionic, orientational, and space-charge polarizability.

$$\alpha = \alpha_e + \alpha_i + \alpha_o + \alpha_s$$

$$\alpha = 4\pi\epsilon_o R^3 + \frac{e^2}{\omega_o^2} \left[\frac{1}{M} + \frac{1}{m} \right] + \frac{\mu^2}{3KT} + 0$$

$$\alpha = 4\pi\epsilon_o R^3 + \frac{e^2}{\omega_o^2} \left[\frac{1}{M} + \frac{1}{m} \right] + \frac{\mu^2}{3KT}$$

We have total polarization $P = Ne\alpha$, then,

$$P = Ne \left[4\pi\epsilon_o R^3 + \frac{e^2}{\omega_o^2} \left[\frac{1}{M} + \frac{1}{m} \right] + \frac{\mu^2}{3KT} \right]$$

This equation is called as Langevin – Debye equation. The Langevin – Debye equation from total polarizability α due to all types of polarizations.

4.11. FERRO, PIEZO, PYRO ELECTRIC MATERIALS

Ferroelectric, piezoelectric, and pyroelectric materials are specialized groups of dielectrics that link electrical properties with mechanical stress, temperature changes, or external electric fields. They play vital roles in modern sensors, actuators, and memory system.

Ferroelectric Materials:

- Ferroelectric materials are a special class of dielectric materials that possess spontaneous electric polarization².
- This polarization arises due to the asymmetric arrangement of positive and negative ions inside the crystal.
- A key characteristic of ferroelectrics is that their polarization can be reversed by applying an external electric field, a property known as ferroelectricity.

² They remain electrically polarized even without any external electric field.

- These materials typically exhibit a hysteresis loop in their polarization-electric field (P-E) relationship, see figure 4.11.
- Ferroelectric materials are widely used in memory devices (FeRAM), sensors, actuators, capacitors, and nonlinear optics.
- Eg: Barium Titanate (BaTiO_3) and Rochelle salt.
- The dielectric constant of Ferroelectric materials is very high and is of the order of 1000 to 10,000. The dielectric constant (ϵ) changes with temperature (T) according to Curie-Weiss law given by, $\epsilon = \frac{C}{T - T_C}$ ($T > T_C$), where C is called the Curie's constant and T_C is the Curie temperature³.

Applications of Ferro electric materials:

- ✚ BaTiO_3 based Ferro electric ceramics are used in the manufacture of capacitors to store electric charge in electrical/electronic circuitry.
- ✚ *Ferroelectric memories* - Ferroelectric materials polarize on cooling below T_C . Data is stored by localized polarization switching in the microscopic regions of ferroelectric thin film in FRAMs [Ferroelectric Random Access Memory⁴].
- ✚ Ferro electric semiconductors such as $\text{BaTiO}_3\text{-SrTiO}_3$, $\text{BaTiO}_3\text{-PbTiO}_3$ and $\text{SrTiO}_3\text{-PbTiO}_3$ are used to make posistors⁵.

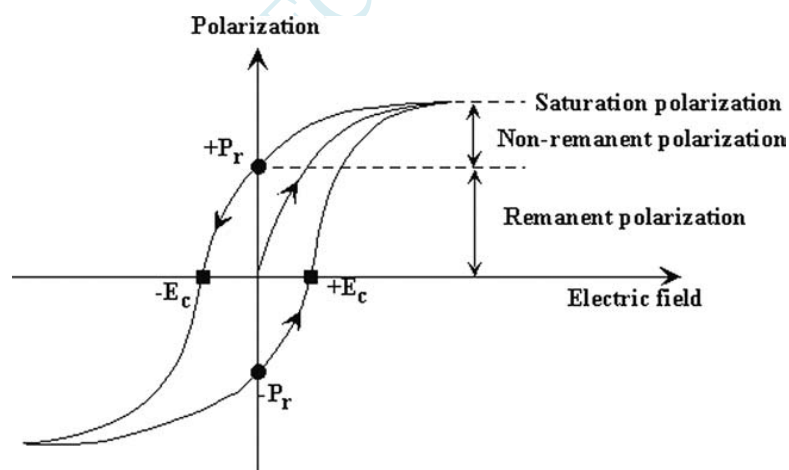


Figure 4.13: P-E loop.

Piezoelectric Materials:

- Piezoelectric materials generate electric charges on their surfaces when subjected to mechanical stress such as compression, stretching, or bending.
- This phenomenon is called the piezoelectric effect. Conversely, these materials also deform when an electric field is applied-known as the converse piezoelectric effect.
- Piezoelectric strains are very small, but the resulting electric fields can be quite large. This

³ Curie temperature is the specific point where a ferroelectric material changes into a paraelectric material.

⁴ A type of non-volatile memory which uses a thin ferroelectric material layer that changes polarity when an electric current is applied. When the current is shut off, the layer retains that last polarity and the chip holds the data.

⁵ Positive temperature coefficient thermistor; used for circuit protection and temperature sensing applications.

effect occurs in crystals that lack a center of symmetry in their structure.

- Piezoelectric materials are used in microphones, ultrasound transducers, sensors, actuators, quartz watches, and energy harvesting devices.
- Eg: Quartz (SiO_2), Zinc Oxide (ZnO) and PVDF (Polyvinylidene Fluoride).

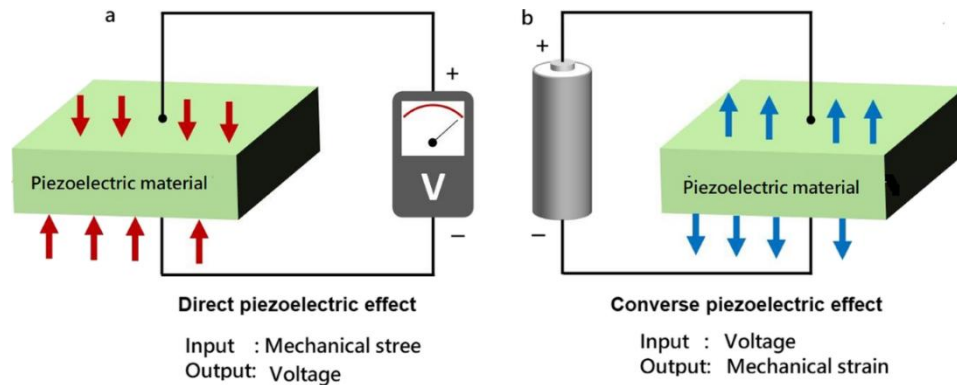


Figure 4.14: Illustration of direct and converse piezoelectric effect.

Applications of Piezo electric effect:

1. Sensors and Actuators

- ✚ *Pressure Sensors:* Used in automotive engines, industrial machinery, and weather instruments to measure pressure changes.
- ✚ *Accelerometers:* Employed in mobile phones, aircraft, and seismic instruments to measure acceleration and vibrations.
- ✚ *Force Sensors:* Found in robotics, medical devices, and weighing scales.
- ✚ *Ultrasonic Transducers:* Used in medical imaging (e.g., ultrasound), cleaning devices, and sonar systems.

2. Medical Devices

- ✚ *Ultrasound Imaging:* Piezoelectric crystals generate high-frequency sound waves for non-invasive imaging of internal body structures.
- ✚ *Lithotripsy:* Used to break kidney stones using focused shockwaves.
- ✚ *Surgical Tools:* Devices like scalpels and drills use ultrasonic vibrations for precision.

3. Consumer Electronics

- ✚ *Piezo Buzzers and Speakers:* Found in alarms, mobile phones, and musical greeting cards to generate sound.
- ✚ *Microphones:* Convert sound waves into electrical signals.
- ✚ *Igniters:* Used in lighters and gas stoves to generate a spark for ignition.

Pyroelectric Materials:

- Pyroelectric materials develop an electric potential (voltage) when their temperature changes.
- This happens because heating or cooling causes slight shifts in the positions of ions inside the crystal structure, leading to a change in the material's electric polarization.

- Since polarization varies with temperature, a voltage is produced across the material when the temperature is not constant.
- Pyroelectric materials are naturally polarized and form a subset of piezoelectric materials.
- They are commonly used in infrared sensors, motion detectors, thermal imaging devices, and fire alarms.
- Eg: Ceramics (LiNbO_3 & LiTaO_3), Synthetic polymers and PZT-based pyroelectric ceramics.

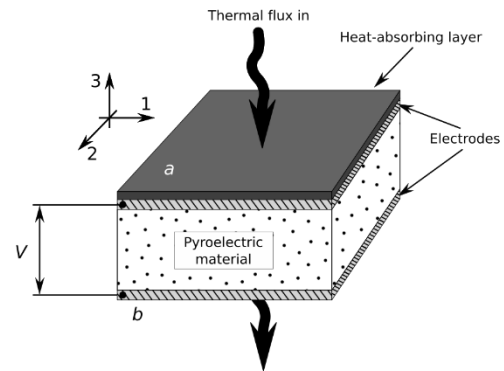


Figure 4.15: Illustration of Pyroelectric material under thermal flux.

Applications of pyroelectric materials:

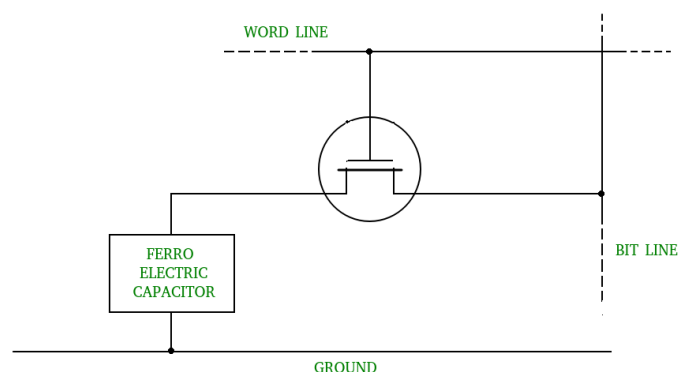
- ✚ **Infrared (IR) Sensors:** Motion Detectors: Pyroelectric materials are used in Passive Infrared (PIR) sensors for detecting human motion in security systems, automatic lighting, and door openers.
- ✚ **Thermal Imaging:** Converts infrared radiation into electrical signals for night vision cameras, security systems, and medical thermography.
- ✚ **Gas Leak Detectors:** Detect heat signatures associated with certain gases, useful in industrial safety.
- ✚ **Medical Applications:** Infrared Thermometers: Pyroelectric sensors measure body temperature from emitted thermal radiation, widely used in non-contact thermometers.
- ✚ **Thermal Imaging for Diagnostics:** Monitors heat patterns in tissues for early detection of conditions such as inflammation or tumors.

4.12. FERROELECTRIC RANDOM-ACCESS MEMORY (FE-RAM)

Ferroelectric Random-Access Memory (FeRAM or FRAM) is a non-volatile memory type that combines fast write and read speeds with data retention when power turns off. It uses a crystal-based ferroelectric layer instead of a standard dielectric layer.

FeRAM Working:

- ⇒ Ferro-electric Random Access Memory (FRAM) is a type of Random Access Memory, which uses a ferro-electric capacitor to achieve its non-volatility⁶.
- ⇒ The di-electric constant of a ferroelectric material usually has a high value. The most widely used ferroelectric material is PZT (Lead Zirconate Titanate). Due to the layer being ferroelectric, it may seem that it contains iron, but it actually does not contain iron.
- ⇒ The basic structure of FRAM is



⁶ Content is not lost when power is turned off.

similar to that of DRAM (Dynamic RAM), the only difference being that the di-electric layer is replaced by a ferroelectric layer.

⇒ To perform read, write and store operations, it consists of a bit-line, a word-line, transistors and a ferroelectric layer.

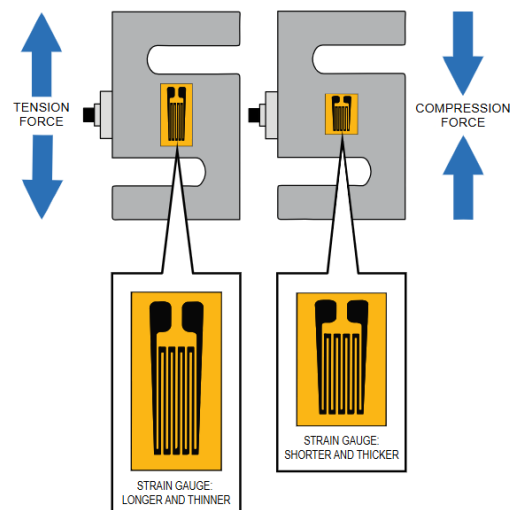
The key features of FRAM are:

1. Consumes less power.
2. Can tolerate radiation.
3. Is more secure than other memory types.
4. Has high endurance for read/write cycles.
5. Is over-writable like SRAM.
6. Less expensive than magnetic memories.

Advantages	Disadvantages
Writes happen much faster than flash memory	Capacities are smaller than traditional flash or DRAM.
It uses very little energy	Chips cost more to build than standard memory
It handles up to 100 trillion write cycles before wearing out	Reading can clear a cell, meaning the system must instantly rewrite the data back.

4.13. LOAD CELL SENSOR

- A load cell is an electro-mechanical sensor used to accurately measure force or weight.
- It is a transducer that is used to convert a force or load into an electrical signal. It is commonly used in various industries and applications to measure force, weight, or pressure.
- Load cells are typically made up of a sensing element or strain gauge, which deforms when a force is applied to it.
- The strain gauge is attached to a metal structure that undergoes deformation under load. This deformation causes a change in the electrical resistance of the strain gauge (Wheatstone Bridge), which is then measured and converted into an electrical signal proportional to the applied force.



4.14. FIRE SENSOR

A fire-sensor is one kind of detector which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. It to chemical or thermal changes using specialized mechanisms, such as an optical photoelectric chamber that detects scattered light from smoke particles, or an ionization chamber that monitors disrupted electrical currents caused by smoke. Other variants use a thermistor to register rapid temperature spikes or an infrared photodiode to

capture the precise light wavelengths emitted by open flames. Ultimately, both devices utilize precise internal shifts to feed critical data back to an automation or safety control system. This type of sensors can be easily damaged to high temperature; so, this sensor can be placed at a certain distance from the flame.

Working Principle:

This sensor/detector can be built with an electronic circuit using a receiver like electromagnetic radiation. This sensor uses the infrared flame flash method, which allows the sensor to work through a coating of oil, dust, water vapor, otherwise ice.

Flame Sensor Module:

The pin configuration of this sensor is shown below. It includes four pins which include the following. When this module works with a microcontroller unit then the pins are,

- Pin1 (VCC pin): Voltage supply ranges from 3.3V to 5.3V
- Pin2 (GND): This is a ground pin
- Pin3 (AOUT): This is an analog output pin (MCU.IO)
- Pin4 (DOUT): This is a digital output pin (MCU.IO).



Figure 4.13: Flame-sensor prototype.

Problems

Problem1: A magnetic material has placed in magnetic field intensity $H=500$ A/m and magnetization $M=2.0 \times 10^4$ A/m. Find susceptibility (χ) m.

Solution: $\chi_m = M/H = 2.0 \times 10^4 / 500 = 40$.

Answer: $\chi_m = 40$.

Problem2: If $\mu_r = 1.05$, calculate χ_m using $\mu_r = 1 + \chi_m$.

Solution: $\chi_m = \mu_r - 1 = 0.05$.

Answer: 0.05.

Problem3: Susceptibility of a magnetic material is $\chi_m = 2.5 \times 10^{-4}$. Find it's relative permeability μ_r .

Solution: $\mu_r = 1 + \chi_m = 1.00025$.

Answer: 1.00025.

Problem4: Magnetic field induction $B=0.80$ T at magnetic field intensity $H=1000$ A/m. Find $\mu = B/H$.

Solution: $\mu = 0.80 / 1000 = 8.0 \times 10^{-4}$ H/m.

Answer: 8.0×10^{-4} H/m.

Problem5: A magnetic material has a magnetization $M=1.2 \times 10^6$ A/m. Neglect applied field and use $B = \mu_0 M$. Find magnetic induction B.

Solution: $B = 4\pi \times 10^{-7} \times 1.2 \times 10^6 = 1.508$ T.

Answer: ≈ 1.51 T.

Problem6: Material A has coercivity 50 A/m; Material B has coercivity 5000 A/m. Which is more suitable as a soft magnetic material?

Solution: Soft magnetic materials require low coercivity for easy magnetization/demagnetization. 50 A/m is much lower.

Answer: Material A.

Problem7: Approximate a hysteresis loop area as $B \times H$. If $B=0.6$ T and $H=800$ A/m, estimate energy loss per unit volume per cycle.

Solution: Energy loss/volume $\approx$ area = $B H = 0.6 \times 800 = 480$ J/m³ per cycle.

Answer: ≈ 480 J/m³/cycle.

Problem8: Capacitance changes from 50 pF in air to 250 pF after inserting a dielectric. Find ϵ_r .

Solution: $\epsilon_r = C/C_0 = 250/50 = 5$.

Answer: $\epsilon_r = 5$.

Problem9: A parallel-plate capacitor has $A=0.01$ m² and $d=1$ mm. Find C in vacuum.

Solution: $C = \epsilon_0 A/d = 8.854 \times 10^{-12} \times 0.01/10^{-3} = 8.854 \times 10^{-11}$ F.

Answer: 88.54 pF.

Problem10: A parallel-plate capacitor has $A=0.01$ m² and $d=1$ mm is filled with $\epsilon_r=6$. Find new Capacitance

Solution: $C = \epsilon_r C_0 = 6 \times 88.54 = 531.24$ pF.

Answer: 531.24 pF.

Problem11: A 2 μ F capacitor is charged to 500 V. Find stored energy.

Solution: $U = \frac{1}{2} CV^2 = 0.5 \times 2 \times 10^{-6} \times (500)^2 = 0.25$ J.

Answer: 0.25 J.

Problem12: The capacitor in Problem 11 is isolated and a dielectric $\epsilon_r=4$ is inserted. Find new energy.

Solution: For isolated capacitor, Q remains constant and C increases 4 times.

$U' = U/\epsilon_r = 0.25/4 = 0.0625$ J.

Answer: 0.0625 J.

Problem13: A 100 pF capacitor is connected to 100 V. $\epsilon_r=5$ dielectric is inserted while connected. Find initial and final energy.

Solution: $U_i = \frac{1}{2} CV^2 = 0.5 \times 100 \times 10^{-12} \times 10^4 = 5 \times 10^{-7}$ J. $C_f = 500$ pF. $U_f = \frac{1}{2} C_f V^2 = 2.5 \times 10^{-6}$ J.

Answer: $U_i = 0.50$ μ J, $U_f = 2.50$ μ J.

Problem14: A capacitor has $C_0=120$ pF and $C=840$ pF after dielectric insertion. It is charged to 200 V. Find ϵ_r and stored energy after insertion.

Solution: $\epsilon_r = 840/120 = 7$. $U = \frac{1}{2} CV^2 = 0.5 \times 840 \times 10^{-12} \times 40000 = 1.68 \times 10^{-5}$ J.

Answer: $\epsilon_r = 7$, $U = 16.8$ μ J.

Problem15: A magnetic material has $M=8.0 \times 10^5$ A/m under $H=2000$ A/m. A separate dielectric capacitor has $C_0=50$ pF and becomes 300 pF with dielectric. Find χ_m , μ_r , and ϵ_r .

Solution: $\chi_m = M/H = 8 \times 10^5/2000 = 400$. $\mu_r = 401$. $\epsilon_r = 300/50 = 6$.

Answer: $\chi_m = 400$, $\mu_r = 401$, $\epsilon_r = 6$.

Review Questions

Short Answer Questions

1. Define magnetic dipole moment.
2. What are soft and hard magnetic materials?
3. Define dielectric polarization.
4. What is hysteresis loss?
5. Name three types of dielectric polarization.
6. What is the Curie-Weiss law?
7. Define piezoelectricity.
8. What is Giant Magnetoresistance (GMR)?
9. Name an application of magnetic hyperthermia.
10. What is Ferroelectric RAM (Fe-RAM)?

Long Answer Questions

1. Classify magnetic materials into diamagnetic, paramagnetic, ferromagnetic, ferrimagnetic, and antiferromagnetic materials based on their properties and magnetic susceptibility.
2. Explain the Weiss domain theory of ferromagnetism. Draw and explain the hysteresis loop (B-H curve) and account for energy losses during magnetization.
3. Differentiate between soft and hard magnetic materials based on their properties, hysteresis loops, and industrial applications (including EV motor magnets).
4. Describe the sol-gel synthesis method for ferrimagnetic materials. Discuss the applications of magnetic nanomaterials in cancer therapy (magnetic hyperthermia).
5. Explain the different mechanisms of dielectric polarization (electronic, ionic, and orientational).
6. Define ferroelectricity, piezoelectricity, and pyroelectricity. Explain their operational mechanisms and working principles in applications like Fe-RAM, load cells, and fire sensors.
7. Explain the Giant Magnetoresistance (GMR) effect and its underlying physics, describing its implementation in modern data storage and sensing devices.
8. Describe the composition, properties, and applications of ferrites in high-frequency electrical components.

Objective Type Questions

Multiple Choice Questions

1. The magnetic moment of an electron arises mainly from its
 - A) Charge and motion
 - B) Temperature only
 - C) Nuclear mass only
 - D) Lattice spacing only
2. The atomic unit of magnetic moment is the
 - A) Bohr magneton
 - B) Debye
 - C) Farad
 - D) Tesla
3. Diamagnetic materials generally have
 - A) Weak negative susceptibility
 - B) Strong positive susceptibility
 - C) Infinite permeability
 - D) Permanent magnetization
4. Paramagnetic materials are generally
 - A) Weakly attracted by a magnetic field
 - B) Strongly repelled
 - C) Permanently magnetized
 - D) Non-magnetic at all temperatures
5. Ferromagnetic materials exhibit
 - A) Spontaneous magnetization
 - B) Only weak repulsion
 - C) No magnetic domains
 - D) Zero susceptibility

6. Regions of spontaneous magnetization in ferromagnets are called
 A) Domains B) Grains
 C) Qubits D) Voids
7. Weiss domain theory is associated with
 A) Ferromagnetism
 B) Dielectric breakdown
 C) Laser action
 D) Fibre communication
8. The curve showing magnetization against applied magnetic field is called the
 A) Hysteresis loop B) Bragg curve
 C) I-V curve D) Dispersion curve
9. The residual magnetization after removal of the applied field is called
 A) Retentivity B) Coercivity
 C) Susceptibility D) Permeability
10. The field required to reduce magnetization to zero is called
 A) Coercive field B) Retentive field
 C) Saturation field only D) Critical angle
11. Soft magnetic materials are characterized by
 A) Low coercivity
 B) High coercivity
 C) Very high hysteresis loss
 D) Permanent magnetization
12. Hard magnetic materials are suitable for
 A) Permanent magnets
 B) Transformer cores only
 C) AC electromagnets only
 D) Low-loss switching only
13. Ferrimagnetic materials have opposing magnetic moments that are
 A) Unequal
 B) Equal and opposite
 C) Always parallel
 D) Completely absent
14. The sol-gel method can be used for the synthesis of
 A) Ferrimagnetic materials
 B) Only gases
 C) Classical bits
 D) Optical fibres only
15. Magnetic materials are used in
 A) Motors and transformers
 B) Only glass windows
 C) Only capacitors
 D) Only optical fibres
16. A dielectric is essentially an
 A) Electrical insulator capable of polarization
 B) Perfect conductor
 C) Magnetic metal only
 D) Semiconductor only
17. The ability of a dielectric to polarize in an electric field is related to its
 A) Permittivity B) Coercivity
 C) Retentivity D) Magnetic moment
18. The relative permittivity is defined as
 A) ϵ/ϵ_0 B) ϵ_0/ϵ
 C) $\epsilon+\epsilon_0$ D) $\epsilon-\epsilon_0$
19. Electronic polarization is due to
 A) Displacement of electron cloud relative to nucleus
 B) Movement of magnetic domains
 C) Rotation of a crystal
 D) Photon emission
20. Ionic polarization results from
 A) Relative displacement of positive and negative ions
 B) Electron diffraction
 C) Magnetic saturation
 D) Population inversion
21. Orientational polarization is associated with
 A) Permanent dipoles
 B) Free electrons only
 C) Nuclei only
 D) Crystal vacancies
22. Ferroelectric materials exhibit
 A) Spontaneous polarization
 B) Zero dielectric response
 C) Only diamagnetism
 D) No dipole moments
23. Piezoelectric materials produce an electrical response when
 A) Mechanical stress is applied
 B) Magnetic field is removed
 C) Light is absorbed
 D) Pressure is zero
24. Pyroelectric materials are associated with polarization changes due to
 A) Temperature changes
 B) Magnetic fields only
 C) X-ray diffraction
 D) Crystal vacancies
25. Fe-RAM is an application associated with
 A) Ferroelectric materials
 B) Hard magnetic materials
 C) Optical fibers
 D) Ruby lasers

Multiple Choice Questions

1. The magnetic moment unit associated with an electron is the _____ magneton.
2. Materials weakly repelled by a magnetic field are generally called _____ materials.
3. Materials weakly attracted by a magnetic field are called _____ materials.
4. Materials showing spontaneous magnetization are called _____ materials.
5. The regions of uniform spontaneous magnetization are called magnetic _____.
6. Weiss proposed the concept of magnetic _____ in ferromagnets.
7. The magnetization curve that shows magnetic history is called the _____ loop.
8. The residual magnetization after removing the field is called _____.
9. The field needed to demagnetize a material is called the _____ field.
10. Materials with low coercivity are called _____ magnetic materials.
11. Materials with high coercivity are called _____ magnetic materials.
12. Hard magnetic materials are commonly used to make _____ magnets.
13. In ferrimagnetism, opposing magnetic moments are _____ in magnitude.
14. The _____-gel method can be used to synthesize ferrimagnetic materials.
15. Magnetic materials are used in motors, generators and _____.
16. A dielectric is an electrical _____ that can be polarized.
17. The ratio ϵ/ϵ_0 is called relative _____.
18. The formation or alignment of electric dipoles in a dielectric is called _____.
19. Electronic polarization involves displacement of the electron _____.
20. Ionic polarization involves relative displacement of positive and negative _____.
21. Alignment of permanent dipoles produces _____ polarization.
22. Ferroelectric materials have spontaneous electric _____.
23. Mechanical stress producing electrical polarization is the _____ effect.
24. Polarization changes caused by temperature variation are associated with the _____ effect.
25. Ferroelectric Random-Access Memory is abbreviated as _____.

MCQs: 1-A, 2-A, 3-A, 4-A, 5-A, 6-A, 7-A, 8-A, 9-A, 10-A, 11-A, 12-A, 13-A, 14-A, 15-A, 16-A, 17-A, 18-A, 19-A, 20-A, 21-A, 22-A, 23-A, 24-A, 25-A

Fill in the blanks: 1. Bohr; 2. diamagnetic; 3. paramagnetic; 4. ferromagnetic; 5. domains; 6. domains; 7. hysteresis; 8. retentivity; 9. coercive; 10. soft; 11. hard; 12. permanent; 13. unequal; 14. sol; 15. transformers; 16. insulator; 17. permittivity; 18. polarization; 19. cloud; 20. ions; 21. orientational; 22. polarization; 23. piezoelectric; 24. pyroelectric; 25. Fe-RAM.

UNIT – 5: LASERS & FIBER OPTICS

Contents: - Lasers: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, CO₂ laser, semiconductor diode laser, applications: Bar code scanner, LIDAR for autonomous vehicle.

Fiber optics: Introduction to fibre optics, total internal reflection, construction of optical fibre, acceptance angle, numerical aperture, classification of optical fibres, losses in optical fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

5.1. INTRODUCTION TO LASER

The light emitted from a conventional light source (e.g. Sodium lamp) is incoherent because the radiation emitted from different atoms do not have a definite phase relationship with each other. However, a laser is a device that creates a narrow, focused beam of intense light through optical amplification based on the stimulated emission of radiation.

LASER is the acronym for *Light Amplification by Stimulated Emission of Radiation*. Lasers are the light sources having high monochromaticity, high intensity, high directionality and high degree of coherence. Laser emits pure sine waves having constant phase difference with respect to space and time.

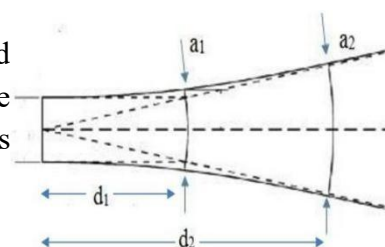
5.2. CHARACTERISTICS OF LASER

Laser differs from ordinary light through four primary features: monochromaticity (a single precise wavelength/color), exceptional directionality (minimal beam spread or divergence), strong coherence (all waves locked in phase), and immense intensity/brightness concentrated in a tiny area.

Key Characteristics of Laser Light:

1. **Highly Directional** – Laser is highly directional which is expressed in terms of divergence (zero angular spread or scattering). The degree of divergence of laser for laser spots of radii a_2 and a_1 for distances d_2 and d_1 from the laser source is given by

$$\phi = \frac{a_2 - a_1}{2(d_2 - d_1)}$$



The laser beam spreads less than 0.01 mm for a distance of one meter, whereas search light spreads about one meter distance travelled. Hence, laser is highly directional.

2. **Highly Monochromatic** – Laser is highly monochromatic (single wavelength) compared to the conventional monochromatic light because of the characteristic nature of stimulated emission of laser. The degree of monochromaticity is nearly 10^{-3} .
3. **Highly Coherent** – The correlation of the phase and amplitude at any one point with any other point in a beam of light or wave train is called coherence. In a laser beam if you consider a wave front, all the wave trains are having same phase and amplitude. There are temporal coherence and spatial coherence

4. **Highly Intense and bright** - Because lasers emit energy in a narrow beam, that energy is concentrated into a tiny area, resulting in immense brightness. Due to coherence and directionality, laser light can be intensely concentrated. If 'a' is the amplitude of a single wave train then its intensity is given by $I = n^2 a^2$.

****Summary of key characteristics:**

Coherent	All waves in phase (same frequency/amplitude).
Directional	Extremely low beam divergence (parallel).
Intensity	High power concentrated in a small spot.
Monochromatic	Single, pure colour/wavelength.

Stimulated Absorption: An electron in a lower energy state (E_1) absorbs an incoming photon's energy and jumps to a higher energy state (E_2). This transition is known as stimulated absorption.

Spontaneous Emissions: An excited electron in a higher energy level (E_2) falls back to a lower energy level (E_1) naturally, without external influence, emitting a photon of energy. This photon emission is known as spontaneous emission.

Stimulated Emissions: An incident photon energy interacts with an already excited atom (E_2), stimulating it to drop to the lower level (E_1) before it would naturally. It releases a second photon that is an exact replica of the incident photon. This phenomenon of forced emission of photons is called stimulated emission.

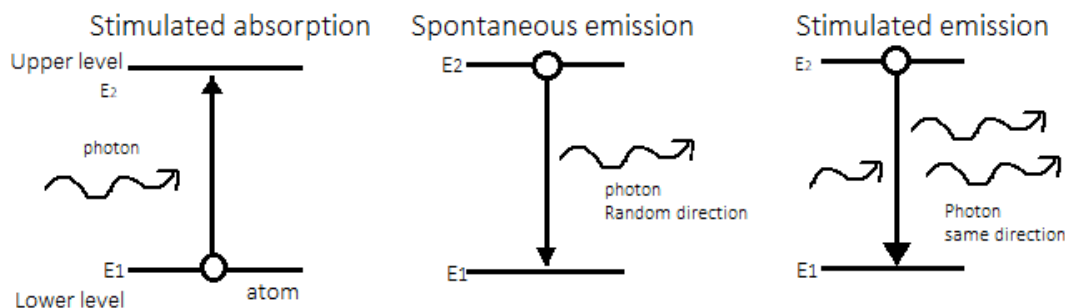


Figure 5.1: Radiation interaction with the matter.

5.3. PRINCIPLE AND WORKING OF LASER

The basic requirement to the laser action is that there must be a greater number of atoms in the metastable state than in the ground state. When this is achieved, there will be more stimulated emissions of the atoms in the metastable state to the ground state. The steps in the laser action are: 1. Existence of metastable state, 2. Pumping and 3. Population Inversion.

- 1. Metastable States:** The energy state in which atoms stay for longer time than mean life time of the order of 10^{-8} sec. large number of excited atoms at this level, thus creating a favorable condition for stimulation. In order to achieve population inversion, we must have an energy state which has a longer lifetime and such a state is called as a metastable state. Therefore, for achieving lasing action one must identify systems with metastable states.
- 2. Pumping Methods:** The process of supplying energy to the medium to transfer it into the state of population inversion is known as Pumping.

The most commonly used pumping techniques are: Optical pumping, Electrical discharge and Direct conversion.

- a. **Optical pumping:** A light source such as flash discharge tube is used. Ex: Solid state lasers.
 - b. **Electrical Discharge:** The electric field causes ionization of the medium and raises it to the excited state. Ex: Gas lasers. 1) Optical pumping- used in ruby Laser 2) Electric Discharge-Used in gas lasers 3) Chemical reaction- used in chemical lasers 4) Injection current- Used in semiconductor lasers.
 - c. **Chemical Pumping:** In certain systems, chemical reactions (e.g., $\text{H}_2 + \text{F}_2 = 2\text{HF}$) occur, releasing energy that excites the molecules, creating an inversion.
 - d. **Direct conversion:** In semiconductor laser diodes, the forward biased P-N junction diode create inversion due to uneven concentrations.
3. **Population Inversion:** Population inversion is the condition of the material in which population of atoms in the upper energy level N_2 is made more than that in the lower energy level N_1 and necessary condition to achieve lasing action. A system in which population inversion is achieved is called as active system for lasing.

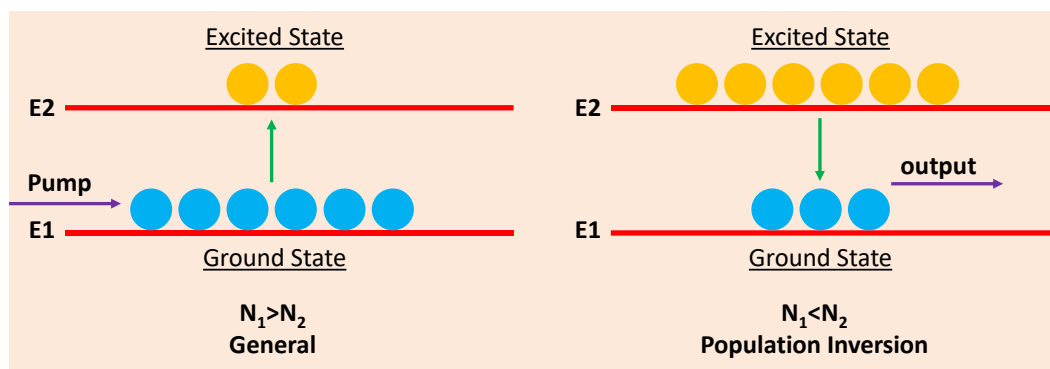


Figure 5.2: The process of population inversion.

5.4. LASING ACTION

Population inversion though is the primary condition, but this itself is not sufficient for producing a laser. As there are certain losses of the emitted photons within the material itself in addition to spontaneous emission, one has to think about the geometry that can overcome these losses and there is overall gain. This requires an optical cavity or resonator.

The principle behind the laser is like this: Let we produced a large number of atoms that all are in excited states, out of all, if one of the atoms emitted spontaneously, then the emitted photon would stimulate other atoms to emit. These emitted photons would, in turn, stimulate further emission. The result would be an intense burst of coherent radiation as shown in Fig. 5.3.

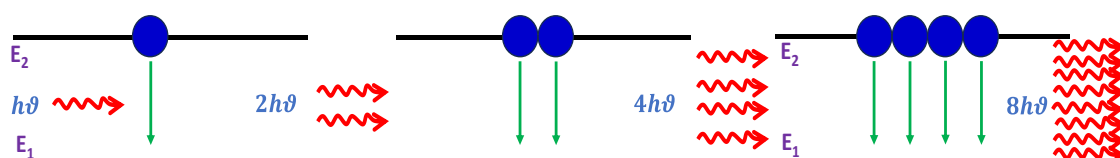


Figure 5.3: Lasing Action.

DIFFERENT LEVEL SCHEMES

Three level schemes: In a three-level system the atoms are pumped to E₂ a fast-decaying stage and from there the atoms will decay to level E₁, metastable state. If E₁ is a metastable state large number of atoms will be accumulated at E₁ resulting in a population inversion between E₀ and E₁. The transition from E₁ to E₀ results in LASER production. The laser so produced will be pulsed wave in nature.

Four Level scheme: In a four-level scheme, atoms are pumped from E₀ level to the highest level E₃. E₃ to E₂ is a fast transition and level E₂ is a metastable state. Transition from E₂ to E₁ gives the laser radiation. If the transition from E₁ to E₀ is fast, this scheme will give continuous wave output.

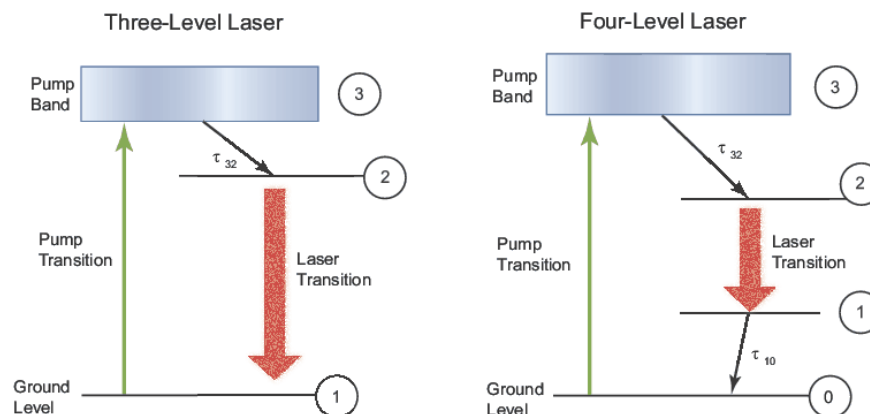


Figure 5.3: Different energy level scheme.

5.5. EINSTEIN COEFFICIENTS

Let us consider N_1 and N_2 be the population inversions with energy levels E_1 and E_2 in a system of atoms at thermal equilibrium temperature (T), respectively.

Absorption (B_{12}): Atom absorbs a photon and moves from E_1 to E_2 .

- Probability of an atom absorbing a photon ($h\nu$) and moving from E_1 to E_2 .

Equation (Rate): $P_{abs} = B_{12}N_1 \rho(\nu)$ (where $\rho(\nu)$ is the energy density of the radiation)

Spontaneous Emission (A_{21}): Excited atom drops from E_2 to E_1 naturally, emitting a photon randomly.

- Probability of an atom in E_2 falling to E_1 voluntarily, emitting a photon.

Equation (Rate): $P_{spo} = A_{21}N_2$

Stimulated Emission (B_{21}): Incoming photon forces an excited atom (E_2) to drop to E_1 , emitting a second photon identical to the first.

- Probability of an incident photon ($h\nu$) stimulating an atom in E_2 to fall to E_1 .

Equation (Rate): $P_{sti} = B_{21}N_2 \rho(\nu)$

We assume the system is in thermal equilibrium, meaning the rate of upward transitions equals the rate of downward transitions:

$\therefore \text{Absorption rate} = \text{Spontaneous emission rate} + \text{Stimulated emission rate}$

$$B_{12}N_1 \rho(\nu) = A_{21}N_2 + B_{21}N_2 \rho(\nu)$$

Rearrange the above equation for $\rho(\nu)$,

$$\begin{aligned} B_{12}N_1 \rho(\nu) - B_{21}N_2 \rho(\nu) &= A_{21}N_2 \\ \rho(\nu)[B_{12}N_1 - B_{21}N_2] &= A_{21}N_2 \\ \Rightarrow \rho(\nu) &= \frac{A_{21}N_2}{[B_{12}N_1 - B_{21}N_2]} \\ \Rightarrow \rho(\nu) &= \frac{A_{21}N_2}{B_{21}N_2 \left[\frac{B_{12}N_1}{B_{21}N_2} - 1 \right]} \\ \Rightarrow \rho(\nu) &= \frac{A_{21}}{B_{21} \left[\frac{B_{12}N_1}{B_{21}N_2} - 1 \right]} \quad \text{--- (1)} \end{aligned}$$

Apply Boltzmann Distribution, then the population ratio is given by,

$$\frac{N_1}{N_2} = e^{\frac{h\nu}{kT}} \quad \text{--- (2)}$$

Substitute the equation (2) in (1), we get,

$$\rho(\nu) = \frac{A_{21}}{B_{21}} \frac{1}{\left[\frac{B_{12}}{B_{21}} \left[e^{\frac{h\nu}{kT}} \right] - 1 \right]} \quad \text{--- (3)}$$

Compare eqn. (3) with the Planck's Law, where the Planck's law is

$$\rho(\nu) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{kT}} - 1}$$

By comparison, we get Einstein's coefficients:

$$\begin{aligned} \frac{A_{21}}{B_{21}} &= \frac{8\pi h\nu^3}{c^3} \quad (\text{Ratio of spontaneous to stimulated emission}) \\ B_{21} &= B_{12} \quad (\text{Stimulated absorption is equal to stimulated emission}) \end{aligned}$$

5.6. TYPES OF LASERS

There are many types of lasers available for research, medical, industrial and commercial uses. Lasers are mainly classified by their lasing medium. The main types are,

1. Solid-state laser: Ruby laser, Nd: YAG laser.
2. Gas laser: CO₂ laser, Argon-ion laser, He-Ne laser.
3. Semiconductor (diode) laser: InP laser, GaAs laser.

5.7. RUBY LASER

The Ruby laser is first constructed and demonstrated in 1960 by T.H. Maiman. It is a solid-state laser that uses a synthetic ruby crystal as its active medium to produce a pulsed beam of coherent red light.

Construction of Ruby Laser:

Active Medium: A synthetic cylindrical Ruby Rod (Al₂O₃ doped with 0.05% of Cr₂O₃).

The length is typically 2–10 cm. The Cr^{3+} ions are responsible for the laser action and give the crystal its pink/red colour.

Pumping Source: A helical **Xenon Flash** Tube surrounds the ruby rod. This provides high-intensity light (optical pumping) to excite the chromium ions.

Optical Resonator: The ends of the ruby rod are polished and silvered. One end is **fully silvered** (100% reflective), while the other is **partially silvered** (about 90% reflective), acting as the output coupler.

Cooling System: Since the flash lamp generates significant heat, the rod is often cooled using liquid nitrogen or circulating water/air to maintain efficiency.

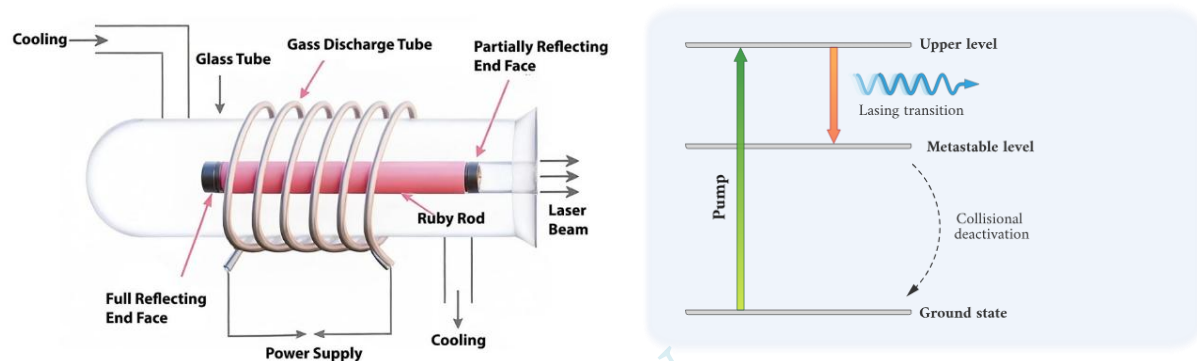


Figure 5.4: Ruby laser and 3-level laser.

Working of Ruby laser:

The Ruby laser operates as a three-level laser system. The process involves several distinct energy transitions.

- ❖ When the Xenon lamp flashes, it emits white light. The Cr^{3+} ions absorb photons at wavelengths around 550 nm (green/yellow) and are excited from the Ground State (E_1) to the Excited State (E_3).
- ❖ The ions in E_3 are very unstable (lifetime $\sim 10^{-8}$ seconds). They quickly lose some energy as heat and drop to a lower energy level called the Metastable State (E_2). This transition is non-radiative, meaning no light is emitted.
- ❖ The Metastable State (E_2) has a much longer lifetime ($\sim 3 \times 10^{-3}$ seconds). Because ions stay here longer than they do in E_1 , they begin to accumulate. When there are more ions in E_2 than in E_1 , Population Inversion is achieved.
- ❖ A few ions eventually drop from E_2 to E_1 spontaneously, emitting a photon of wavelength 694.3 nm. This photon travels through the rod, triggering other excited ions to drop and emit identical photons (Stimulated Emission).
- ❖ These photons bounce back and forth between the two mirrors, causing a "cascade" effect that amplifies the light. Once the intensity is high enough, a powerful, coherent pulse of red-light escapes through the partially silvered mirror.

Applications: Holography, removal of tattoos, and early distance measurement (LIDAR), toys for children, drilling holes in hard materials, High-brightness holographic camera systems.

5.8. HE-NE LASER

The Helium-Neon (He-Ne) laser is one of the most common gas lasers. Ali Javan et al., fabricated the first He-Ne laser at Bell Telephone laboratory in USA in the year 1961. It uses a four-level scheme for population inversion.

Construction of He-Ne laser:

The construction of a He-Ne laser involves a discharge tube, a gas mixture, and an optical resonator.

Discharge Tube: A narrow quartz or glass tube (capillary tube) typically 20–100 cm long.

Gas Mixture: The tube is filled with a mixture of Helium (He) and Neon (Ne) gases, usually in a ratio of 10:1 (10 parts Helium to 1 part Neon) at low pressure. Helium acts as the "pumping" gas, while Neon is the "active" gas that produces the laser light.

Pumping Source: A high-voltage DC power supply is connected to two electrodes (anode and cathode) at the ends of the tube to initiate an electric discharge.

Optical Resonator: Two mirrors are placed at the ends of the tube. One is fully reflective, and the other is partially reflective (allowing about 1% of light to pass through as the output beam).

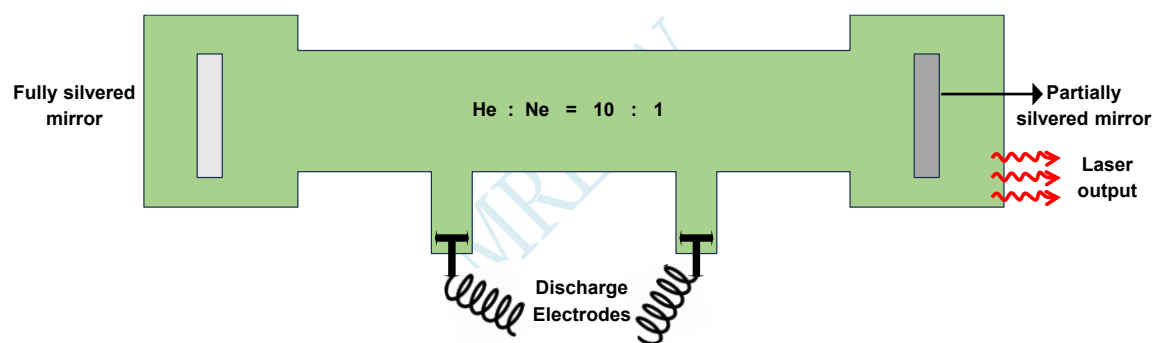
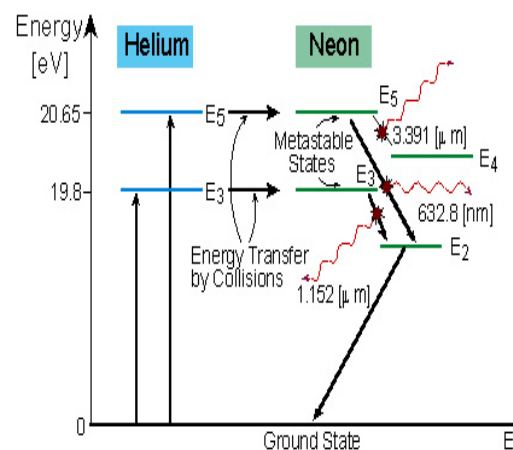


Figure 5.5: He-Ne laser and 4-level laser

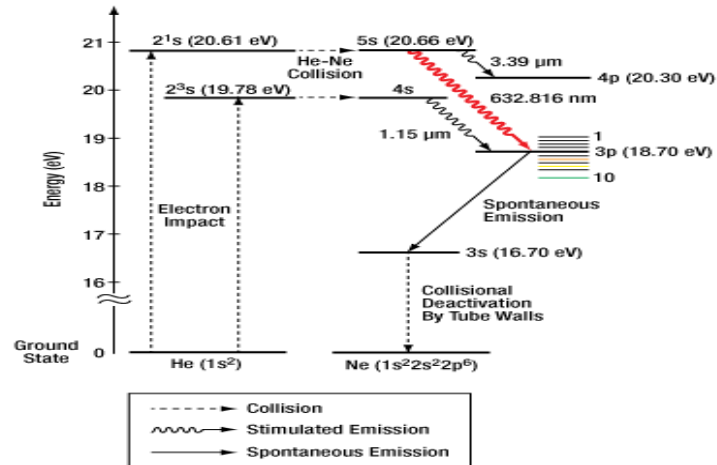
Working of He-Ne Laser:

The working principle is based on stimulated emission after achieving population inversion.

- ❖ The high voltage excites helium atoms (He) to higher metastable energy levels (let's say E_2 and E_3) through electron collisions.
- ❖ Excited helium atoms collide with ground-state neon atoms (Ne), transferring their energy, causing the neon atoms to populate higher energy levels (E_6 and E_4).
- ❖ The number of excited Ne atoms becomes higher than the lower energy levels, achieving a population inversion state.



- ❖ Excited Ne atoms drop down to lower levels (E_5 or E_3) and emit photons (specifically 629.3 nm red light for $E_6 \rightarrow E_3$).
- ❖ These photons travel back and forth within the resonator, stimulating other excited Ne atoms to emit similar photons, resulting in a continuous, amplified, highly collimated beam.
- ❖ The He-Ne laser is favoured for its stability, high coherence, and low cost.



Advantages:

1. He-Ne laser emits continuous laser radiation.
2. Due to the setting of end windows at Brewster's angle, the output laser is linearly polarized.
3. Gas lasers are more monochromatic and directional when compared with the solid-state laser.

Disadvantages:

1. The output power is moderate
2. The mirrors should be replaced regularly since they are usually eroded by the gas discharge.

Applications of He-Ne Laser:

He-Ne laser is used widely in commercial and industrial applications, including barcode scanners, Tool alignment, non-contact measuring and monitoring, Blood analysis, Particle counting and food sorting, etc.

5.9. CARBON DIOXIDE (CO₂) LASER

The carbon dioxide (CO₂) laser is a laser based on a gas mixture of carbon dioxide (CO₂), nitrogen (N₂), and helium (He) at the ratios of 1:2:3 acts as medium. The first gas laser was developed by Indian born American Scientist Prof. C.K.N. Patel. It is a four-level molecular gas laser. In this laser, transition takes place between vibrational energy states of carbon dioxide molecules. It is a very efficient laser compared with other types of lasers.

Active medium: A gas mixture of CO₂, N₂ and helium.

Pumping method: Electrical discharge method,

Optical resonator: It is formed with two concave mirrors.

Power output: 10 kW.

Nature of output: Continuous wave or pulsed wave.

Wavelength of output: 9.6 μm and 10.6 μm

Construction:

It consists of a quartz discharge tube 5 m long and 2.5 cm in diameter. This discharge tube is filled with the gas mixture of CO₂, nitrogen (N₂) and helium (He) with suitable partial pressures. The terminals of the discharge tube are connected to D.C. power supply. The ends of this tube are fitted with NaCl Brewster windows so that the laser light generated is plane polarized. The optical resonator is formed with two concave mirrors one fully reflecting M₁ with the other partially reflecting M₂.

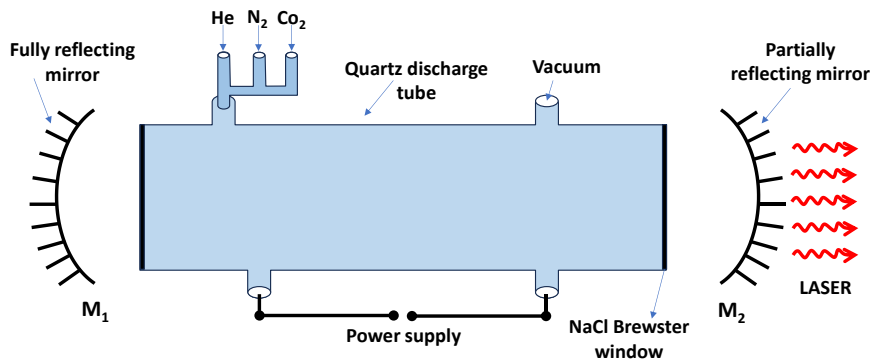
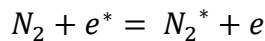


Figure 5.6: Schematic sketch of CO₂ Laser.

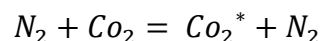
Construction:

The energy level diagram of nitrogen and carbon dioxide molecules is shown in Fig 5.7. When the electrical discharge occurs in gas mixture, the electrons collide with nitrogen molecules and they are raised to excited energy states. This process is represented by the equation,



N₂ → Nitrogen molecule in ground state, e*
→ Electron with high energy, N₂* →
Nitrogen molecule in excited state and e →
Same electron with lesser energy.

Since excited energy level of nitrogen is very close to E₅ energy level of CO₂ molecules, CO₂ molecules are excited by energy transfer and population inversion is achieved. This process is represented by the equation,



N₂* → Nitrogen molecule in excited state, CO₂ → Carbon dioxide molecule in ground state,
CO₂* → Carbon dioxide molecule in excited state and N₂ → Nitrogen molecule in ground state.

Any of the spontaneously emitted photon triggers laser action in the tube. There are two possible types of laser transition,

(i). Transition E₅-E₄: This transition produces a laser beam of wavelength 10.6 μm.

(ii). Transition E₅ – E₃: This transition produces a laser beam of wavelength 9.6 μm. Normally

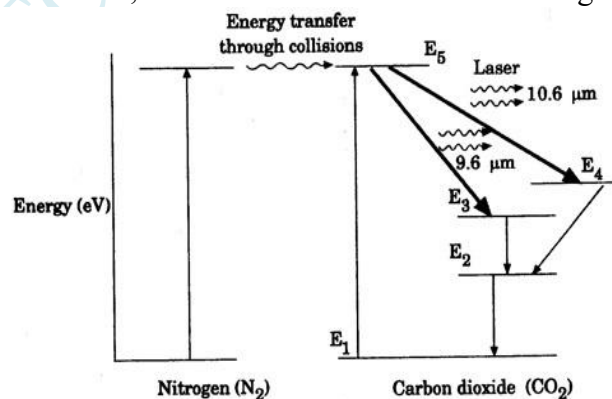


Figure 5.7: Energy level of CO₂ Laser.

10.6 μm transition is more intense than 9.6 μm transition. The power output from this laser is 10 kW.

Note: The helium gas is used to conduct heat generated in the central region of the discharge tube to the walls of the discharge tube.

Advantages

- The construction of CO₂ laser is simple.
- The output from this laser is continuous.
- It has high efficiency.
- It has very high output power.
- The output power can be increased by increasing the length, of discharge tube.

Disadvantages

- The contamination of oxygen by carbon monoxide has some effect on laser action.
- The operating temperature plays an important role in determining the output power of the laser.
- Corrosion may occur at the surfaces of the discharger tube.
- Due to high power laser light, accidental exposure may damage eyes, since it is invisible (infra-red region) to our eyes.

Applications

- High-power CO₂ laser finds application in material processing, welding, drilling, cutting, soldering, etc.
- The low atmospheric attenuation (10.6 μm) makes CO₂ laser suitable for open air communication.
- It is used in remote sensing.
- It is used in the treatment of liver and lung diseases.
- It is mostly used in neurosurgery and general surgery.
- It is used to perform microsurgery and bloodless operations.

5.10. SEMICONDUCTOR DIODE LASER

A laser diode is a laser where the active medium is a semiconductor similar to Light emitting diode. The laser diode is formed from a PN junction and is powered by injected electric current. These devices are referred to as injection laser diodes to distinguish them from optically pumped laser diodes.

Broadly there are two types of semiconductor diode lasers. They are known as,

1. Homo-junction Semiconductor laser and
2. Hetero- junction laser.

1. Homo-Junction Semiconductor Laser:

It is a diode laser. When PN junction is formed with single crystalline material (i.e., either side of the junction has the same base material), then it is called as Homo junction semiconductor laser.

Principle: When PN junction diode is forward biased, recombination of electrons and holes occurs and photons are emitted. These emitted photons stimulate further recombination and a laser output is obtained.

Construction:

- P-N Junction diode laser is formed by heavily doped GaAs with Ge (P-type) and GaAs with T (N-type).
- The diode is extremely small in size with $500\ \mu\text{m}$ long and about $100\ \mu\text{m}$ wide and thick. The top and bottom face are provided with ohmic contact.
- The front and rear faces are polished parallel to each other and perpendicular to the plane of the junction. The polished faces constitute an optical resonator as in Fig. 5.8.

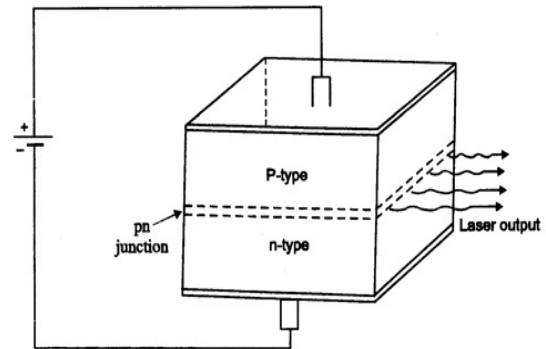


Figure 5.8: Schematic of homo-junction diode laser

Working:

When a high forward bias voltage is applied, electrons are injected from the n-region, and holes from the p-region into the narrow junction region. The high carrier concentration creates a population inversion, where higher energy states have more electrons than lower energy states. Electrons drop from the conduction band to the valence band, recombining with holes and emitting photons. These photons trigger further recombination (stimulated emission), generating a coherent beam at the polished junction faces. The emitted wavelength is typically in the infrared region, e.g. ~ 8400 to $8600\ \text{\AA}$ for GaAs.

Drawbacks Of Homo-Junction Lasers:

- Threshold current density is very large ($400\ \text{Amp/mm}^2$)
- Only pulsed mode output is obtained.
- Laser output has large beam divergence.
- Poor coherence and poor stability.
- Electromagnetic field confinement is poor.

2. Multi-Junction (Heterojunction) Semiconductor Laser:

A diode laser which makes use of different semiconductor materials on the two sides of the junction is known as a Hetero-junction diode laser.

Principle: When PN junction diode is forward biased, recombination of electrons and holes occurs and photons are emitted. These emitted photons stimulate between the polished surfaces of the junction and finally a laser output is obtained from the diode.

Construction:

In the hetero-junction semiconductor laser Fig. 5.9 (a, b), p-GaAs layer is sandwiched between two layers of GaAlAs. The material GaAlAs (p-type and n-type) has a wider energy gap and lower high refractive index. P-GaAs is the active region where the laser action takes place. The front and the rear face are polished parallel to each other and perpendicular to the plane of the

junction. The polished face will act as an optical resonator.

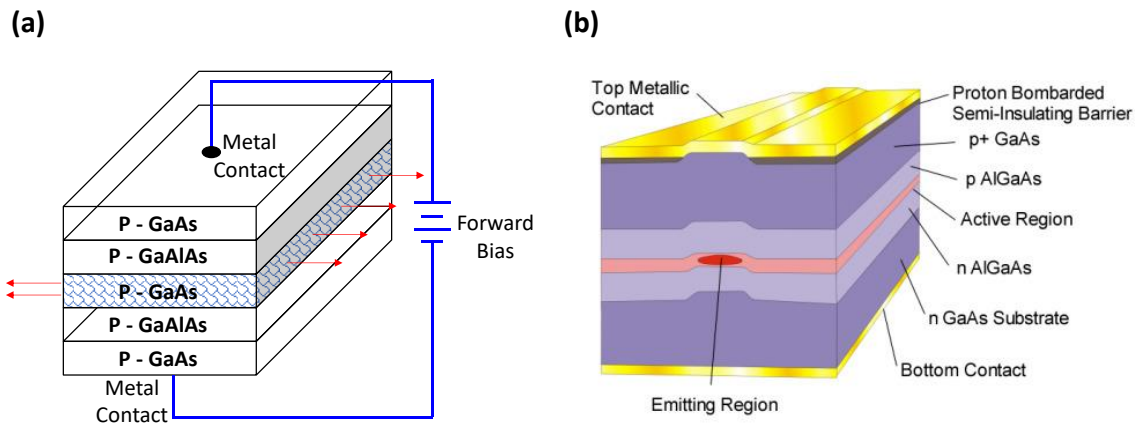


Figure 5.9: (a, b) Schematic diagram of Hetero junction diode laser.

Working:

- When the forward bias is not connected, then the energy diagram will be as shown in figure below, no electrons and holes are present in the depletion region as in Fig. 5.8(a).
- When a small forward bias is given to the pn junction then small number of electrons and holes will be injected into the depletion region from respective regions as shown in Fig.4.8 (b).
- When relatively large current is passed through the junction then large number of electrons and holes will be injected into depletion region and direct recombination process takes place efficiently Fig. 4.8 (c).

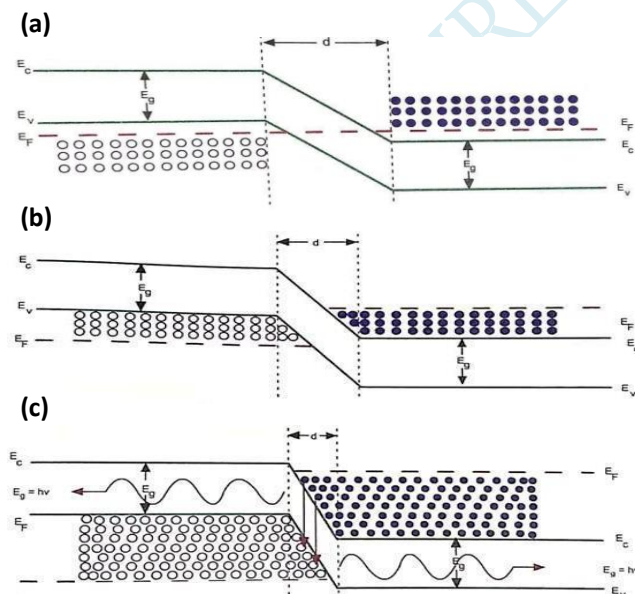


Figure 5.8: (a) At equilibrium (Unbiased condition), (b) With a small forward bias is applied and (c) When large bias is applied.

Further, the emitted photons increase the rate of recombination. Thus, a greater number of photons are produced. Hence emitted photons from induced recombination are having the same frequency as that of original induced photons. When the donor and acceptor concentrations are about 10^{24} atoms per unit volume the emitted wavelengths are $9020 \text{ \AA}$, $8425 \text{ \AA}$ and $8370 \text{ \AA}$.

Advantages of Hetero-Junction Semiconductor Laser:

- Threshold current density is very low (10 Amp/mm^2) at room temperature.

- Continuous wave operation is possible.
- Laser beam divergence is reduced.
- High output power (10 MW) can be achieved even with low operating current ($< 500\text{mA}$)
- Very narrow beam with high coherence and monochromacy is achieved.
- It is highly stable with longer life.

Applications:

- Hetero-junction diode lasers are used as optical sources in optical fiber communication.

5.11. APPLICATIONS OF LASERS: BAR CODE SCANNER

A laser barcode scanner uses a low-power semiconductor laser diode (typically emitting visible red light at 650 nm) to read alternating dark and light bars. An internal moving mirror sweeps the beam across the code, a sensor measures the reflected light intensity, and internal electronics decode the pattern into digital data.



Figure 5.10: Laser Bar code Scanner

Working:

A laser barcode scanner works by projecting a focused red beam of light across a series of dark and light bars. The white spaces reflect the light back strongly, while the black bars absorb it. A photodiode sensor measures this reflected pattern, converts it into electrical signals, and decodes it into readable digital data.

Main Components

Laser Diode: Emits the initial coherent beam of visible red light (typically 650 nm).

Oscillating/Rotating Mirror: Rapidly sweeps the stationary laser beam back and forth to create a visible scan line across the code.

Photodiode Sensor: Captures the varying intensity of light bouncing off the barcode.

Decoder Chip: Converts analog electrical pulses into digital strings of characters that match the item's identification number.

Step-by-Step Working Principle

Illumination: The internal diode projects a thin laser beam onto the target barcode via moving mirrors. White or light spaces reflect the beam back, while black or dark bars absorb the light energy.

Optics & Detection: The reflected light hits the scanner optics, directing the raw data into the photodiode sensor which registers high and low signal fluctuations. An analog-to-digital converter (ADC) turns the wave changes into binary values (ones and zeros).

Output: The internal decoder reads the binary sequence, validates the checksum, and transmits text data to the connected computer or POS system.

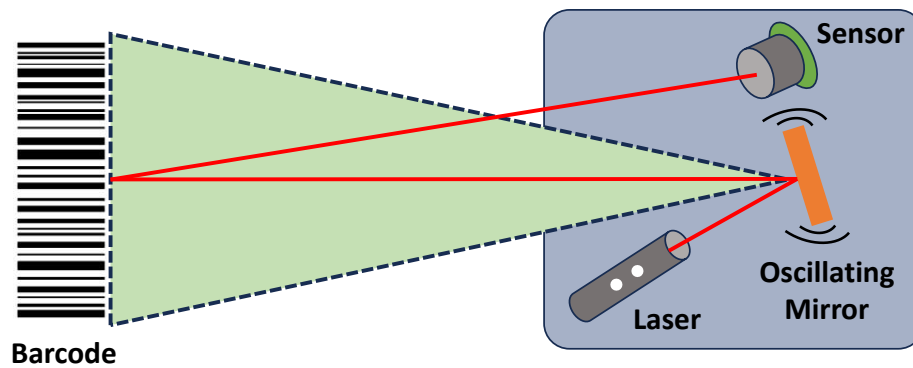
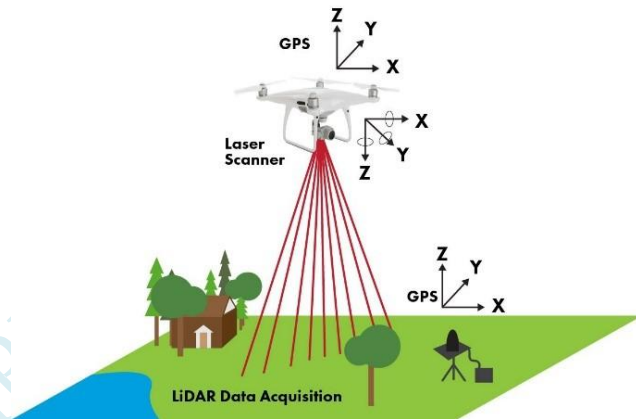


Figure 5.11: Barcode scanner working principle.

5.12. LIDAR (LIGHT DETECTION AND RANGING)

LiDAR, which stands for *Light Detection and Ranging*, is a remote sensing method used for measuring the exact distance of an object on the earth's surface.

LiDAR uses pulsing laser beams to measure distances and create precise, real-time 3D "point cloud" maps of a vehicle's environment. It serves as a core technology for autonomous vehicles, allowing self-driving systems to "see" and navigate safely around obstacles in dynamic road conditions.



Core Components

Laser Transmitter: Sends infrared light pulses (commonly at 905 nm or 1550 nm wavelengths).

Photodetector Receiver: Captures the light bounced back from nearby surfaces.

Scanner / Processing Unit: Translates spatial echo data into digital depth points and coordinates.

Uses:

- **3D Environmental Mapping:** Emits hundreds of thousands of lasers pulses every second to build a live, 360-degree topographical layout of the surroundings.
- **Obstacle Detection:** Identifies and tracks road hazards like pedestrians, animals, other vehicles, debris, and potholes.
- **Distance and Speed Calculation:** Uses Time-of-Flight (ToF) or Frequency Modulated Continuous Wave (FMCW) principles to calculate exact proximity and vector movements of surrounding objects.
- **Active Night Vision:** Operates independently of ambient light, ensuring high-accuracy tracking at night or in low-visibility environments.

Applications:

Agriculture & Archaeology, Atmospheric Physics, Military, Meteorology, Oceanography, etc.

5.13. FIBER OPTICS

5.13.1. Introduction

Fiber optics is a branch of physics based on the transmission of light through transparent fibers of glass or plastic. These fibers can carry light over distances ranging from few meters to hundreds of kilometers. **John Tyndall**, a British physicist demonstrated that light can be guided along a curved stream of water, employing the phenomena of **Total Internal Reflection**. Modern electronic communications such as telegraphy, telephony and radio communication use radio waves and microwaves to carry information over copper wires and co-axial cables. The information carrying capacity of these wires is restricted. The use of light in place of radio waves and microwaves enhances tremendously the number of signals that can be transmitted simultaneously.

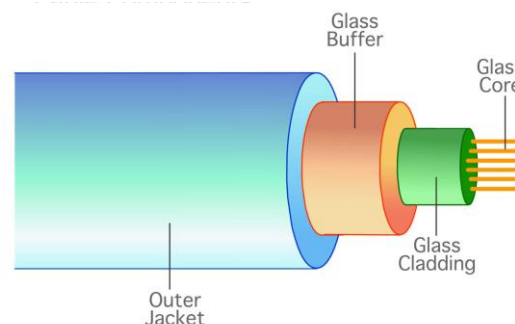


Figure 5.12: Optical fiber cable layout and its internal components.

The fundamental principle behind the operation of an optical fiber is 'Total Internal Reflection (TIR)'. This phenomenon allows light to be trapped and transmitted over long distances through a flexible glass or plastic without escaping through the sides.

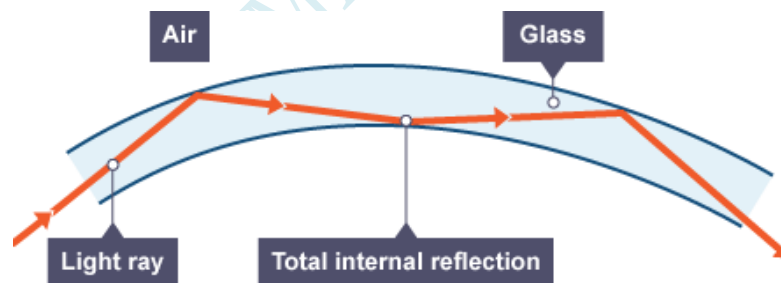


Figure 5.13: Total internal reflection in optical fiber.

5.14. TOTAL INTERNAL REFLECTION

The *total internal reflection* is the complete reflection of a light ray within the medium (air, water glass, etc.). The total internal reflection occurs when light travels from a **denser medium to a less dense medium** and the *angle of incidence must be greater than the critical angle*, as shown in figure 5.14. The phenomenon of total internal reflection has a wide application in optical devices such as telescopes, binoculars, periscopes, etc.

- Consider the case below.
 - A beam of light travels from a watery medium (n_1) to one of air (n_2). The light ray will be refracted at the point where the two mediums meet. The refracted light beam bends away from the normal as it passes from a medium with a higher refractive

index to one with a lower refractive index.

- The incident beam of light is refracted in such a way that it travels over the water's surface at a certain angle of incidence, called critical angle (θ_c).
- The incident ray is reflected back to the medium when the angle of incidence is larger than the critical angle. This event is referred to as total internal reflection. The fiber's core (the inner part) must have a *higher refractive index* (n_1) than the cladding (the outer layer, n_2).

The formula for total internal reflection is stated as:

$$\frac{n_1}{n_2} = \frac{\sin \theta_1}{\sin \theta_2}$$

where, θ_1 is the angle of incidence, θ_2 is the angle of refraction, n_1 is the denser medium & n_2 is the rarer medium.

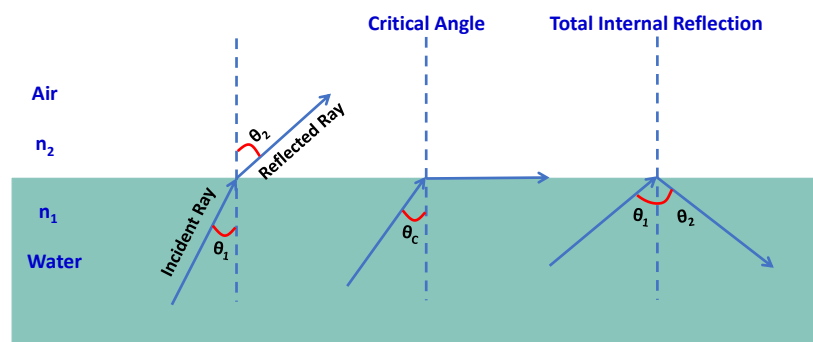


Figure 5.14: Total internal reflection.

Critical Angle: Critical angle is defined as the largest angle of incidence for which refraction of light inside the medium still can be possible. Therefore, the angle of incidence, when the angle of refraction is 90 degrees and the ray which is refracting to total internal refraction is called the critical angle.

Mathematically, the critical angle for a total internal reflection is defined as,

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

5.15. ACCEPTANCE ANGLE

The maximum incidence angle of an optical fiber at which the light can enter the core is known as the acceptance angle, as shown in Fig. 5.15(a). So, any ray with an incident angle less than the acceptance angle can be guided through the fiber.

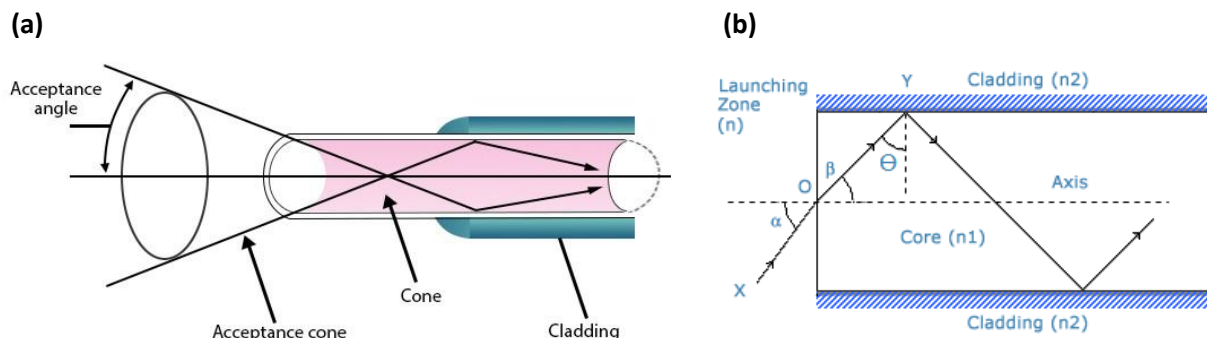


Figure 5.15: Acceptance angle of fiber optical cable.

Consider the light is launched from air where refractive index is n . The light ray enters the core at an angle, α (incidence) and refracts at an angle, β (refractive). The ray suffers a total internal reflection at 'Y'.

Angle $\beta = 90 - \theta$.

The Snell's law,

$$\begin{aligned}\therefore n \sin \alpha &= n_1 \sin \beta \\ \Rightarrow \frac{\sin \alpha}{\sin \beta} &= \frac{n_1}{n} \\ \Rightarrow \sin \alpha &= \frac{n_1}{n} \sin \beta \\ \Rightarrow \sin \alpha &= \frac{n_1}{n} \cos \theta_c \quad (\text{where } \beta = 90 - \theta_c) \quad \text{----- (1)}\end{aligned}$$

Using the trigonometric identity, the solution for acceptance angle,

$$\begin{aligned}\sin^2 \theta_c + \cos^2 \theta_c &= 1 \\ \cos^2 \theta_c &= 1 - \sin^2 \theta_c \\ \therefore \cos \theta_c &= \sqrt{1 - \sin^2 \theta_c} \\ \cos \theta_c &= \sqrt{1 - \frac{n_2^2}{n_1^2}} \quad \text{----- (2)}\end{aligned}$$

Substitute equation (2) in equation (1), we get

$$\sin \alpha = \frac{n_1}{n} \sqrt{1 - \frac{n_2^2}{n_1^2}} = \frac{n_1}{n} \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}} = \frac{\sqrt{n_1^2 - n_2^2}}{n}$$

If the medium surrounding the fiber is air, then $n = 1$.

$$\begin{aligned}\sin \alpha &= \sqrt{n_1^2 - n_2^2} \\ \alpha &= \sin^{-1} \sqrt{n_1^2 - n_2^2}\end{aligned}$$

This is the final expression of **acceptance angle**.

5.16. NUMERICAL APERTURE

Acceptance angle is related to the numerical aperture (NA) of an optical fiber. *Numerical Aperture is a dimensionless quantity that determines the ability of a system to gather light.* It is the measure of the range of angles over which the system can accept light or the angular acceptance of a system for the incoming light.

$$NA = n \sin \theta$$

Here n is the refractive index of the medium from which light is entering the fiber and θ is the acceptance angle.

We have another equation for the numerical aperture as,

$$NA = n \sin \theta = \sqrt{n_1^2 - n_2^2}$$

Where n_1 is the refractive index of the core and n_2 is the refractive index of the cladding.

From the above equation, the acceptance angle can be calculated as,

$$\theta = \sin^{-1} \sqrt{\frac{n_1^2 - n_2^2}{n^2}}$$

5.17. CLASSIFICATION OF OPTICAL FIBRES

Optical fibers are classified into three categories based on materials, modes, refractive index.

Materials: 1. Glass fibers and 2. Plastic fibers

No. of Modes: 1. Single mode fibers and 2. Multimode fibers

Refractive index profile: 1. Step index fiber and 2. Graded index fiber.

Single mode fiber- Single mode fiber typically has a core diameter of about $9 \mu\text{m}$, which allows only one mode of light to propagate. Single mode fiber is typically used with laser light sources at wavelengths such as 1310 nm and 1550 nm . Single mode fiber, by contrast, is better suited for long-distance transmission, with typical reaches ranging from several kilometers to tens of kilometers.

Multimode fiber- Multimode fiber has a larger core, usually $50 \mu\text{m}$ or $62.5 \mu\text{m}$, enabling multiple light modes to travel simultaneously. Multimode fiber is most commonly used with VCSEL light sources at 850 nm , while 1300 nm wavelengths and LED sources may still appear in some legacy applications. Multimode fiber is generally used for short-reach links, with supported distances varying by data rate, transceiver type, and fiber grade, typically ranging from tens to hundreds of meters.

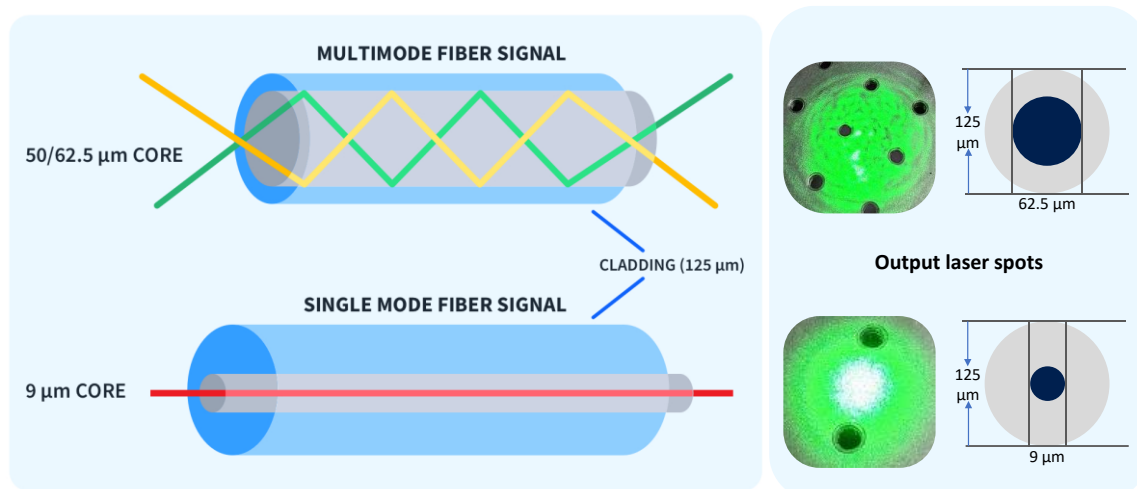


Figure 5.16: Schematic comparison of light propagation in multimode and single-mode optical fibers, along with their output laser spot distributions.

Optical fibers are classified by **refractive index profile** into Step-Index and Graded-Index. By **modes of propagation**, they are classified as Single Mode and Multimode.

Step Index Multimode Fiber: Step index multimode fiber was the first fiber design. The core of step index multimode fiber is made completely of one type of optical material and the cladding is another type with different optical characteristics. It has higher attenuation and is too slow for many uses, due to the dispersion caused by the different path lengths of the various modes traveling in the core. When a digital signal is allowed to pass through it, it may choose a zig-zag path or a straight-line path along the axis or parallel to the axis. The signal pulse that has chosen the straight-line path will come first and the other ray will follow with a time lag resulting in a signal broadening. This is known as **inter model dispersion**. This can be avoided by graded index fiber.

Graded index Multimode Fiber: Graded index multimode fiber uses variations in the composition of the glass in the core to compensate for the different path lengths of the modes. It offers hundreds of times more bandwidth than step index fiber - up to about 4 gigahertz/km.

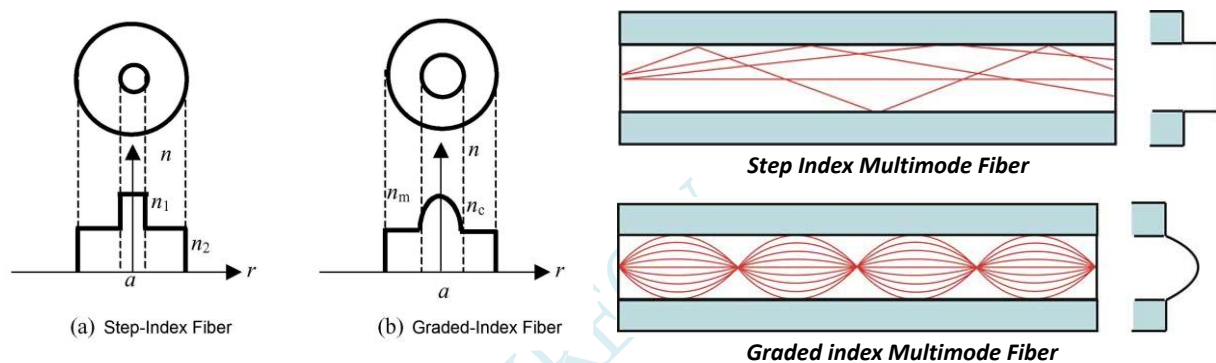


Figure 5.17: Step and graded multimode fibers.

Single Mode Fiber: Supports only one mode (path) of propagation, usually the fundamental mode. It has a very small core (typically 8-12 μm) and is used for long-distance, high-bandwidth applications due to low dispersion.

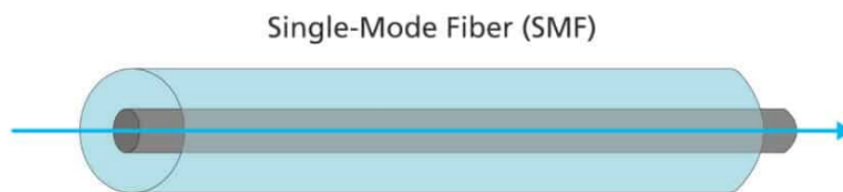


Figure 5.18: Single-mode fiber.

5.18. CONSTRUCTION OF OPTICAL FIBRE

An optical fiber is a thin, flexible, transparent strand made of high-quality glass (silica) or plastic designed to transmit data using light pulses. Its engineering relies on the principle of total internal reflection, requiring a dense inner layer and a less dense outer layer to guide light waves without signal escape

An optical fiber is typically fabricated with a multi-layered concentric structure designed to guide light efficiently via total internal reflection.

Core: The innermost central region, typically made of high-purity silica glass (SiO_2) or plastics.

The core has a higher refractive index (n_1) to carry the light signal along the fiber axis. Its diameter varies depending on the type,

- **Single-Mode Fiber (SMF):** Microscopic core, usually $\sim 9\ \mu\text{m} - 10\ \mu\text{m}$ in diameter, allowing only a single mode of light to travel.
- **Multimode Fiber (MMF):** Larger core, typically $50\ \mu\text{m}$ or $62.5\ \mu\text{m}$ in diameter, allowing multiple light rays/modes to propagate simultaneously.

Cladding: The outer optical layer surrounding the core, made of pure silica or doped glass with a lower refractive index ($n_2 < n_1$). The step change or gradual variation in refractive index between the core and cladding confines light within the core through total internal reflection. Regardless of core size, the outer cladding diameter is standardly $125\ \mu\text{m}$.

Primary Coating (Buffer Coating): A protective layer of acrylate resin, silicone, or polyimide applied directly over the cladding (typically bringing the outer diameter to $250\ \mu\text{m}$). It protects the glass against micro-abrasions, moisture, and mechanical damage.

Secondary Buffer / Tight Buffer (Optional): A secondary layer of hard plastic (such as PVC or Nylon, typically $900\ \mu\text{m}$ in diameter) added to individual fibers for mechanical isolation and extra physical protection.

Outer Jacket: The outermost protective sheath, extruded from tough polymer materials like Polyurethane (PU), PVC, or LSZH (Low Smoke Zero Halogen). It shields the internal components from environmental factors, chemical exposure, UV degradation, and physical crushing.

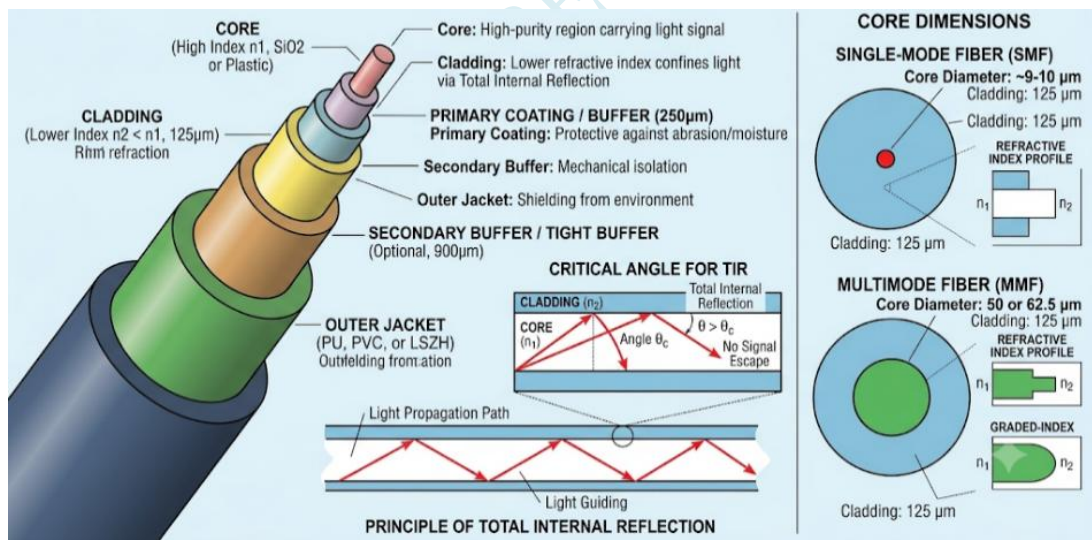


Figure 5.19: Comprehensive overview of optical fiber construction and total internal reflection (TIR).

5.19. OPTICAL FIBER COMMUNICATION SYSTEM

An optical fiber communication system transmits information by converting electrical signals into light pulses, which travel through a fiber optical cable via **total internal reflection**. The system consists of a (i) transmitter (source), (ii) an optical fiber (channel), and (iii) a receiver (detector) to achieve high-speed, long-distance communication.

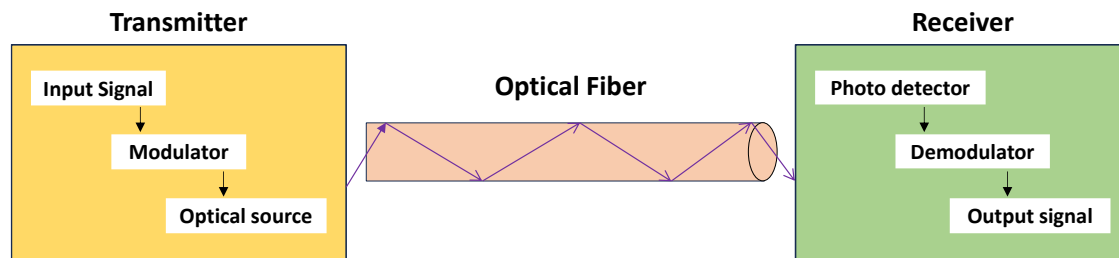


Figure 5.20: Block diagram of optical fiber in communication system.

Transmitter: Converts electrical signals (voice, data, video) into light pulses using a light-emitting diode (LED) or laser diode.

Optical Fiber: A flexible fiber of glass or plastic that guides the light pulses using Total Internal Reflection.

Receiver: Contains a photodetector (such as a PIN or APD diode) that converts the received light pulses back into electrical signals.

Advantages: (i) Extremely high data transmission capacity, (ii) Low loss of signal over long distances and (iii) Free from electromagnetic interference.

5.20. LOSSES IN OPTICAL FIBER

The intensity of the light at the receiving end of the fiber is always found to be less than the intensity at the input end. This is known as attenuation or loss in optical fiber. It is measured in decibels per kilometer (dB/km), with relation:

$$\alpha = \frac{10}{L} \log \left(\frac{P_o}{P_i} \right)$$

Where, L is Length of the fiber (km), P_i , P_o are input and output powers.

The main causes of losses are, a) Scattering loss, b) Absorption loss c) Bending loss and d) Loss due to dispersion.

a) Scattering loss: It happens because of microscopic variations in the density of the glass. The light is not absorbed, but sent into another direction. When light hits these tiny "bumps" or inconsistencies, it scatters in all directions. Some of that light escapes the core entirely, leading to signal loss.

b) Absorption loss: This occurs when the material of the fiber (silica glass) absorbs light energy and converts it into heat.

a. Intrinsic Absorption: Caused by the natural properties of the pure glass itself. Even the clearest glass absorbs some light at specific wavelengths.

b. Extrinsic Absorption: Caused by impurities trapped in the glass during manufacturing, most notably hydroxyl (OH^-) ions (water vapor). These create "water peaks" where attenuation is significantly higher.

c) Bending loss:

a. Macro bending: Large-scale bends that you can see with your eyes (like a sharp kink in the cable). If the bend is too sharp, the light hits the cladding at an angle smaller than the

critical angle and leaks out.

b. Micro bending: Microscopic bends in the fiber axis caused by improper cable coating or mechanical stress (pressure) during installation.

d) Loss due to dispersion: While dispersion technically refers to signal distortion, it can contribute to the overall degradation of the signal quality. Additionally, coupling losses occur at joints, splices, or connectors where two fibers meet.

Fiber loss in a network can be minimized by following these procedures: Polish and clean the ends, minimize end gaps, Connect the same-size fibers, Avoid unnecessary bending, Protection of the connection.

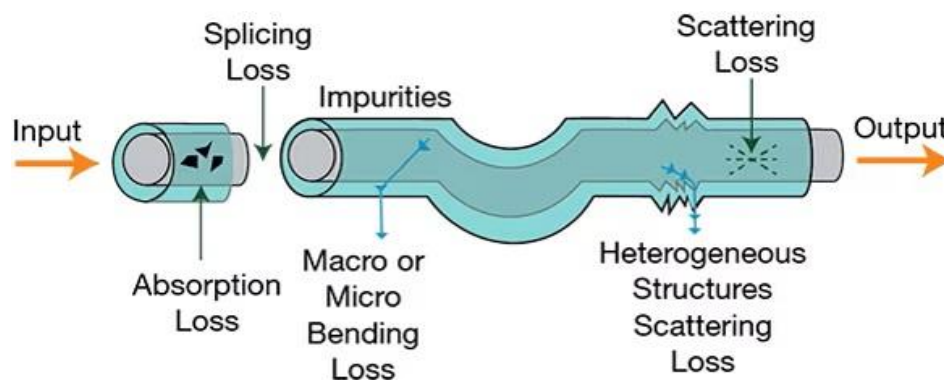


Figure 5.21: Different types of losses in optical fiber.

5.21. APPLICATIONS OF OPTICAL FIBER

1. Optical Fibers in Communication Systems

- **Telecommunications and Internet Backbones:** Form the core global network infrastructure, transmitting massive high-speed data streams across long-haul subsea and terrestrial cables with minimal attenuation.
- **Fiber to the Home (FTTH) / Last-Mile Access:** Delivers gigabit broadband, high-definition television, and VoIP directly to residential and commercial end-users via passive optical networks (PON).
- **Data Center Interconnects (DCI):** Provides ultra-low latency, high-bandwidth interconnectivity between server racks and data centers to handle high-density cloud computing and AI workloads.
- **Cellular Mobile Backhaul:** Connects 5G base stations and cell towers to central processing hubs, supporting low-latency data traffic and dense network architectures.
- **Secure and Local Area Networks (LANs):** Used in enterprise, military, and aerospace environments where high security, immunity to electromagnetic interference (EMI), and long-reach connectivity are critical.

2. Optical Fiber Sensors for Structural Health Monitoring (SHM)

- **Strain and Stress Monitoring:** Embedded directly into concrete structures, steel

bridges, and composite airplane wings (using Fiber Bragg Grating or distributed sensing) to detect mechanical stress and micro-deformations.

- **Bridge and Civil Infrastructure Safety:** Tracks load distribution, expansion joint displacement, and structural degradation in bridges, tunnels, and dams over long spans.
- **Pipeline Integrity Management:** Continuous distributed temperature and acoustic sensing (DTS/DAS) along oil, gas, and water pipelines to pinpoint leaks, ground movement, or unauthorized intrusion.
- **Geotechnical and Slope Stability Sensing:** Monitors ground deformation, landslide risks, and subsidence near railway lines, highways, and retaining walls.
- **Wind Turbine and Aerospace Diagnostics:** Real-time monitoring of bending strain, fatigue, and structural integrity in wind turbine blades and aircraft airframes under dynamic operating loads.

Problems

Problem1: Find the frequency of laser radiation whose wavelength is 632.8 nm. ?

Solution: $f=c/\lambda=3\times 10^8/(632.8\times 10^{-9})=4.74\times 10^{14}$ Hz.

Answer: $\approx 4.74\times 10^{14}$ Hz.

Problem2: Find the energy of photon whose wavelength 632.8 nm.

Solution: $E=hc/\lambda=(6.626\times 10^{-34}\times 3\times 10^8)/(632.8\times 10^{-9})=3.14\times 10^{-19}$ J.

Answer: $\approx 3.14\times 10^{-19}$ J.

Problem3: A laser emits 2×10^{18} photons/s at a wave length 650 nm. Find power.

Solution: $E_{ph}=hc/\lambda\approx 3.06\times 10^{-19}$ J. $P=N E=2\times 10^{18}\times 3.06\times 10^{-19}\approx 0.612$ W.

Answer: ≈ 0.612 W.

Problem4: A 5 mW laser emits at 632.8 nm. Find photons emitted per second.

Solution: $E_{ph}\approx 3.14\times 10^{-19}$ J. $N=P/E=0.005/(3.14\times 10^{-19})=1.59\times 10^{16}$ photons/s.

Answer: $\approx 1.59\times 10^{16}$ s⁻¹.

Problem5: $n_1=1.48$, $n_2=1.46$. Find NA in air.

Solution: $NA=\sqrt{(n_1^2-n_2^2)}=\sqrt{(2.1904-2.1316)}=\sqrt{0.0588}=0.2425$.

Answer: $NA\approx 0.243$.

Problem6: $n_1=1.50$, $n_2=1.45$. Find acceptance angle in air.

Solution: $NA=\sqrt{(2.25-2.1025)}=0.3839$. $\theta_a=\sin^{-1}(0.3839)\approx 22.57^\circ$.

Answer: $\approx 22.6^\circ$.

Problem7: A fibre has $NA=0.22$. Find acceptance angle in air.

Solution: $\theta_a=\sin^{-1}(0.22)=12.71^\circ$.

Answer: $\approx 12.7^\circ$.

Problem8: $n_1=1.50$ and $NA=0.25$. Find n_2 .

Solution: $NA^2=n_1^2-n_2^2$. $n_2=\sqrt{(2.25-0.0625)}=\sqrt{2.1875}=1.479$.

Answer: $n_2 \approx 1.48$.

Problem9: A 5 km fibre has attenuation 2 dB/km. Input power is 10 mW. Find output power.

Solution: Loss = $2 \times 5 = 10$ dB. $P_{out} = P_{in} \times 10^{(-10/10)} = 10 \times 0.1 = 1$ mW.

Answer: 1 mW.

Problem10: Input power is 2 mW and output is 0.5 mW after 4 km. Find total loss and attenuation/km.

Solution: Loss = $10 \log_{10}(P_{in}/P_{out}) = 10 \log_{10}(4) = 6.02$ dB. Attenuation = $6.02/4 = 1.505$ dB/km.

Answer: 6.02 dB total, 1.51 dB/km.

Problem11: Acceptance angle is 12° in air. Find NA.

Solution: $NA = \sin 12^\circ = 0.2079$.

Answer: ≈ 0.208 .

Problem12: $n_1 = 1.50$, $n_2 = 1.47$, length = 10 km, attenuation = 0.3 dB/km. Find NA, acceptance angle and output/input power ratio.

Solution: $NA = \sqrt{(2.25 - 2.1609)} = 0.2985$. $\theta_a = \sin^{-1}(0.2985) = 17.36^\circ$. Loss = 3 dB.

Ratio = $10^{(-3/10)} = 0.501$.

Answer: $NA \approx 0.299$, $\theta_a \approx 17.4^\circ$, $P_{out}/P_{in} \approx 0.501$.

Problem13: A 10 mW laser has 60% coupling efficiency. Fibre length = 8 km, attenuation = 0.4 dB/km. Find output power.

Solution: $P_{in, fibre} = 10 \times 0.60 = 6$ mW. Loss = $0.4 \times 8 = 3.2$ dB. $P_{out} = 6 \times 10^{(-3.2/10)} = 2.87$ mW.

Answer: ≈ 2.87 mW.

Problem14: A diode laser operates at 850 nm and emits 3 mW. Find photon energy and photon rate.

Solution: $E_{ph} = hc/\lambda = 2.34 \times 10^{-19}$ J. $N = P/E = 0.003/(2.34 \times 10^{-19}) = 1.28 \times 10^{16}$ photons/s.

Answer: $E_{ph} \approx 2.34 \times 10^{-19}$ J; $N \approx 1.28 \times 10^{16} \text{ s}^{-1}$.

Problem15: A 1550 nm laser launches 5 mW into a 20 km fibre with attenuation 0.25 dB/km. Connector loss is 2 dB. Find total loss, received power and photon energy.

Solution: Fibre loss = $0.25 \times 20 = 5$ dB. Total loss = 7 dB. $P_{out} = 5 \times 10^{(-7/10)} = 0.998$ mW.

Photon energy = $hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8)/(1550 \times 10^{-9}) = 1.28 \times 10^{-19}$ J.

Answer: total loss = 7 dB, $P_{out} \approx 1.0$ mW, $E_{ph} \approx 1.28 \times 10^{-19}$ J.

Review Questions

Short Answer Questions (1 Mark)

1. What is stimulated emission?
2. Define population inversion in lasers.
3. What are Einstein's coefficients?
4. Name the key characteristics of laser light.
5. Define acceptance angle of an optical fiber.
6. What is Numerical Aperture (NA)?

7. State the basic principle behind light propagation in an optical fiber.
8. Name two types of losses in optical fibers.
9. Mention an application of LIDAR.
10. Distinguish between single-mode and multi-mode optical fibers.

Long Answer Questions (10 Marks)

1. Explain the principle of laser action. Derive the relation between Einstein's A and B coefficients for spontaneous and stimulated emissions.
2. Describe the construction, energy level diagram, and working principle of a Helium-Neon (He-Ne) laser.
3. Explain the construction and operational principle of a Semiconductor Diode laser and a CO₂ laser.
4. Derive expressions for the acceptance angle and Numerical Aperture (NA) of a step-index optical fiber.
5. Classify optical fibers based on refractive index profile (step-index, graded-index) and modes of propagation (single-mode, multi-mode).
6. Compare their characteristics. Describe the various attenuation/loss mechanisms in optical fibers (absorption, scattering, and bending losses).
7. Explain the working principle, block diagram, and advantages of an Optical Fiber Communication (OFC) system.
8. Discuss practical applications of lasers in technology, focusing on barcode scanners and LIDAR systems for autonomous vehicles.
9. Explain how fiber optic sensors work, detailing their application in structural health monitoring.
10. Describe the conditions necessary to achieve laser action: metastable states, pumping mechanisms, optical resonators, and active media.

Objective Type Questions

Multiple Choice Questions

1. LASER stands for

A) Light Amplification by Stimulated Emission of Radiation	emission
B) Light Absorption by Spontaneous Emission of Radiation	C) Spontaneous emission
C) Light Amplification by Solar Emission of Radiation	D) Reflection
D) Linear Absorption of Stimulated Energy Radiation	
2. Laser radiation is highly

A) Incoherent	B) Coherent
C) Random	D) Divergent
3. Laser light is highly directional because of its

A) Low intensity	B) Low divergence
C) Random phase	D) Large bandwidth
4. The process in which an atom absorbs a photon and moves to a higher energy state is

A) Absorption	B) Stimulated
---------------	---------------
5. Emission triggered by an incident photon is called

A) Stimulated emission	B) Absorption
C) Spontaneous absorption	D) Diffraction
6. Emission that occurs without external stimulation is called

A) Stimulated emission	B) Spontaneous emission
C) Population inversion	D) Pumping
7. A long-lived excited state in a laser medium is called a

A) Metastable state	B) Ground state
C) Valence state	D) Forbidden band

Optics

8. Population inversion means
 - A) More atoms in the upper laser level than the lower level
 - B) More atoms in the ground state
 - C) No excited atoms
 - D) Equal numbers at all levels
9. Pumping is used to
 - A) Supply energy to the active medium
 - B) Remove photons
 - C) Reduce coherence
 - D) Increase fiber loss
10. The optical resonator in a laser generally consists of
 - A) Two mirrors
 - B) Two lenses only
 - C) Two prisms only
 - D) One capacitor
11. Ruby laser is a
 - A) Solid-state laser
 - B) Gas laser
 - C) Liquid laser
 - D) Fiber detector
12. The active medium in a ruby laser is
 - A) Ruby crystal
 - B) Helium gas
 - C) Neon gas only
 - D) CO₂ gas
13. He–Ne laser is a
 - A) Gas laser
 - B) Solid laser
 - C) Semiconductor detector
 - D) Fiber laser
14. CO₂ laser is classified as a
 - A) Gas laser
 - B) Solid-state laser
 - C) Diode detector
 - D) Magnetic laser
15. A semiconductor laser commonly uses a
 - A) p–n junction
 - B) Grain boundary
 - C) Dielectric capacitor
 - D) Magnetic domain
16. A heterojunction semiconductor laser contains
 - A) Two semiconductor materials of different band gaps
 - B) Only one metal
 - C) Two optical fibers only
 - D) No active region
17. One application of lasers mentioned in the syllabus is
 - A) Bar-code scanning
 - B) Crystal packing
 - C) Transformer winding
 - D) Dielectric polarization
18. LIDAR systems for autonomous vehicles use
 - A) Laser radiation
 - B) X-rays
 - C) Sound waves only
 - D) Gamma rays
19. Optical fiber communication primarily uses
 - A) Light signals
 - B) Mechanical waves
 - C) Magnetic domains
 - D) Nuclear radiation
20. Light propagation in an optical fiber is based on
 - A) Total internal reflection
 - B) Photoelectric effect
 - C) Compton scattering
 - D) Nuclear resonance
21. For total internal reflection, the core refractive index must be
 - A) Greater than the cladding index
 - B) Less than the cladding index
 - C) Equal to zero
 - D) Independent of cladding
22. The central light-guiding part of an optical fiber is the
 - A) Core
 - B) Cladding
 - C) Coating
 - D) Jacket only
23. The optical layer surrounding the core is the
 - A) Cladding
 - B) Detector
 - C) Resonator
 - D) Active medium
24. Numerical aperture indicates the fiber's
 - A) Light-gathering ability
 - B) Electrical conductivity
 - C) Magnetic moment
 - D) Mechanical hardness
25. A fiber supporting only one propagation mode is called
 - A) Single-mode fiber
 - B) Multimode fiber
 - C) Step-only fiber
 - D) Plastic-only fiber

Fill in the Blanks

1. LASER stands for Light Amplification by Stimulated Emission of _____.
2. Laser radiation is highly _____.
3. Laser radiation has high spatial and temporal _____.
4. Absorption involves transition from a lower energy state to a _____ energy state.
5. Emission caused by an incident photon is called _____ emission.
6. Emission without external stimulation is called _____ emission.

7. A long-lived excited state is called a _____ state.
8. The condition required for laser action in which upper-level population exceeds lower-level population is called population _____.
9. Supplying energy to the active medium is called _____.
10. The pair of mirrors forming the optical resonator is also called the laser _____.
11. Ruby laser is a _____-state laser.
12. The active medium of ruby laser is a ruby _____.
13. He–Ne laser is a _____ laser.
14. CO₂ laser is a _____ laser.
15. Semiconductor diode lasers commonly use a _____ junction.
16. Semiconductor lasers may use homo-junction or _____-junction structures.
17. A laser application used to read product codes is the _____ scanner.
18. LIDAR for autonomous vehicles uses _____ radiation.
19. An optical fiber transmits information using _____ signals.
20. The guiding principle of an optical fiber is total _____ reflection.
21. The central region of an optical fiber is called the _____.
22. The layer surrounding the core is called the _____.
23. The angle at which light can enter and still be guided is the _____ angle.
24. The light-gathering ability of a fiber is described by numerical _____.
25. A fiber supporting only one mode is called a _____-mode fiber.

MCQs: 1-A, 2-A, 3-A, 4-A, 5-A, 6-A, 7-A, 8-A, 9-A, 10-A, 11-A, 12-A, 13-A, 14-A, 15-A, 16-A, 17-A, 18-A, 19-A, 20-A, 21-A, 22-A, 23-A, 24-A, 25-A

Fill in the blanks: 1. Radiation; 2. coherent; 3. coherence; 4. higher; 5. stimulated; 6. spontaneous; 7. metastable; 8. inversion; 9. pumping; 10. cavity; 11. solid; 12. crystal; 13. gas; 14. gas; 15. p-n; 16. hetero; 17. bar-code; 18. laser; 19. light; 20. internal; 21. core; 22. cladding; 23. acceptance; 24. aperture; 25. single.



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B. Tech I Year I SEM (R-26) Model Question Paper - 2026

Branch :
Academic Year : 2026-2027
Date of Exam : Dec- 2026

Subject Name: Advanced Engineering Physics
Max. Marks: 60 Marks
Time of Exam: 3 Hours

Name: -----

Hall Ticket No.

					A		
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PART – A

Note: This Part consists of 10 Questions. Answer all questions

(10 × 1 = 10 Marks)

S. No	Questions	Marks	BTL
1	a Define the Atomic Packing Factor (APF).	1M	Analyze
	b Give an example of a surface defect in crystalline solids.	1M	Understand
	c What is Heisenberg's uncertainty principle?	1M	Understand
	d What is the physical origin of energy band gaps in solids?	1M	Understand
	e Differentiate between a classical bit and a quantum bit.	1M	Classify
	f What is a Hadamard gate?	1M	Understand
	g What are soft and hard magnetic materials?	1M	Explain
	h What is Giant Magnetoresistance (GMR)?	1M	Explain
	i Define population inversion in lasers.	1M	Understand
	j Distinguish between single-mode and multi-mode optical fibers.	1M	Classify

PART – B

(5 × 10 = 50M)

- Note:**
1. This Part consists of 10 QUESTIONS
 2. Answer any 1 question from each Section. Each question carries 10 Marks.
 3. Illustrate your answers with NEAT sketches wherever necessary.

SECTION - I

2.A	Explain unit cell. Compare SC, BCC and FCC structures with respect to atoms/unit cell, coordination number and atomic packing factor.	5M	Explain
2.B	Explain Miller indices and derive the expression for inter-planar distance in a cubic crystal.	5M	Explain

(OR)

3.B	State Bragg's law and explain the powder X-ray diffraction method.	5M	Understand
3.B	Explain the working of XRD.	5M	Understand

SECTION - II

4.A	Explain de Broglie hypothesis, Heisenberg uncertainty principle and physical significance of the wave function.	5M	Explain
4.B	Derive the time-independent Schrödinger equation for a particle in a one-dimensional infinite potential box and obtain the allowed energy levels.	5M	Analyze

(OR)

5.A	Discuss operators, eigenvalues, eigenfunctions and expectation value.	5M	Understand
5.B	Explain discrete energy levels and quantum confinement.	5M	Explain

SECTION - III

6.A	Explain Dirac bra-ket notation, Hilbert space and the Bloch sphere representation of a single qubit.	5M	Explain
6.B	Explain Hadamard, Pauli-X and CNOT gates with suitable state transformations.	5M	Explain

(OR)

7.A	Discuss quantum measurement and entanglement.	5M	Understand
7.B	Discuss the basic principles and applications of any two of Deutsch–Jozsa, Shor and Grover algorithms.	5M	Understand

SECTION - IV

8.A	Classify magnetic materials and explain the origin of magnetic moment.	5M	Classify
8.B	Describe Weiss domain theory and hysteresis. Differentiate soft and hard magnetic materials with applications.	5M	Understand

(OR)

9.A	Explain electronic, ionic and orientational polarization. Discuss ferroelectric, piezoelectric and pyroelectric materials	5M	Explain
9.B	Describe the applications of ferroelectric materials including Fe-RAM, load cell and fire sensor	5M	Understand

SECTION - V

10.A	Explain interaction of radiation with matter using Einstein coefficients. Discuss metastable state, population inversion, pumping and lasing action.	5M	Explain
10.B	Explain Ruby or He–Ne laser.	5M	Explain

(OR)

11.A	Explain optical fiber construction and total internal reflection. Derive expressions for acceptance angle and numerical aperture.	5M	Explain
11.B	Discuss fiber types, losses and applications in communication and structural-health monitoring.	5M	Understand

**** BEST OF LUCK ****



MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)

Accredited by NAAC with 'A+' Grade | Programmes Accredited by NBA
National Ranking by NIRF Innovation – Rank band (151-300), MHRD, Govt. of India
Approved by AICTE, Affiliated to JNTUH, ISO 9001:2015 Certified Institution
Maisammaguda, Dhulapally, Secunderabad 500100.



B. Tech I Year I SEM (R-26) Model Question Paper - 2026

Branch : AIML,DS,CS, IT

Subject Name: Advanced Engineering Physics

Academic Year : 2026-2027

Max. Marks: 60 Marks

Date of Exam : Dec- 2026

Time of Exam: 3 Hours

Name: -----

Hall Ticket No.

PART – A

					A				

Note: This Part consists of 10 Questions. Answer all questions

(10 × 1 = 10 Marks)

S. No	Questions	Marks	BTL
1	a Define Unit Cell and Bravais lattice.	1M	BTL 1
	b State Bragg's law and write the expression.	1M	BTL 1
	c What is the physical significance of the wave function?	1M	BTL 1
	d Define eigenvalue and eigenfunction.	1M	BTL 1
	e What is meant by Hilbert space in quantum computation?	1M	BTL 1
	f Define quantum gates?	1M	BTL 1
	g Define magnetic dipole moment.	1M	BTL 1
	h What is Ferroelectric RAM (Fe-RAM)?	1M	BTL 1
	i Define population inversion and metastable state.	1M	BTL 1
	j Define the principle on which Optical Fibers works.	1M	BTL 1

PART – B

(5 × 10 = 50M)

- Note:**
1. This Part consists of 10 QUESTIONS
 2. Answer any 1 question from each Section. Each question carries 10 Marks.
 3. Illustrate your answers with NEAT sketches wherever necessary.

SECTION - I

2.A	Explain Simple Cubic, Body Centered Cubic and Face Centered Cubic crystal structures and compare their atomic packing factors.	5M	BTL 2
2.B	Derive Bragg's law for X-ray diffraction. Explain how XRD data is used to determine crystallite size and lattice parameter.	5M	BTL 2

(OR)

3.B	Explain the concept of nanomaterials with respect to the surface-to-volume ratio. Discuss how the surface-to-volume ratio changes when the particle size is reduced to the nanoscale	5M	BTL 2
3.B	Find the Miller indices of the atomic planes having intercepts a, 2a and ∞ on x, y and z axes respectively.	5M	BTL 3

SECTION - II

Advanced Engineering Physics**Model Question Papers**

4.A	Derive one dimensional time independent Schrodinger wave equation.	5M	BTL 3
4.B	Describe the Kronig-Penney model qualitatively and explain how it leads to the formation of allowed and forbidden energy bands in solids using the E-k diagram.	5M	BTL 2

(OR)

5.A	Explain quantum confinement in nanomaterials and analyze its effect on discrete energy levels.	5M	BTL 4
5.B	Compute the De-Broglie wavelength of an electron whose kinetic energy is 10 eV.	5M	BTL 3

SECTION - III

6.A	Explain Dirac's Bra-Ket notation and its properties used in quantum computation.	5M	BTL 2
6.B	Differentiate classical bits and qubits and explain quantum entanglement.	5M	BTL 2

(OR)

7.A	State Pauli matrices. Apply Pauli matrices on the states $ 0\rangle$ and $ 1\rangle$	5M	BTL 2
7.B	Describe the Deutsch-Jozsa algorithm in detail. Explain how it demonstrates quantum speedup over classical algorithms.	5M	BTL 4

SECTION - IV

8.A	Explain the Weiss domain theory of ferromagnetism. Draw and explain the hysteresis loop (B-H curve) for ferromagnetic material.	5M	BTL 2
8.B	Describe the sol-gel synthesis method for ferrimagnetic materials. Discuss the applications of magnetic nanomaterials in cancer therapy (magnetic hyperthermia).	5M	BTL 2

(OR)

9.A	Explain the different mechanisms of dielectric polarization (electronic, ionic, and orientational)	5M	BTL 2
9.B	A magnetic material has placed in magnetic field intensity $H=500$ A/m and magnetization $M=2.0 \times 10^4$ A/m. Find susceptibility (χ).	5M	BTL 4

SECTION - V

10.A	Explain Einstein's coefficients and their relations and discuss their significance in laser action.	5M	BTL 2
10.B	Explain the construction and working of CO ₂ laser with neat and labelled diagram.	5M	BTL 3

(OR)

11.A	Derive expressions for the acceptance angle and Numerical Aperture (NA) of an optical fiber.	5M	BTL 3
11.B	What is the numerical aperture of an optical fibre cable with a clad index of 1.378 and a core index of 1.546.	5M	BTL 4

**** BEST OF LUCK ****



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B. Tech I Year I SEM (R-26) Model Question Paper - 2026

Branch : _____ Subject Name: Advanced Engineering Physics
Academic Year : 2026-2027 Max. Marks: 60 Marks
Date of Exam : July- 2027 Time of Exam: 3 Hours

Name: _____

Hall Ticket No.

PART – A

					A				
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Note: This Part consists of 10 Questions. Answer all questions

(10 × 1 = 10 Marks)

S. No	Questions	Marks	BTL
1	a What is a Schottky defect?	1M	Understand
	b Explain the surface-to-volume ratio.	1M	Analyze
	c Calculate the de-Broglie wavelength of an electron moving with a velocity of 5×10^6 m/s.	1M	Apply
	d State the Heisenberg uncertainty principle	1M	Remember
	e Define quantum entanglement	1M	Understand
	f Define bit and qubit.	1M	Analyze
	g Define Bohr magneton.	1M	Remember
	h Define polarizability.	1M	Understand
	i Write the characteristics of laser.	1M	Remember
	j What is numerical aperture of an optical fiber?	1M	Analyze

PART – B

(5 × 10 = 50M)

- Note:
1. This Part consists of 10 QUESTIONS
 2. Answer any 1 question from each Section. Each question carries 10 Marks.
 3. Illustrate your answers with NEAT sketches wherever necessary.

SECTION - I

2.A	What is space lattice and Basis? Find the packing fractions for SCC, BCC and FCC crystals.	5M	Understand
2.B	A plane cuts intercepts of 2, 2/3 along x and y axes of a crystal. Determine Miller Indices of that plane parallel to z-axis.	5M	Apply

(OR)

3.A	Derive an expression for the inter-planar spacing in case of cubic crystal.	5M	Remember
3.B	Derive Bragg's law. Describe the construction and working of Debye-Scherrer (powder) method	5M	Understand

SECTION - II

Advanced Engineering Physics

Model Question Papers

4.A	Derive the Schrodinger time-independent wave equation.	5M	Apply
4.B	Draw E-K curves for an electron in a periodic potential and explain how it leads to the formation of energy bands in solids.	5M	Remember

(OR)

5.A	Discuss the postulates of quantum mechanics. Explain the physical significance of wave function.	5M	Understand
5.B	Calculate the wavelength associated with an electron raised to a potential 1600 V.	5M	Apply

SECTION - III

6.A	What are the advantages of quantum computing over classical computation?	5M	Understand
6.B	Write a note on the Bloch sphere representation of a qubit	5M	Remember

(OR)

7.A	Explain Dirac's bra-ket notation and give the properties and operators of bra and kets.	5M	Remember
7.B	Discuss the Deutsch-Jozsa algorithm.	5M	Apply

SECTION - IV

8.A	Explain the Weiss theory of ferromagnetism and distinguish soft and hard magnetic materials based in terms of hysteresis behavior.	5M	Apply
8.B	Find the relative permeability of a ferromagnetic material if a field of strength 220 amp/meter produces a magnetization 3300 amp/meter in it	5M	Apply

(OR)

9.A	Discuss the synthesis of ferromagnetic materials using sol-gel method.	5M	Analyze
9.B	Explain the piezo and pyroelectric materials briefly and give the applications.	5M	Understand

SECTION - V

10.A	Derive the relation between the probabilities of spontaneous emission and stimulated emission in terms of Einstein's coefficients.	5M	Remember
10.B	Explain the construction, working, and energy level diagram of the He-Ne laser.	5M	Understand

(OR)

11.A	Distinguish between light propagation in step-index optical fiber and graded-index Optical fiber.	5M	Understand
11.B	Calculate the numerical aperture and acceptance angle for an optical fiber with core and cladding refractive indices being 1.48 and 1.45 respectively.	5M	Apply

**** BEST OF LUCK ****

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year II SEM (R-25) Previous Question Paper - June 2026

Branch: Common to EEE, ECE, CSE (AI&ML, DS, CS)

Subject Name: Advanced Engineering Physics

Max. Marks: 60 Marks

Time of Exam: 3 Hours

Name: -----

Hall Ticket No.

PART – A

						A			
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Note: This Part consists of 5 Questions. Answer all questions

(5 × 2 = 10 Marks)

S. No	Questions	Marks	BTL
1	a Write the classification of nano materials.	1M	Understand
	b Write Heisenberg uncertainty principle.	1M	Analyze
	c Define qubit in quantum computing.	1M	Apply
	d Write the differences between Ferro and Para magnetic materials.	1M	Remember
	e Write about population inversion and pumping.	1M	Understand

PART – B

(5 × 10 = 50M)

- Note:**
1. This Part consists of 10 QUESTIONS
 2. Answer any 1 question from each Section. Each question carries 10 Marks.
 3. Illustrate your answers with NEAT sketches wherever necessary.

SECTION - I

2.A	What are different crystal systems in three dimensions?	5M	Understand
2.B	Explain Bragg's Law of X-ray diffraction.	5M	Apply

(OR)

3.A	Write the principle and working of XRD with neat block diagram.	5M	Remember
3.B	Write the applications of nano materials.	5M	Understand

SECTION - II

4.A	Discuss the de-Broglie hypothesis of matter waves and derive the wave length equation of matter waves.	5M	Apply
4.B	Write the postulates of quantum mechanics.	5M	Remember

(OR)

5.A	Obtain Eigen values and Eigen functions of a particle in one-dimensional box.	5M	Understand
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5.B	Write the classification of solids based on band theory.	5M	Apply
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SECTION - III

6.A	Write Dirac Bra and Ket notation and its properties.	5M	Understand
6.B	What is Hilbert's space and explain Bloch's sphere?	5M	Remember

(OR)

7.A	Discuss the concept quantum computing for information processing.	5M	Remember
7.B	Explain the concept of Entanglement.	5M	Apply

SECTION – IV

8.A	Write the classification of magnetic materials and write the difference between soft and hard magnetic materials.	5M	Apply
8.B	Write the applications of different magnetic materials.	5M	Apply

(OR)

9.A	Discuss electronic and ionic polarizations.	5M	Analyze
9.B	Write the applications of ferroelectric materials.	5M	Understand

SECTION – V

10.A	Write the construction and working of Ruby laser.	5M	Remember
10.B	Discuss the applications of lasers.	5M	Understand

(OR)

11.A	Explain the terms total internal reflection, acceptance angle and numerical aperture.	5M	Understand
11.B	Write the applications of Optical fibers in communication.	5M	Apply

**** BEST OF LUCK ****